Thermodynamics of Materials

A Study Guide and Lecture Notes

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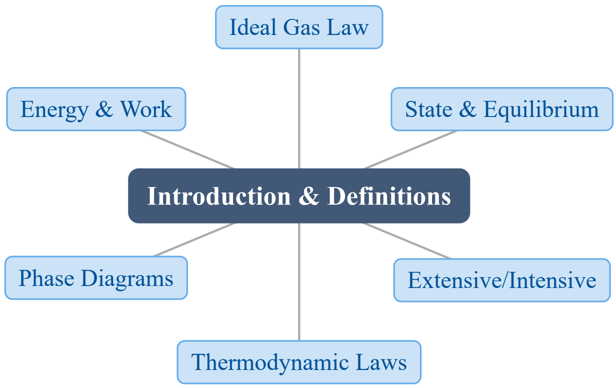
1. Chapter Overview



Thermodynamics is the **macroscopic** science of energy transformation and the direction and equilibrium of processes. It makes no assumptions about atoms, so its conclusions are universally valid. This chapter fixes the vocabulary — state functions, equilibrium, extensive/intensive variables, and the four laws — and the ideal-gas equation of state that anchors all later work.

2. Learning Objectives

After studying this chapter, students should be able to:

- Define a **state function** and use the exact-differential / zero cyclic-integral test to recognize one.
- State the ideal-gas equation $PV = nRT$ and use it for P, V, T, or n calculations.
- Distinguish **extensive** from **intensive** variables and form molar (intensive) quantities.
- Define **equilibrium** and explain why all driving forces vanish in it.
- State the four laws of thermodynamics and the SI units of energy, work, and pressure.

3. Why This Matters?

These definitions are not trivia — they are the operating system of every later calculation:

- **Process design** — furnaces, heat exchangers, and reactors are sized with P, V, T balances and energy accounting.
- **Alloy development** — phase diagrams (built on the phase/component ideas here) decide which phase is stable at a given T and composition.
- **Materials processing** — annealing, sintering, and solidification all seek an equilibrium or a controlled approach to it.
- **Metallurgy & extraction** — the first law ($\Delta U = q - w$) is the energy budget behind any thermal step.

Getting "extensive vs intensive" wrong is the single most common source of unit and scaling errors downstream.

4. Intuition First

Everyday picture A cup of hot coffee left on a desk **spontaneously cools** until it matches the room — it never reheats itself. Macroscopic properties (temperature, here) stop changing: that is **equilibrium**.

The mental model Think of the system as being fully described by a short list of dials — pressure P, volume V, temperature T, amount n. If you know the dials, you know the **state**, and any **state function** (like internal energy U) is fixed regardless of how you got there.

The concept A **state function** depends only on the current state, not the path. Its differential is **exact**: the round trip sums to zero.

The math For $Z = Z(x, y)$ a state function,

$$dZ = \frac{\partial Z}{\partial x}|_y \, dx + \frac{\partial Z}{\partial y}|_x \, dy$$

and $\frac{\partial^2 Z}{\partial x \partial y} = \frac{\partial^2 Z}{\partial y \partial x}$.

5. Visual Explanation

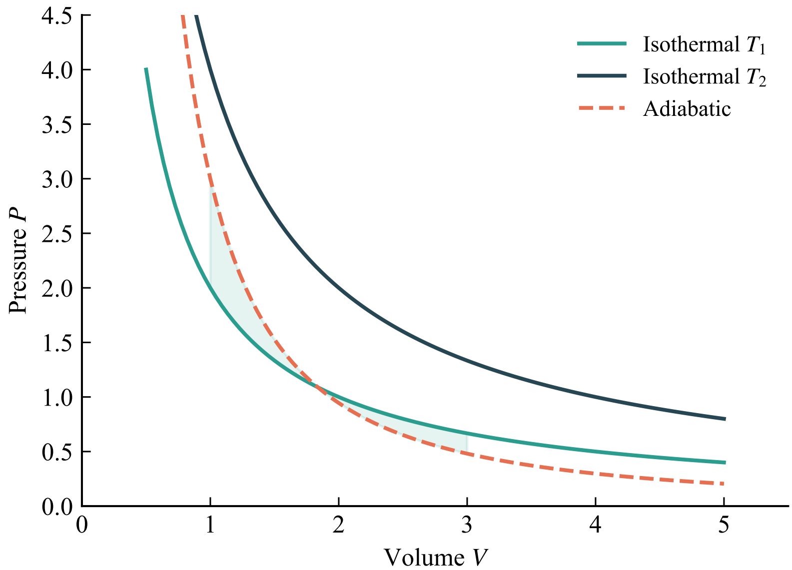


Fig. 1: Ideal gas P-V behavior: at fixed T pressure falls as volume rises (Boyle), and at fixed P volume rises with T (Charles).

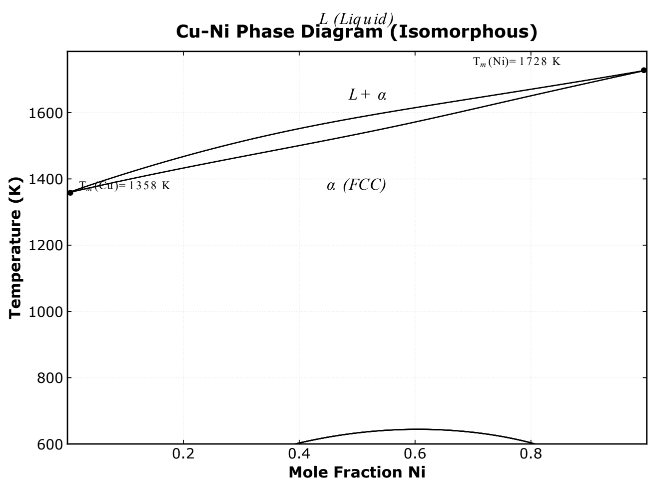


Fig. 2: Equilibrium phase diagram in temperature–composition space — which phase is stable and where phase transformations occur.

6. Memory Box

State function vs path function — heat Q and work W are path functions (written δQ, δW); U, H, S are state functions (written dU, dS).

Exact-differential test: the cyclic integral $\oint dZ = 0$ for a state function; $\oint \delta Q \neq 0$ in general.

Extensive scales with size (V, U, n, S); **intensive** does not (T, P, ρ, μ).

Molar quantity (intensive) = extensive ÷ amount: $v = V/n$, $u = U/n$, $s = S/n$.

Units: $1\ \ \mathrm{J} = 1\ \ \mathrm{N} \cdot \mathrm{m} = 1\ \ \mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-2}$; $1\ \ \mathrm{Pa} = 1 \frac{\mathrm{N}}{\mathrm{m}^2}$.

7. Common Mistakes

- **Treating Q and W as state functions.** They depend on the path; only their difference ΔU is path-independent ($\Delta U = q - w$).
- **Writing dQ or dW.** Use inexact differentials δQ, δW — this notation is required, not optional.
- **Mixing up R units.** $8.314\ \ \mathrm{J} \cdot \mathrm{mol}^{-1} \cdot \mathrm{K}^{-1}$ vs $0.08206\ \ \mathrm{L} \cdot \mathrm{atm} \cdot \mathrm{mol}^{-1} \cdot \mathrm{K}^{-1}$ — pick one and keep P, V units consistent with it.
- **Calling a** molar **quantity extensive.** $v = V/n$ is intensive; only the raw V is extensive.
- **Assuming equilibrium means "equal temperature everywhere."** Full equilibrium also needs zero pressure and chemical-potential differences, not just thermal balance.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$PV = nRT$	Ideal-gas equation of state	P, V, n, T, R	Pa, m³, mol, K	Low P, high T
$R = 8.314\ \ \mathrm{J} \cdot \mathrm{mol}^{-1} \cdot \mathrm{K}^{-1}$	Ideal-gas constant	R	"J/ (mol·K)"	SI value
$1\ \ \mathrm{J} = 1\ \ \mathrm{N} \cdot \mathrm{m} = 1\ \ \mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-2}$	Energy / work unit	—	J	SI
$1\ \ \mathrm{Pa} = 1 \frac{\mathrm{N}}{\mathrm{m}^2}$	Pressure unit	P	Pa	SI
$\frac{\partial^2 Z}{\partial x \partial y} = \frac{\partial^2 Z}{\partial y \partial x}$	Exact-differential criterion	Z, x, y	—	If Z is a state function
$V_{\text{total}} = V_1 + V_2$	Additivity of extensive vars	V	m³	Per phase at equilibrium
$v = V/n$	Molar (intensive) quantity	V, n	"m³/ mol"	Any extensive ÷ n

9. Worked Examples

Example 1 — Ideal gas in a closed vessel

Problem A closed 25 L container holds 2 mol of N_2 at 300 K. Find the pressure. If the volume is fixed and the gas is heated to 400 K, find the new pressure. Finally, if instead the gas is isothermally compressed to 10 L, what is the pressure?

Thinking Process Use $PV = nRT$. Keep one unit system throughout ($R = 0.08206 \ \ \mathrm{L \cdot atm \cdot mol^{-1} \cdot K^{-1}}$).

Solution

$$P_1 = \frac{nRT_1}{V} = \frac{2 \cdot 0.08206 \cdot 300}{25} = 1.969 \ \ \mathrm{atm}$$

At constant V, heating to 400 K:

$$P_2 = \frac{nRT_2}{V} = \frac{2 \cdot 0.08206 \cdot 400}{25} = 2.626 \ \ \mathrm{atm}$$

Isothermal compression to 10 L:

$$P_3 = \frac{nRT_1}{10} = \frac{2 \cdot 0.08206 \cdot 300}{10} = 4.924 \ \ \mathrm{atm}$$

Answer $P_1 = 1.969 \ \ \mathrm{atm}$, $P_2 = 2.626 \ \ \mathrm{atm}$, $P_3 = 4.924 \ \ \mathrm{atm}$.

Interpretation Charles's law appears as $P_2/P_1 = T_2/T_1 = \frac{400}{300} = 1.333$ at fixed V; Boyle's law appears as $P_3/P_1 = V_1/V_3 = \frac{25}{10} = 2.5$ at fixed T.

Example 2 — Classify extensive vs intensive

Problem Classify as extensive (E) or intensive (I): temperature T, pressure P, volume V, density ρ, internal energy U, molar volume v, entropy S, chemical potential μ.

Thinking Process Extensive variables add when subsystems combine; intensive variables are equal in every part at equilibrium and do not add.

Solution

Extensive (E)	Intensive (I)
V (volume)	T (temperature)
U (internal energy)	P (pressure)
S (entropy)	ρ (density)
n (amount)	v (molar volume)
—	μ (chemical potential)

Answer Extensive: V, U, S, n. Intensive: T, P, ρ, v, μ.

Interpretation Dividing an extensive variable by the total amount gives the corresponding **intensive** molar quantity: $v = V/n$, $u = U/n$, $s = S/n$.

Example 3 — Testing whether a differential is exact

Problem A quantity $F(x, y)$ has differential $dF = (2xy)dx + (x^2 + 3y^2)dy$. Is F a state function?

Thinking Process Apply the exact-differential criterion: check whether the mixed partials are equal, $(\partial M/\partial y)_x = (\partial N/\partial x)_y$ with $M = 2xy$, $N = x^2 + 3y^2$.

Solution

$$M = \frac{\partial F}{\partial x}|_y = 2xy, \quad N = \frac{\partial F}{\partial y}|_x = x^2 + 3y^2$$

$$\frac{\partial M}{\partial y}|_x = 2x, \quad \frac{\partial N}{\partial x}|_y = 2x$$

The two are equal, so dF is an exact differential; F is a state function (indeed $F = x^2y + y^3$).

Answer Yes — dF is exact, so F is a state function and its cyclic integral is zero.

Interpretation This is the same test used to decide that dU is exact but δQ and δW are not — the mathematical backbone of the first law.

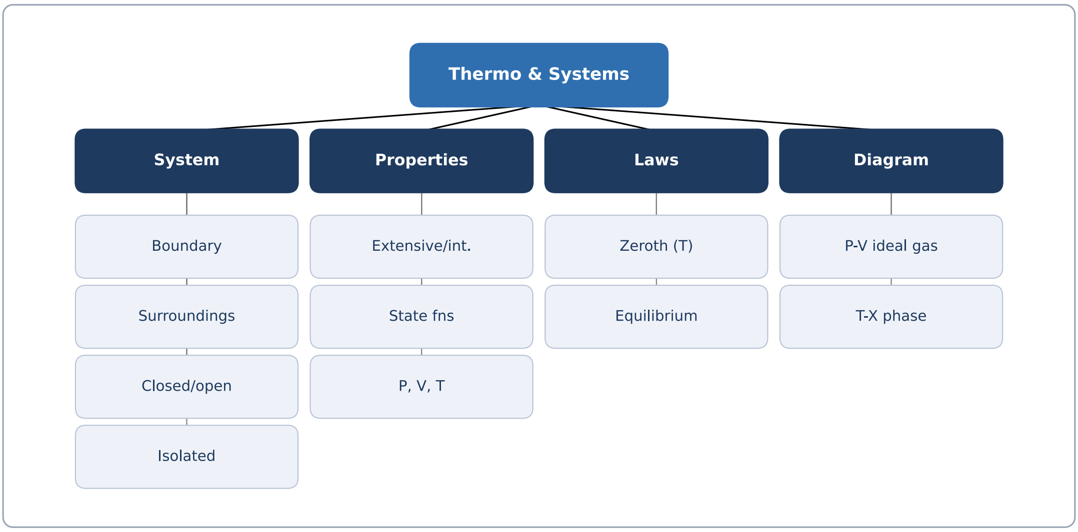
10. Quick Check

1. Is internal energy U a state function or a path function?
2. Write the exact-differential criterion for a function $Z(x, y)$.
3. True or False: $\Delta U = q - w$ holds only for reversible processes.
4. A 1 mol gas at 273 K occupies 22.4 L. What is its pressure in atm ($R = 0.08206$)?
5. Is molar volume $v = V/n$ extensive or intensive?

Answers: 1. State function — its value depends only on the current state. 2. $\frac{\partial^2 Z}{\partial x \partial y} = \frac{\partial^2 Z}{\partial y \partial x}$. 3. False — the first law is general; $\Delta U = q - w$ for any process. 4. $P = nRT/V = \frac{1 \cdot 0.08206 \cdot 273}{22.4} \approx 1.00$ atm (standard molar volume). 5. Intensive — it is an extensive quantity divided by amount.

11. Chapter Summary

- Thermodynamics is the macroscopic science of energy transformation, independent of microscopic assumptions.
- A state function depends only on the current state; its differential is exact and its cyclic integral is zero.
- Equilibrium is the state with no net change: temperature, pressure, and chemical-potential differences all vanish.
- The ideal-gas law $PV = nRT$ combines Boyle's and Charles's laws; use consistent R units.
- Extensive variables add with size; intensive do not; molar quantities are intensive.
- Four laws: Zeroth (temperature), First (energy), Second (entropy), Third ($S = 0$ at 0 K).



12. Flash Cards

Q What is a state function?

A A property fixed by the current state alone; its change is path-independent ($\oint dZ = 0$).

Q What is equilibrium?

A A state whose macroscopic properties no longer change with time; all driving forces (ΔT, ΔP, Δμ) are zero.

Q Ideal gas equation?

A $PV = nRT$, valid at low pressure and high temperature.

Q Extensive vs intensive?

A Extensive scales with system size (V, U, n); intensive does not (T, P, ρ).

Q How do you test for an exact differential?

A Check $(\partial M/\partial y)_x = (\partial N/\partial x)_y$ for $dZ = Mdx + Ndy$.

Q Zeroth law in one line?

A If $T_A = T_B$ and $T_B = T_C$, then $T_A = T_C$ — basis for the thermometer.

Q SI unit of energy and of pressure?

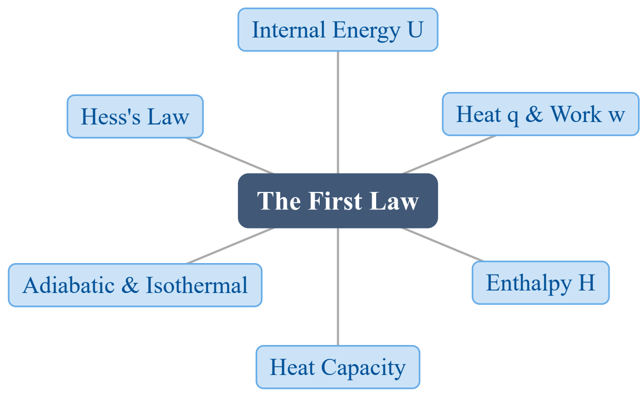
A $1 \text{ J} = 1 \text{ N} \cdot \text{m}$; $1 \text{ Pa} = 1 \frac{\text{N}}{\text{m}^2}$.

13. Exam Tips

- **High priority:** the exact-differential test and the $PV = nRT$ algebra — both appear constantly in later chapters.
- Always note your unit system for R before plugging numbers; a wrong R unit is the fastest way to lose marks.
- Memorize the **definitions** (state function, equilibrium, extensive/intensive) verbatim — exam questions probe them directly.
- Use the four laws as a checklist: Zeroth (temperature), First (energy), Second (entropy), Third ($S = 0$ at 0 K).

2 The First Law of Thermodynamics

1. Chapter Overview



Energy cannot be created or destroyed, only transferred as heat (q) or work (w). This gives the conservation law $\Delta U = q - w$, introduces the state function internal energy U, and leads to the enthalpy H and the heat capacities C_v, C_p.

2. Learning Objectives

After studying this chapter, students should be able to:

- State the first law as $\Delta U = q - w$ and explain why U is a state function while q and w are path functions.
- Define enthalpy $H = U + PV$ and use $\Delta H = q_P$ and $\Delta U = q_V$ for constant-pressure and constant-volume processes.
- Relate heat capacity to internal energy and enthalpy: $C_v = (\partial U/\partial T)_V$ and $C_p = (\partial H/\partial T)_P$.
- Derive and apply the reversible adiabatic relation $PV^\gamma =$ const and the isothermal work $w = RT\ln(V_2/V_1)$.
- Apply Hess's law to compute reaction heats from a sum of measured steps.

3. Why This Matters?

The first law is the foundation of every energy balance in materials processing:

- **Calorimetry & thermochemistry** — measuring ΔH of reactions, alloys, and phase changes.

- **Process design** — furnace heats, cooling, and compression work all obey $\Delta U = q - w$.
- **Heat treatment** — predicting temperature changes in adiabatic expansion/compression (quenching, gas liquefaction).
- **Extraction metallurgy** — combining reaction steps via Hess's law to find net heats without direct measurement.

Without it, we could not link a measured heat input to a real change in a material's internal state.

4. Intuition First

Everyday picture Rub your hands together quickly — they warm up. No flame, no heater: the **mechanical** work of friction was converted into heat. The same idea powers a bicycle pump: pump fast and the barrel gets hot because you are doing work on the trapped air.

The concept Heat and work are simply two **forms of energy transfer** between a system and its surroundings. They are not properties the system "has"; they are amounts that flow along a path. What the system **does** possess is an internal store of energy — the kinetic and potential energy of its atoms — called the **internal energy** U.

The math Joule's paddle-wheel experiment made this quantitative: a falling weight does work that stirs water, and the water's temperature rise tells us the heat equivalent. The result is the **mechanical equivalent of heat**,

$$1 \text{ cal } (15°C) \approx 4.184 \text{ J} \quad (0.239 \text{ cal/J})$$

so any work w can be expressed as a heat q: $q \propto w$. The first law is the bookkeeping rule that ties the net transfer to the change in the internal store: $\Delta U = q - w$.

5. Visual Explanation

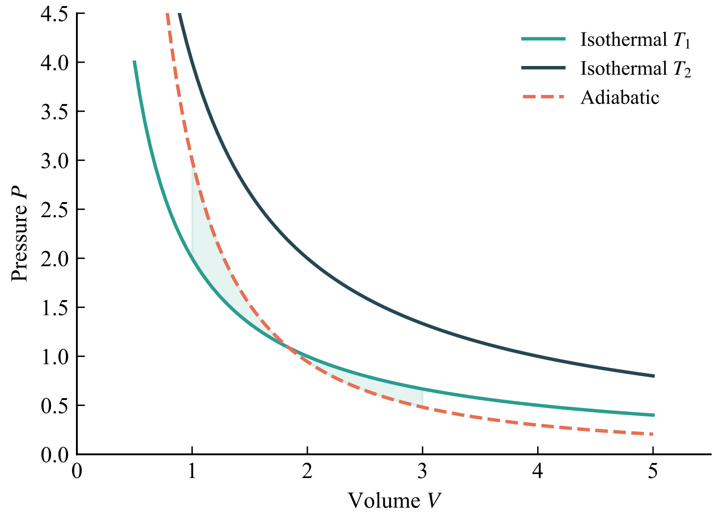


Fig. 3: A P–V diagram of reversible expansion/compression paths; the area under the curve is the PdV work done by the system.

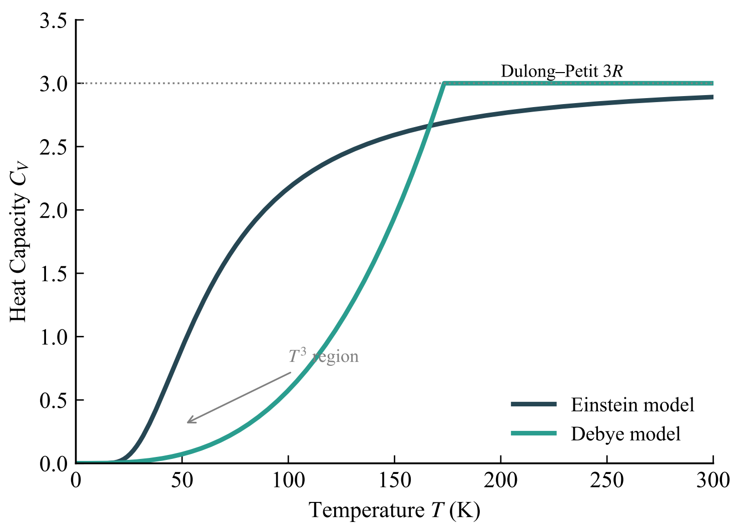


Fig. 4: Variation of the heat capacities c_p and c_v with temperature (dashed lines mark ideal-gas limits).

6. Memory Box

Core identities:

Quantity	Definition	Key relation
U	Internal energy (state function)	$\Delta U = q - w$
H	$U + PV$	$\Delta H = q_P$
C_v	$\partial U/\partial T$ at const V	$dU = C_v dT$
C_p	$\partial H/\partial T$ at const P	$dH = C_p dT$
γ	$\frac{c_p}{c_v}$	$PV^\gamma = \text{const}$

Sign convention: $q > 0$ when the system **absorbs** heat; $w > 0$ when the system **does work** on the surroundings. δ marks an inexact differential; d marks an exact one.

7. Common Mistakes

- **Treating q or w as state functions.** They depend on the path; only ΔU and ΔH are fixed by the end states.
- **Writing $\Delta H = q$ for every process.** Enthalpy change equals heat only at **constant pressure** ($\Delta H = q_P$); at constant volume the heat is $\Delta U = q_V$.
- **Confusing the adiabatic and isothermal work formulas.** Adiabatic: $q = 0$ and $PV^\gamma = \text{const}$; isothermal (ideal gas): $w = RT \ln(V_2/V_1)$. Do not swap them.
- **Forgetting δ vs d.** Writing dq or dw is wrong; heat and work are inexact differentials (δq, δw).
- **Assuming $c_p - c_v = R$ for any substance.** It holds for an ideal gas; real condensed phases have other relations.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$\Delta U = q - w$	First law; U is a state function	q, w	J	Closed system
$dU = \delta q - \delta w$	Differential first law	$\delta q, \delta w$	J	δ inexact
$H = U + PV$	Enthalpy	U, P, V	J	State function
$\Delta H = q_P$	Constant-pressure heat effect	q_P	J	Const P only
$\Delta U = q_V$	Constant-volume heat effect	q_V	J	Const V only
$C_v = (\partial U/\partial T)\vert_V$	Const-volume heat capacity	U, T	J/K	Any system

Formula	Meaning	Variables	Units	Conditions / Limits
$C_p = (\partial H/\partial T)\|_P$	Const-pressure heat capacity	H, T	J/K	Any system
$c_p - c_v = R$	Mayer's relation	c_p, c_v	J/mol·K	1 mol ideal gas
$PV^\gamma = \text{const}$	Reversible adiabatic path	P, V, γ	—	$q = 0$, ideal gas
$w = RT\ln(V_2/V_1)$	Reversible isothermal work	R, T, V	J	1 mol ideal gas

9. Worked Examples

Example 1 — Reversible isothermal expansion

Problem 1 mol of an ideal gas at $T = 1000$ K is expanded reversibly and isothermally from $P_1 = 20$ atm to $P_2 = 4$ atm. Find the work done and the heat absorbed.

Thinking Process For an ideal gas $dT = 0$, so $dU = 0$ and the first law gives $\delta w = \delta q$. Work equals the integral of $PdV = RT/V, dV$, which becomes $RT\ln(V_2/V_1)$. Use $PV = RT$ to replace volumes by pressures: $V_2/V_1 = P_1/P_2$.

Solution

$$w = RT\ln\left(\frac{V_2}{V_1}\right) = RT\ln\left(\frac{P_1}{P_2}\right)$$

$$w = (8.314 \ \ \mathrm{J/(mol{\cdot}K)})(1000 \ \ \mathrm{K})\ln\left(\frac{20}{4}\right)$$

$$w = 8314 \times \ln(5) = 8314 \times 1.6094 \approx 1.338 \times 10^4 \ \ \mathrm{J}$$

$$q = w \approx 13.38 \ \ \mathrm{kJ}$$

Answer $w = q \approx 13.4$ kJ (work done **by** the gas).

Interpretation All the supplied heat is converted into work — the reversible isothermal limit. Any real expansion does strictly less work.

Example 2 — Final state of reversible adiabatic expansion

Problem 2 mol of an ideal gas ($c_v = 3/2R$) starts at $P_1 = 5$ atm, $T_1 = 400$ K and expands reversibly and adiabatically to $P_2 = 1$ atm. Find the final temperature T_2 and the work w.

Thinking Process Adiabatic means $q = 0$, so from the first law $w = -\Delta U = -nc_v(T_2 - T_1)$. We need T_2. Use the adiabatic temperature–pressure relation $TP^{\frac{1-\gamma}{\gamma}} = \text{const}$, i.e. $T_2 = T_1(P_2/P_1)^{\frac{\gamma-1}{\gamma}}$ with $\gamma = c_p/c_v = (\frac{5}{2}R)/(\frac{3}{2}R) = \frac{5}{3}$.

Solution

$$\gamma = \frac{5/2}{3/2} = 5/3 = 1.667, \quad (\gamma - 1)/\gamma = \frac{0.667}{1.667} = 0.400$$

$$T_2 = 400\left(\frac{1}{5}\right)^{0.4} = 400(0.2)^{0.4} = 400 \times 0.525 \approx 210 \ \ \text{K}$$

$$w = -nc_v(T_2 - T_1) = -2\left(\frac{3}{2} \times 8.314\right)(210 - 400)$$

$$w = -(24.94)(-190) = 4.74 \times 10^3 \ \ \text{J} = 4.74 \ \ \text{kJ}$$

Answer $T_2 \approx 210$ K and $w \approx 4.74$ kJ done **by** the gas.

Interpretation Adiabatic expansion cools the gas (400 K $\to$ 210 K); this cooling is the basis of gas liquefaction and refrigeration cycles.

Example 3 — Hess's law for CO formation

Problem Given (1) $C(\text{graphite}) + O_2 \to CO_2$, $\Delta H_1 = -393.5$ kJ/mol, and (2) $CO + \frac{1}{2}O_2 \to CO_2$, $\Delta H_2 = -283.0$ kJ/mol, find ΔH_3 for $C(\text{graphite}) + \frac{1}{2}O_2 \to CO$.

Thinking Process Hess's law: because H is a state function, the total enthalpy change of a reaction equals the sum of the enthalpy changes of any path between the same end states. Subtract reaction (2) from reaction (1) to leave the desired reaction.

Solution

$$(1) - (2): \quad C(\text{graphite}) + O_2 - CO - \frac{1}{2}O_2 \to CO_2 - CO_2$$

$$C(\text{graphite}) + \frac{1}{2}O_2 \to CO$$

$$\Delta H_3 = \Delta H_1 - \Delta H_2 = -393.5 - (-283.0) = -110.5 \quad \text{kJ/mol}$$

Answer $\Delta H_3 = -110.5$ kJ/mol.

Interpretation The heat of a hard-to-measure reaction is recovered from measurable combustion data alone — the direct payoff of enthalpy being a state function.

10. Quick Check

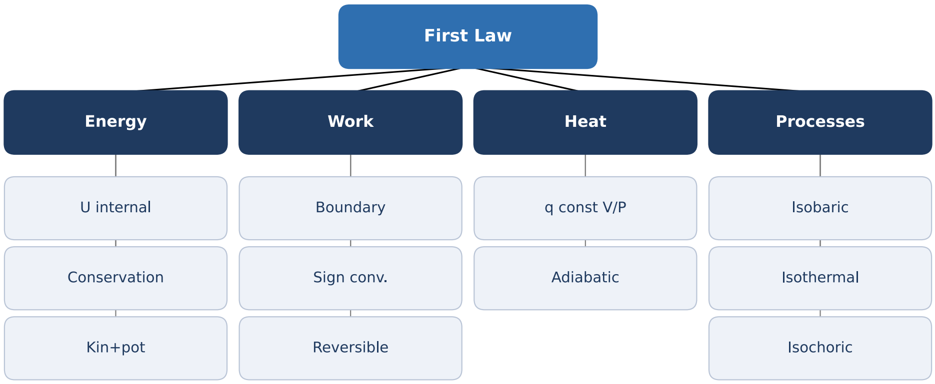
1. Write the first law in the form relating ΔU to q and w.
2. True or False: heat q is a state function.
3. At constant pressure, which enthalpy relation holds: $\Delta H = q_P$ or $\Delta H = q_V$?
4. For an ideal gas, what is Mayer's relation between c_p and c_v?
5. In a reversible adiabatic expansion of an ideal gas, state the invariant relation between P, V, and γ.

Answers: 1. $\Delta U = q - w$ 2. False — q depends on the path, so it is not a state function 3. $\Delta H = q_P$ (constant pressure) 4. $c_p - c_v = R$ for 1 mol of ideal gas 5. $PV^\gamma = \text{const}$ (with $q = 0$)

11. Chapter Summary

- Internal energy U is a state function; the first law is $\Delta U = q - w$ with q, w path-dependent.
- Constant volume: $\Delta U = q_V$; constant pressure defines enthalpy $H = U + PV$ with $\Delta H = q_P$.
- Heat capacities $C_v = (\partial U/\partial T)_V$ and $C_p = (\partial H/\partial T)_P$; $dU = C_v dT$, $dH = C_p dT$.
- Mayer's relation $c_p - c_v = R$ holds for 1 mol of an ideal gas.
- Reversible adiabatic: $PV^\gamma = \text{const}$; reversible isothermal (ideal gas): $w = RT \ln(V_2/V_1)$.

- Hess's law uses the state-function property of H to find reaction heats from a sum of steps.



12. Flash Cards

Q State the first law.

A $\Delta U = q - w$; energy is conserved, U is a state function.

Q How do q and w differ from U?

A They are path functions (inexact, δ); U is a state function (exact, d).

Q Define enthalpy H.

A $H = U + PV$; at constant pressure $\Delta H = q_P$.

Q What is the mechanical equivalent of heat?

A $1\ \text{cal}\ (15°C) \approx 4.184\ \ \text{J}$; work and heat are interchangeable.

Q Reversible adiabatic relation?

A $PV^\gamma = \text{const}$, $q = 0$; also $TV^{\gamma-1} = \text{const}$.

Q Reversible isothermal work (ideal gas)?

A $w = RT\ln(V_2/V_1) = RT\ln(P_1/P_2)$; $dU = 0$.

Q State Hess's law.

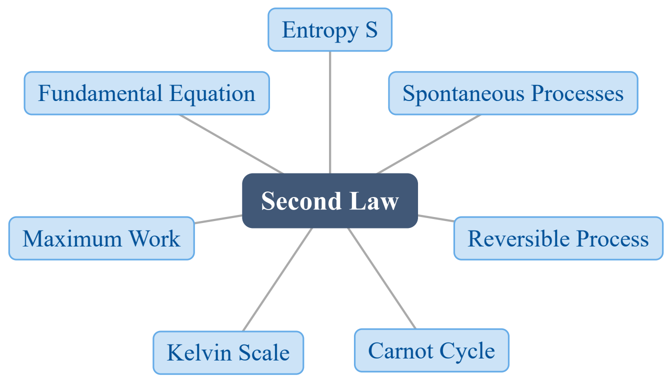
A Reaction enthalpy is path-independent; sum the ΔH of steps to any path.

13. Exam Tips

- **High priority:** the sign convention ($w > 0$ = work done **by** the system) and the exact vs inexact differential notation (ΔU vs δq).
- Memorize the two heat-effect shortcuts: $\Delta U = q_V$ and $\Delta H = q_P$ — most exam problems start here.
- For adiabatic/isothermal ideal-gas work, always state $q = 0$ or $dT = 0$ **first**, then pick $PV^\gamma = \text{const}$ or $w = RT\ln(V_2/V_1)$ — never blend them.
- Hess's law shows up as "combine these reactions": flip or scale steps and track the sign of each ΔH.
- Know Mayer's relation and that $\gamma = \frac{5}{3}$ (monatomic), $\frac{7}{5}$ (diatomic) at room temperature.

3 The Second Law of Thermodynamics

1. Chapter Overview



The first law conserves energy but says nothing about direction. The second law introduces a new state function — entropy S — that fixes the direction of spontaneous change and sets an upper bound on useful work. From the Carnot cycle we derive the absolute temperature scale and the fundamental thermodynamic equation $dU = TdS - PdV$.

2. Learning Objectives

After studying this chapter, students should be able to:

- Define entropy S and state that $dS = \delta q_{\mathrm{rev}}/T$, with S a state function.
- Distinguish spontaneous, reversible, and irreversible processes and relate them to $\Delta S_{\mathrm{total}}$.
- Compute the entropy change of an ideal gas for reversible and free expansion.
- Derive the Carnot efficiency $\eta = 1 - T_c/T_h$ and explain the Kelvin temperature scale.
- State the principle of maximum work and write the combined fundamental equation $dU = TdS - PdV$.

3. Why This Matters?

The second law is the reason a dropped glass never reassembles and why no engine is 100% efficient. It governs every real materials process:

- **Metallurgy & extraction** — Ellingham diagrams rank oxide stability by $\Delta G° = \Delta H° - T\Delta S°$.
- **Heat engines & power plants** — the Carnot limit sets the maximum achievable efficiency.
- **Phase stability** — at constant U, V equilibrium is the state of maximum entropy.
- **Materials design** — entropy drives mixing, ordering, and defect equilibria.

Entropy is also the quantitative measure of irreversibility and energy "quality" degradation.

4. Intuition First

Everyday picture: a hot cup of coffee cools on its own, but a cold cup never spontaneously heats up; a deck of cards scatters when shuffled but never reorders itself.

Mental picture: imagine a divided box of gas molecules. Removing the partition, the molecules spread to fill all space — there are vastly more **disordered** arrangements than **ordered** ones, so spreading is overwhelmingly more probable.

Concept: entropy S measures that disorder / number of accessible microstates. A process runs spontaneously toward the state with more microstates.

Math: for a reversible step, the entropy gained equals the heat shared divided by the temperature at which it is shared:

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

For the universe, $\Delta S_{\text{total}} \geq 0$, with equality only for a reversible process.

5. Visual Explanation

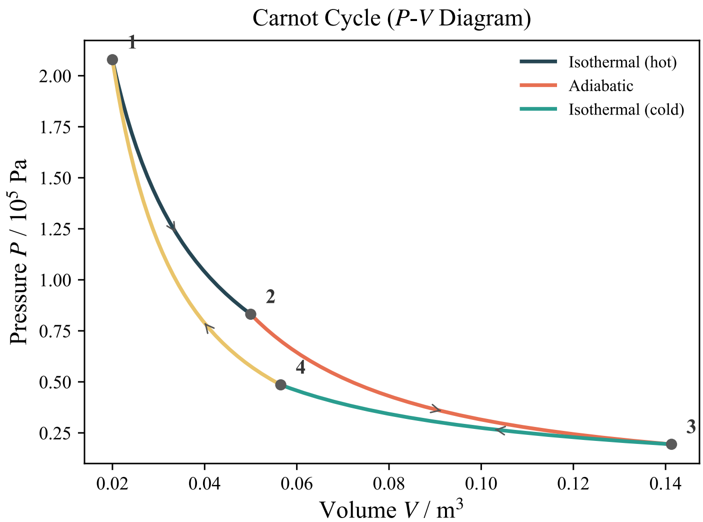


Fig. 5: Carnot cycle on a P-V diagram: isothermal expansion (absorb q_h), adiabatic expansion, isothermal compression (reject q_c), and adiabatic compression.

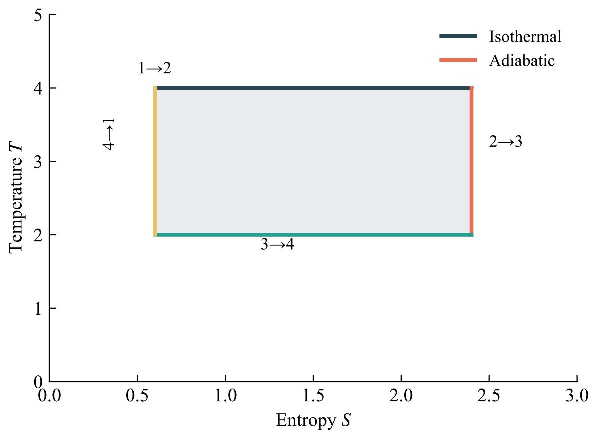


Fig. 6: Carnot cycle on a T-S diagram: a rectangle between T_h and T_c; the enclosed area is the net work $w = q_h - q_c$.

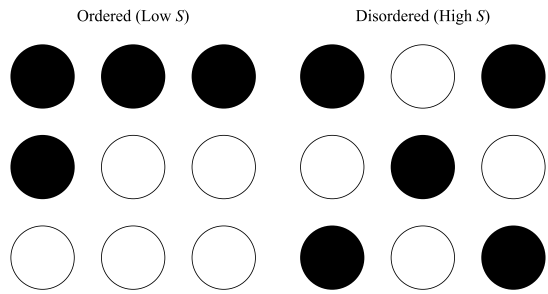


Fig. 7: More microstates (right) correspond to higher entropy; spontaneous change favors the larger number of accessible arrangements.

6. Memory Box

Entropy sign rules (the heart of the chapter):

Process	ΔS_{system}	ΔS_{total}
Reversible	$= q_{\text{rev}}/T$	$= 0$
Irreversible (spontaneous)	$= q/T + \sigma, \sigma > 0$	> 0
Free expansion	$= nR\ln(V_2/V_1)$	> 0

Carnot: $\eta = 1 - q_c/q_h = 1 - T_c/T_h$ (no engine can exceed this). **Fundamental equation:** $dU = TdS - PdV$ is the combined first + second law for a closed system doing only PV work.

7. Common Mistakes

- **$\Delta S = q/T$ always.** This holds **only for a reversible** path. For an irreversible process $\Delta S > q/T$; always compute ΔS via a reversible path between the same states.
- **Reversible means fast.** A reversible process is an idealized quasi-static limit at equilibrium at every instant; real processes are irreversible and give $\Delta S_{\text{total}} > 0$.
- **Free expansion has $\Delta S_{\text{total}} = 0$.** The gas gains entropy but the surroundings gain none ($q = 0$), so $\Delta S_{\text{total}} = nR\ln(V_2/V_1) > 0$.
- **Efficiency $\eta = 1 - T_c/T_h$ for any engine.** This is the **Carnot** (maximum) efficiency; real engines have strictly lower η.
- **The fundamental equation applies everywhere.** $dU = TdS - PdV$ is valid only for a closed system with PV-only work; add $\mu_i dn_i$ or other work terms otherwise.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$dS = \frac{\delta q_{\text{rev}}}{T}$	Entropy definition (rev.)	q, T	J/K	Reversible path
$\Delta S = nR\ln(V_2/V_1)$	Ideal-gas isothermal ΔS	n, R, V	J/K	Const T
$\eta = 1 - T_c/T_h$	Carnot efficiency	T_h, T_c	—	Reversible engine
$w \le T\Delta S - \Delta U$	Maximum-work principle	T, S, U	J	Upper bound on w
$(q_c/q_h) = (T_c/T_h)$	Kelvin scale relation	q, T	—	Carnot cycle
$dU = TdS - PdV$	Fundamental equation	U, S, V, P, T	J	Closed, PV only

9. Worked Examples

Example 1 — Entropy change of reversible isothermal expansion

Problem 2 mol of an ideal gas expands isothermally and reversibly from 10 L to 30 L at 300 K. Find ΔS_{sys}, ΔS_{surr}, and ΔS_{total}.

Thinking Process For an ideal gas isothermal change $\Delta U = 0$, so $q = w = nRT\ln(V_2/V_1)$. Because the process is reversible, $\Delta S_{\text{sys}} = q_{\text{rev}}/T$ and the surroundings lose the same heat.

Solution

$$w = q = nRT\ln(V_2/V_1) = (2)(8.314)(300)\ln(30/10) = 5483 \ \ \text{J}$$

$$\Delta S_{\text{sys}} = q/T = 5483/300 = 18.28 \ \ \text{J/K}$$

$$\Delta S_{\text{surr}} = -q/T = -18.28 \ \ \text{J/K}$$

$$\Delta S_{\text{total}} = 18.28 - 18.28 = 0$$

Answer $\Delta S_{\text{sys}} = 18.28$ J/K, $\Delta S_{\text{surr}} = -18.28$ J/K, $\Delta S_{\text{total}} = 0$.

Interpretation A reversible process has zero total entropy change, consistent with $\Delta S_{\text{universe}} = 0$.

Example 2 — Carnot engine efficiency

Problem A Carnot engine operates between 500 K and 300 K. (a) Find its efficiency. (b) If 100 kJ of heat enters at the hot end, what is the work output? (c) If T_h rises to 600 K, by how much does η increase?

Thinking Process Use the Carnot efficiency $\eta = 1 - T_c/T_h$; then $w = \eta q_h$. Compare the two temperatures for (c).

Solution (a) $\eta = 1 - 300/500 = 0.40 = 40\%$ (b) $w = \eta q_h = 0.40(100) = 40 \text{ kJ}$ (c) $\eta' = 1 - 300/600 = 0.50 = 50\%$; increase $= 10\%$

Answer Efficiency is 40%, work output 40 kJ, and raising T_h to 600 K raises efficiency to 50% (+10%).

Interpretation Higher T_h widens the temperature gap and improves efficiency; no engine between these two reservoirs can beat the Carnot 50% limit.

Example 3 — Free expansion (irreversible)

Problem The same 2 mol gas free-expands from 10 L to 30 L at 300 K into vacuum. Find ΔS_{total} and compare the work done with the reversible case.

Thinking Process State is the same as Example 1, so ΔS_{sys} is identical; but free expansion has $q = w = 0$, hence $\Delta S_{\text{surr}} = 0$. Compute ΔS_{sys} via the reversible path between the same endpoints.

Solution

$$\Delta S_{\text{sys}} = nR\ln(V_2/V_1) = (2)(8.314)\ln 3 = 18.28 \text{ J/K}$$

$$\Delta S_{\text{surr}} = 0 \quad \text{since} \quad q = 0$$

$$\Delta S_{\text{total}} = 18.28 > 0$$

Reversible work $w_{\text{rev}} = 5483 \text{ J}$; free expansion does $w = 0$.

Answer $\Delta S_{\text{total}} = 18.28$ J/K > 0; free expansion delivers zero work versus 5483 J reversibly.

Interpretation Both paths reach the same final state (same ΔS_{sys}), but only the reversible path keeps $\Delta S_{\text{total}} = 0$; irreversibility destroys the potential to do work.

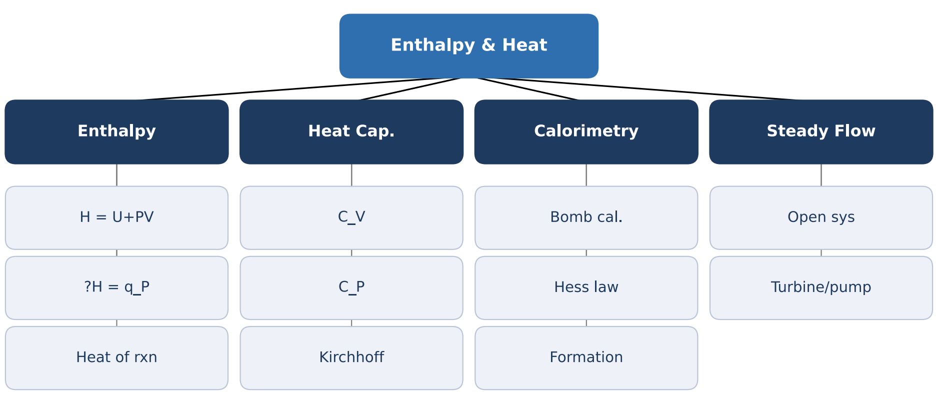
10. Quick Check

1. True or False: $\Delta S = q/T$ for any process.
2. For an isolated system at equilibrium, entropy is at a (maximum / minimum).
3. A Carnot engine between 400 K and 200 K has efficiency ___ .
4. In free expansion of an ideal gas, $\Delta S_{\text{surroundings}}$ equals ___ .
5. The combined law for a closed PV-only system is $dU =$ ___ .

Answers: 1. False — only for a reversible path; irreversible gives $\Delta S > q/T$ 2. Maximum 3. $1 - 200/400 = 50\%$ 4. Zero (no heat exchange) 5. $TdS - PdV$

11. Chapter Summary

- Entropy S is a state function; for a reversible process $dS = \delta q_{\text{rev}}/T$.
- A reversible process has $\Delta S_{\text{total}} = 0$; a spontaneous (irreversible) process has $\Delta S_{\text{total}} > 0$.
- For an ideal gas isothermal change $\Delta S = nR\ln(V_2/V_1)$; free expansion gives $\Delta S_{\text{total}} > 0$.
- The Carnot efficiency $\eta = 1 - T_c/T_h$ is the maximum possible and defines the Kelvin scale.
- Maximum work: $\delta w \le TdS - dU$, reached only in a reversible process.
- The fundamental equation $dU = TdS - PdV$ combines the first and second laws for a closed system.



12. Flash Cards

Q What is entropy?

A S is a state function measuring disorder / number of microstates; $dS = \delta q_{\text{rev}}/T$.

Q Reversible vs irreversible?

A Reversible: $\Delta S_{\text{total}} = 0$; irreversible (spontaneous): $\Delta S_{\text{total}} > 0$.

Q Carnot efficiency?

A $\eta = 1 - T_c/T_h$, the maximum possible for any heat engine.

Q Entropy change of free expansion?

A $\Delta S_{\text{sys}} = nR\ln(V_2/V_1) > 0$, with $\Delta S_{\text{surr}} = 0$.

Q Maximum-work principle?

A $\delta w \leq TdS - dU$; equality holds only for a reversible process.

Q Fundamental equation?

A $dU = TdS - PdV$ — the combined first and second laws (closed, PV work).

Q Kelvin temperature scale?

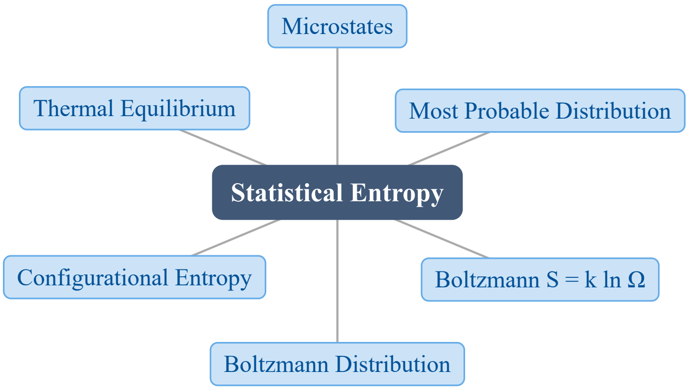
A Defined by $q_c/q_h = T_c/T_h$, independent of working substance.

13. Exam Tips

- **High priority:** computing ΔS for ideal-gas expansion via a reversible path, and classifying processes by ΔS_{total} sign.
- Memorize the Carnot efficiency and that it is an **upper bound** — real engines are lower.
- When asked for entropy, never use the actual irreversible q/T; construct the equivalent reversible path between the same initial and final states.
- State constraints before applying $dU = TdS - PdV$ — it assumes a closed system with PV-only work.
- Free expansion is the classic irreversible example: same ΔS_{sys} as reversible, but $\Delta S_{\text{surr}} = 0$.

4 The Statistical Interpretation of Entropy

1. Chapter Overview



Classical thermodynamics tells us that entropy exists and increases, but not **what it is**. Statistical thermodynamics answers this: a macrostate (fixed U', V', N) contains an enormous number of equally probable **microstates**, and the equilibrium state is simply the **most probable** distribution. Counting those microstates Ω gives the Boltzmann equation $S' = k_B \ln \Omega$, the exponential Boltzmann distribution, and a microscopic meaning for temperature and heat flow.

2. Learning Objectives

After studying this chapter, students should be able to:

- Distinguish a **macrostate** from a **microstate**, and state why every microstate is equally probable.
- Count microstates Ω for particles on energy levels and for configurational (mixing) problems.
- State and apply the **Boltzmann equation** $S' = k_B \ln \Omega$ and interpret entropy as disorder.
- Derive the **Boltzmann distribution** $n_i = ne^{-\beta\varepsilon_i}/Z$ using Lagrange multipliers.
- Compute **configurational entropy** $\Delta S = -R \sum X_i \ln X_i$ and relate it to alloy stability.
- Explain thermal equilibrium ($T_A = T_B$) as the maximization of $\Omega_A \Omega_B$.

3. Why This Matters?

The Boltzmann equation $S' = k_B \ln \Omega$ is the bridge between the atomic world and macroscopic thermodynamics — it is engraved on Boltzmann's tombstone for good reason. In materials science it explains:

- **Why solids mix.** Configurational entropy $\Delta S_{\text{conf}} = -R \sum X_i \ln X_i$ lowers ΔG_{mix} and drives the formation of solid solutions.
- **High-entropy alloys.** Adding more principal elements raises mixing entropy ($R \ln n$ at equimolar composition), stabilizing a single solid-solution phase.
- **Why heat flows hot → cold.** The combined microstate count $\Omega_A \Omega_B$ is overwhelmingly larger after heat flows down the temperature gradient.
- **The direction of time.** "Spontaneous" simply means "toward the vastly more probable distribution."

4. Intuition First

Example first. Put three particles on distinguishable sites A, B, C with total energy $U' = 3u$ and levels $\varepsilon = 0, u, 2u, 3u$. Three distributions are allowed: all three at level 1 (1 arrangement), one at level 3 & two at level 0 (3 arrangements), or one each at levels 0, 1, 2 ($3! = 6$ arrangements). Of the 10 total microstates, the "spread-out" distribution owns 6 — so it is the most probable.

Picture. Every allowed arrangement is one microstate; all are equally likely. Nature does not "prefer" order — it just lands in whichever distribution has the **most** microstates.

Concept. Equilibrium = most probable distribution. **Math.** For 10^{23} particles the most probable distribution utterly dominates ($\Omega \approx \Omega_{\text{max}}$), and its count defines entropy via $S' = k_B \ln \Omega$.

5. Visual Explanation

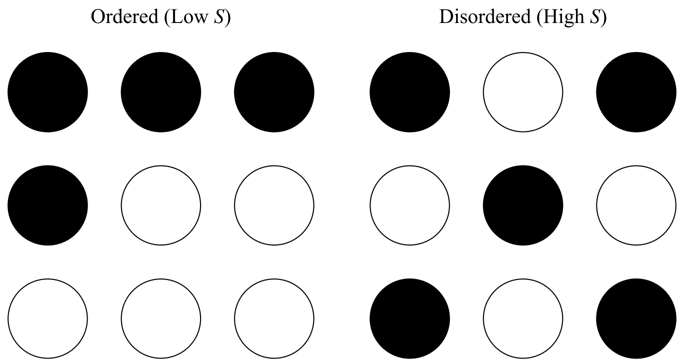


Fig. 8: Microstates of a fixed macrostate: distinct microscopic arrangements of particles over energy levels that all correspond to the same U', V', N. The distribution with the most microstates is the equilibrium (most probable) state.

6. Memory Box

The whole chapter in five symbols:

Relation	What it says
$S' = k_B \ln \Omega$	Entropy counts microstates (Boltzmann)
$\Omega = n!/(n_{0!}n_{1!}...n_{r!})$	Ways to spread particles over levels
$n_i = ne^{-\beta\varepsilon_i}/Z$	Occupation falls exponentially with energy
$\beta = 1/(k_B T)$	Temperature sets the "steepness"
$\Delta S_{\text{conf}} = -R\sum X_i \ln X_i$	Entropy of mixing

Mnemonic: "$S = k \ln \Omega$" reads as "**S*imply** keep l*ogging arrangements." Higher T $\rightarrow$ smaller β $\rightarrow$ flatter distribution $\rightarrow$ more high-energy occupation.

7. Common Mistakes

- **Confusing macrostate and microstate.** The macrostate (U', V', N) is fixed; it **contains** many microstates. Entropy counts the microstates, not the macrostate.
- **Thinking some microstates are "more likely."** Every individual microstate is **equally** probable — a distribution is favored only because it has **more** microstates.

- **Forgetting the constraints in the derivation.** The Boltzmann distribution follows only after imposing **both** $\sum n_i = n$ and $\sum n_i \varepsilon_i = U'$ (two Lagrange multipliers α, β).
- **Sign errors in ΔS_{conf}.** Since $X_i < 1$, $\ln X_i < 0$, so the mixing entropy $-R \sum X_i \ln X_i$ is **positive**. Dropping the minus sign gives a negative entropy — a red flag.
- **Using Stirling on small numbers.** $\ln x! \approx x \ln x - x$ is only accurate for large x; do not apply it to the toy 3-particle counting problems.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$S' = k_B \ln \Omega$	Boltzmann equation	Ω = number of microstates	J/K	Isolated system
$\Omega = \frac{n!}{n_0! n_1! \ldots n_r!}$	Microstates of n particles over levels	n_i = occupation of level i	—	Distinguishable sites
$\ln \Omega \approx n \ln n - \sum_i n_i \ln n_i$	Stirling-approximated $\ln \Omega$	n, n_i	—	Large n_i
$n_i = \frac{n e^{-\beta \varepsilon_i}}{Z}$	Boltzmann distribution	ε_i, Z	—	Most probable dist.
$Z = \sum_{i=0}^{r} e^{-\beta \varepsilon_i}$	Partition function	ε_i	—	Sum over levels
$\beta = \frac{1}{k_B T}$	Meaning of temperature	$k_B = 1.381 \times 10^{-23}$ J/K	J⁻¹	—
$\Delta S_{\text{conf}} = -R \sum_i X_i \ln X_i$	Configurational (mixing) entropy	X_i = mole fraction	J/(mol·K)	Random, ideal mixing

9. Worked Examples

Example 1 — Counting microstates of a small system

Problem Three identical particles sit on three distinguishable sites with total energy $U' = 3u$ and levels $\varepsilon = 0, u, 2u, 3u$. Find the number of microstates in each allowed distribution and the most probable one.

Thinking Process Enumerate the distributions that satisfy $\sum n_i \varepsilon_i = 3u$ with three particles, then count arrangements with $\Omega = n!/(n_0! n_1! \ldots)$.

Solution Distribution (a): all three at level 1 ($\varepsilon = u$): $\Omega = 3!/3! = 1$.

Distribution (b): one at level 3, two at level 0: $\Omega = 3!/(2!1!) = 3$.

Distribution (c): one each at levels 0, 1, 2: $\Omega = 3!/(1!1!1!) = 6$.

Total microstates $= 1 + 3 + 6 = 10$.

Answer Distribution (c) has 6 microstates and probability $6/10 = 0.6$ — the most probable.

Interpretation Even for just three particles, the "spread-out" distribution dominates. For a mole ($\approx 10^{23}$ particles) the most probable distribution outweighs all others so completely that $\Omega \approx \Omega_{\text{max}}$, which is why equilibrium is sharply defined.

Example 2 — Configurational entropy of a binary alloy

Problem A binary substitutional solid solution of 1 mol of atoms has $X_A = 0.3$, $X_B = 0.7$. Compute the configurational entropy and compare it to the entropy of an ideal-gas isothermal expansion.

Thinking Process Use $\Delta S_{\text{conf}} = -R(X_A \ln X_A + X_B \ln X_B)$, which follows from $S = k_B \ln[N!/(N_{A!}N_{B!})]$ via Stirling's approximation.

Solution

$$\Delta S_{\text{conf}} = -(8.314)(0.3 \ln 0.3 + 0.7 \ln 0.7)$$

$$= -(8.314)(-0.361 - 0.250) = 5.08 \quad \text{J/(mol·K)}$$

For an isothermal ideal-gas expansion $\Delta S = R \ln(V_2/V_1)$; matching 5.08 J/(mol·K) needs $V_2/V_1 = e^{5.08/8.314} = 1.85$.

Answer $\Delta S_{\text{conf}} = 5.08$ J/(mol·K), comparable to a modest ($\times$ 1.85) gas expansion.

Interpretation Configurational entropy is of the **same order** as thermal entropy — it is not negligible. Extending to a ternary equimolar alloy gives $\Delta S_{\text{conf}} = R \ln 3 = 9.13$ J/(mol·K); adding components keeps raising the mixing entropy, the thermodynamic basis of high-entropy alloys.

Example 3 — Populations from the Boltzmann distribution

Problem A two-level system has $\varepsilon_1 = 0$ and $\varepsilon_2 = 2.0 \times 10^{-20}$ J. Find the population ratio n_2/n_1 at 300 K, then at 1000 K.

Thinking Process Apply $n_2/n_1 = e^{-\beta\varepsilon_2}$ with $\beta = 1/(k_B T)$; the partition function cancels in the ratio.

Solution At 300 K: $k_B T = (1.381 \times 10^{-23})(300) = 4.143 \times 10^{-21}$ J, so $\beta = 2.414 \times 10^{20}$ J⁻¹.

$$\frac{n_2}{n_1} = e^{-(2.414\times10^{20})(2.0\times10^{-20})} = e^{-4.83} = 0.0080$$

At 1000 K: $k_B T = 1.381 \times 10^{-20}$ J, $\beta = 7.24 \times 10^{19}$ J⁻¹.

$$\frac{n_2}{n_1} = e^{-1.45} = 0.235$$

Answer $n_2/n_1 \approx 0.008$ at 300 K, rising to ≈ 0.24 at 1000 K.

Interpretation Raising temperature lowers β and flattens the distribution, pushing particles into higher levels. This is the microscopic origin of heat capacity: energy raises T by promoting particles up the ladder.

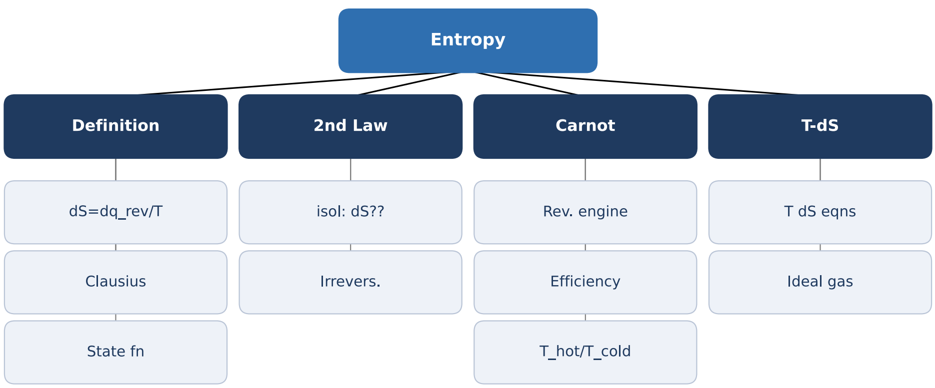
10. Quick Check

1. Define the Boltzmann equation and name every symbol.
2. True or False: within one macrostate, some microstates are more probable than others.
3. What happens to the Boltzmann distribution as temperature increases?
4. For an equimolar A–B alloy, what is ΔS_{conf} in terms of R?
5. What condition on $\Omega_A \Omega_B$ defines thermal equilibrium between two bodies?

Answers: 1. $S' = k_B \ln \Omega$: S' = entropy, k_B = Boltzmann constant, Ω = number of microstates. 2. False — all microstates are equally probable; distributions differ only in their microstate **count**. 3. It flattens (β decreases), so higher energy levels gain relative occupation. 4. $\Delta S_{\text{conf}} = R \ln 2 \approx 5.76$ J/(mol·K). 5. $\Omega_A \Omega_B$ is maximized, which requires $T_A = T_B$.

11. Chapter Summary

- Entropy is statistical: $S' = k_B \ln \Omega$, where Ω counts the microstates of a macrostate.
- The equilibrium state is the **most probable distribution**; for large N, $\Omega \approx \Omega_{\text{max}}$.
- Maximizing $\ln \Omega$ under fixed n and U' gives the Boltzmann distribution $n_i = ne^{-\beta\varepsilon_i}/Z$ with $\beta = 1/(k_B T)$.
- Configurational entropy $\Delta S_{\text{conf}} = -R \sum X_i \ln X_i$ drives mixing and stabilizes (high-entropy) solid solutions.
- Thermal equilibrium ($T_A = T_B$) maximizes $\Omega_A \Omega_B$; irreversible heat flow **creates** entropy.
- Total entropy sums thermal and configurational contributions: $S' = k_B \ln \Omega_{\text{th}} + k_B \ln \Omega_{\text{conf}}$.



12. Flash Cards

Q State the Boltzmann equation.

A $S' = k_B \ln \Omega$ — entropy is k_B times the log of the number of microstates.

Q Macrostate vs. microstate?

A Macrostate = fixed U', V', N; microstate = one specific arrangement. A macrostate holds many equally-probable microstates.

Q Write the Boltzmann distribution.

A $n_i = ne^{-\beta\varepsilon_i}/Z$, with $Z = \sum_i e^{-\beta\varepsilon_i}$ and $\beta = 1/(k_B T)$.

Q What is the partition function Z?

A $Z = \sum_i e^{-\beta\varepsilon_i}$ — the normalizing sum over all energy levels.

Q Configurational entropy of mixing?

A $\Delta S_{\text{conf}} = -R\sum_i X_i \ln X_i$; always positive since $X_i < 1$.

Q Microscopic condition for thermal equilibrium?

A $\Omega_A\Omega_B$ is maximum, i.e. $T_A = T_B$; heat flows to increase the combined microstate count.

Q What does β represent?

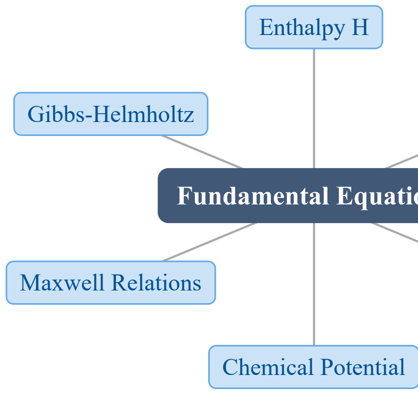
A $\beta = 1/(k_B T)$ — the "steepness" of the energy distribution; small β = high T.

13. Exam Tips

- **High priority:** deriving the Boltzmann distribution by maximizing $\ln\Omega$ under the constraints $\sum n_i = n$ and $\sum n_i\varepsilon_i = U'$ with Lagrange multipliers α and β.
- Memorize $S' = k_B \ln\Omega$ and be able to **count** Ω both for level-occupation and for mixing problems — the two most common exam question types.
- For configurational-entropy problems, go straight to $\Delta S = -R\sum X_i \ln X_i$; check the answer is positive.
- Explain heat flow and the arrow of time as "the system moves to the distribution with far more microstates" — a favorite conceptual question.
- Watch the units: k_B is per particle (J/K), $R = N_O k_B$ is per mole (J/(mol·K)).

5 Fundamental Equations and Their Relationships

1. Chapter Overview



Starting from the combined first and second laws, $dU = TdS - PdV$, we build three new potentials (H, A, G) whose **natural variables** are easier to measure and control. From them follow the four fundamental equations, the chemical potential, and the Maxwell relations.

2. Learning Objectives

After studying this chapter, students should be able to:

- Define enthalpy H, Helmholtz energy A, and Gibbs energy G, and state the **natural variables** of each.
- Write the four fundamental equations for closed and open systems.
- Define the chemical potential μ_i and explain why it drives diffusion and reaction.
- Derive and apply the four Maxwell relations to replace unmeasurable quantities with measurable ones.
- Use the Gibbs-Helmholtz equation to relate the temperature dependence of ΔG to ΔH.

3. Why This Matters?

The Gibbs free energy G is **the** master variable of materials science: minimizing G at constant T, P predicts which phase is stable, whether a reaction proceeds, and where equilibrium lies. This single idea underlies:

- **Phase diagrams** — common-tangent constructions on G-composition curves.
- **Batteries & fuel cells** — cell voltage is $\Delta G = -nFE$.
- **Extraction metallurgy** — Ellingham diagrams plot $\Delta G°$ vs. T.
- **Semiconductors & materials design** — defect and reaction equilibria.

The Maxwell relations are the practical payoff: they turn an **unmeasurable** quantity like $(\partial S/\partial V)_T$ into a **measurable** one like $(\partial P/\partial T)_V$.

4. Intuition First

Why invent H, A, and G at all? The combined law $dU = TdS - PdV$ uses S and V as the independent variables — but **entropy cannot be dialed on a knob**. In the lab we control temperature and pressure. A **Legendre transform** swaps an awkward variable for its convenient partner:

$$U(S, V) \quad \rightarrow \quad H(S, P) \quad \rightarrow \quad G(T, P)$$

Think of it like choosing coordinates: the physics is identical, but $G(T, P)$ lets you compute everything from quantities you can actually set on the bench. Each transform "pays" by adding a PV or subtracting a TS term.

5. Visual Explanation

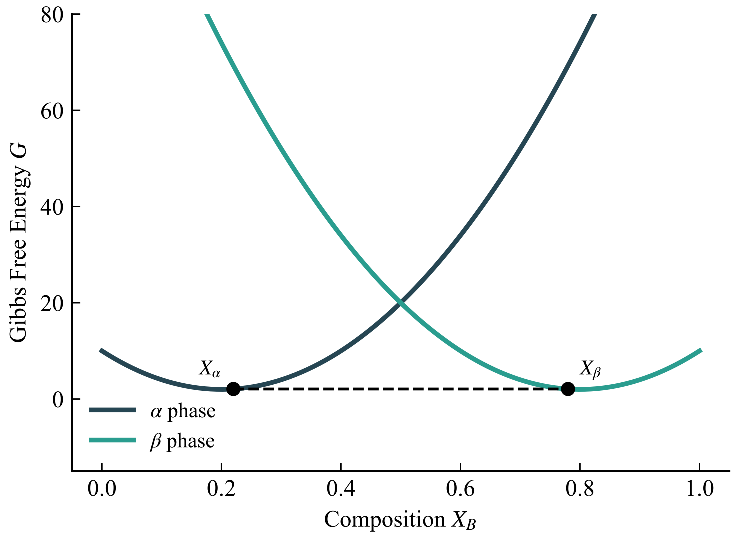
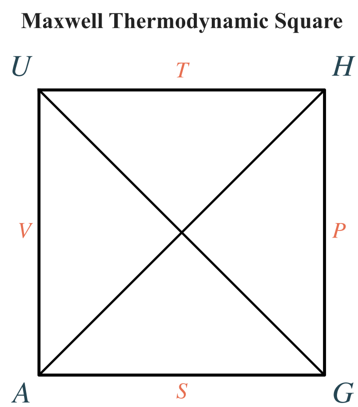


Fig. 9: Relations among the four potentials U, H, A, G and their natural variables T, S, P, V.



$$dU = T\,dS - P\,dV$$

$$dH = T\,dS + V\,dP$$

$$dA = -S\,dT - P\,dV$$

$$dG = -S\,dT + V\,dP$$

Read off: each potential's differential uses its adjacent variables; sign: + if variable lies on the same diagonal as the potential.

Fig. 10: The Maxwell thermodynamic square — a memory device for the four potentials, their natural variables, and the Maxwell relations.

6. Memory Box

Natural variables (what each potential is "happiest" with):

Potential	Definition	Natural vars	Fundamental eqn
U	—	S, V	$dU = TdS - PdV$
H	$U + PV$	S, P	$dH = TdS + VdP$
A	$U - TS$	T, V	$dA = -SdT - PdV$
G	$H - TS$	T, P	$dG = -SdT + VdP$

Square mnemonic: place T-P-V along the top edges and S-$U/H/A/G$ around — "**V*alid** F*acts and **T*heoretical** U*nderstanding **G*enerate** S*olutions to **H*ard** P*roblems."

7. Common Mistakes

- **$\Delta H = q$ always.** Enthalpy change equals heat **only at constant pressure** ($\Delta H = q_P$). At constant volume the heat is $\Delta U = q_V$.
- **Using the wrong equilibrium criterion.** A is minimized at constant T, V; G is minimized at constant T, P. Matching the potential to the constraints is essential.
- **Sign slips in Maxwell relations.** The two relations from U and H carry a minus sign; the two from A and G do not (for $(\partial S/\partial V)_T$). Read them off the square carefully.
- **Assuming ΔH is temperature-independent.** This is only true when $\Delta C_P = 0$; otherwise $\Delta H(T)$ and a plot of $\Delta G°$ vs. T is curved.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$H = U + PV$	Enthalpy; $\Delta H = q_P$	U, P, V	J	Const P for q_P
$A = U - TS$	Helmholtz (work) energy	U, T, S	J	Min at const T, V
$G = H - TS$	Gibbs free energy	H, T, S	J	Min at const T, P
$dG = -SdT + VdP$	Fundamental eqn (closed)	S, T, V, P	J	Only PV work
$\mu_i = \frac{\partial G}{\partial n_i}$	Chemical potential	G, n_i	J/mol	Const T, P, n_j
$\frac{\partial S}{\partial V}\|_T = \frac{\partial P}{\partial T}\|_V$	Maxwell relation	S, V, P, T	—	Closed system

Formula	Meaning	Variables	Units	Conditions / Limits
$\frac{\partial(G/T)}{\partial T}\|_P = -\frac{H}{T^2}$	Gibbs-Helmholtz	G, H, T	—	Const P

9. Worked Examples

Example 1 — Internal pressure from a Maxwell relation

Problem Using a Maxwell relation, express $(\partial U/\partial V)_T$ in measurable quantities, then evaluate it for an ideal gas.

Thinking Process Start from $dU = TdS - PdV$. Hold T constant and divide by dV, then replace the awkward $(\partial S/\partial V)_T$ with $(\partial P/\partial T)_V$ via the Maxwell relation.

Solution

$$\frac{\partial U}{\partial V}|_T = T\frac{\partial S}{\partial V}|_T - P = T\frac{\partial P}{\partial T}|_V - P$$

For an ideal gas $P = RT/V$, so $(\partial P/\partial T)_V = R/V$:

$$\frac{\partial U}{\partial V}|_T = T \cdot \frac{R}{V} - P = \frac{RT}{V} - P = P - P = 0$$

Answer $(\partial U/\partial V)_T = T(\partial P/\partial T)_V - P$, which equals 0 for an ideal gas.

Interpretation The internal energy of an ideal gas depends only on temperature — Joule's classic result — obtained without any calorimetry, purely from the equation of state.

Example 2 — The first TdS equation

Problem Derive $TdS = c_v dT + T(\partial P/\partial T)_V dV$.

Thinking Process Take $S = S(T, V)$, expand its total differential, then identify each partial derivative.

Solution

$$dS = \frac{\partial S}{\partial T}|_V \; dT + \frac{\partial S}{\partial V}|_T \; dV$$

Multiply by T. Use $(\partial S/\partial T)_V = c_v/T$ and the Maxwell relation $(\partial S/\partial V)_T = (\partial P/\partial T)_V$:

$$TdS = c_v dT + T\frac{\partial P}{\partial T}|_V \; dV$$

Answer $TdS = c_v dT + T(\partial P/\partial T)_V dV$.

Interpretation Entropy change is now written entirely in terms of the **measurable** heat capacity c_v and the equation of state — this is what makes the Maxwell relations useful in practice.

Example 3 — Gibbs-Helmholtz consistency

Problem A reaction has $\Delta G^\circ = -200000 + 50T$ (J/mol). Find ΔH° and ΔS°, then verify the result with the Gibbs-Helmholtz equation.

Thinking Process Compare term-by-term with $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$, then check that $(\partial(\Delta G^\circ/T)/\partial T)_P = -\Delta H^\circ/T^2$.

Solution Matching constant and T-terms:

$$\Delta H^\circ = -200000 \quad \text{J/mol}, \quad \Delta S^\circ = -50 \quad \text{J/(mol dot K)}$$

Check: $\Delta G^\circ/T = -200000/T + 50$, so

$$\frac{\partial(\Delta G^\circ/T)}{\partial T} = +\frac{200000}{T^2}$$

and $-\Delta H^\circ/T^2 = -(-200000)/T^2 = +200000/T^2$. The two agree.

Answer $\Delta H^\circ = -200000$ J/mol, $\Delta S^\circ = -50$ J/(mol·K); Gibbs-Helmholtz is satisfied.

Interpretation A **linear** $\Delta G^\circ(T)$ corresponds to temperature-independent ΔH° and ΔS° (i.e. $\Delta C_P = 0$). The intercept gives ΔH°, the negative slope gives ΔS° — a result fully consistent with Gibbs-Helmholtz.

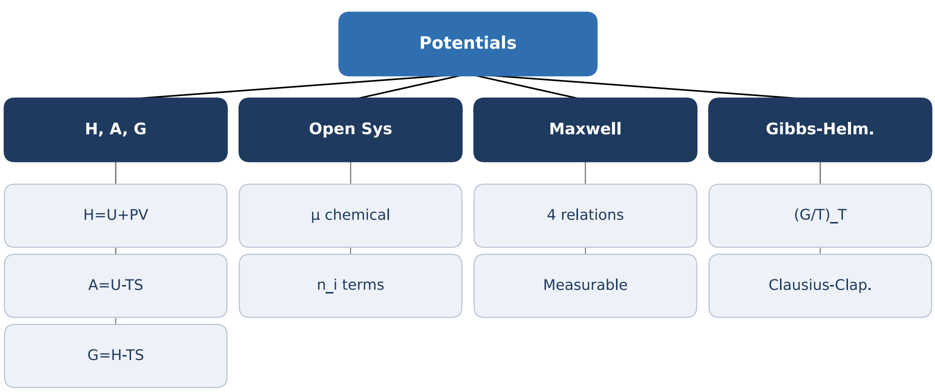
10. Quick Check

1. Which potential is minimized at constant T and P: A or G?
2. True or False: $\Delta H = q$ for any process.
3. Write the Maxwell relation obtained from $dG = -SdT + VdP$.
4. Fill in the blank: the natural variables of enthalpy are *and* .
5. For an ideal gas, what is $(\partial U/\partial V)_T$?

Answers: 1. G 2. False — only at constant pressure, $\Delta H = q_P$ 3. $(\partial S/\partial P)_T = -(\partial V/\partial T)_P$ 4. S and P 5. Zero

11. Chapter Summary

- Enthalpy $H = U + PV$; at constant pressure $\Delta H = q_P$.
- Helmholtz energy $A = U - TS$ is minimized at equilibrium for fixed T, V.
- Gibbs energy $G = H - TS$ is minimized at equilibrium for fixed T, P.
- Chemical potential $\mu_i = (\partial G/\partial n_i)$ enters the four open-system fundamental equations.
- The four Maxwell relations convert unmeasurable derivatives into measurable ones (the TdS equations).
- Gibbs-Helmholtz: $(\partial(G/T)/\partial T)_P = -H/T^2$.



12. Flash Cards

Q What is enthalpy?

A $H = U + PV$; at constant P, $\Delta H = q_P$.

Q What is the Helmholtz free energy and when is it minimized?

A $A = U - TS$; minimized at constant T, V.

Q What is the Gibbs free energy and when is it minimized?

A $G = H - TS$; minimized at constant T, P.

Q Define the chemical potential.

A $\mu_i = (\partial G/\partial n_i)_{T,P,n_j}$ — the rise in G on adding 1 mol of i.

Q State the "measurable" Maxwell relation.

A $(\partial S/\partial V)_T = (\partial P/\partial T)_V$.

Q Gibbs-Helmholtz equation?

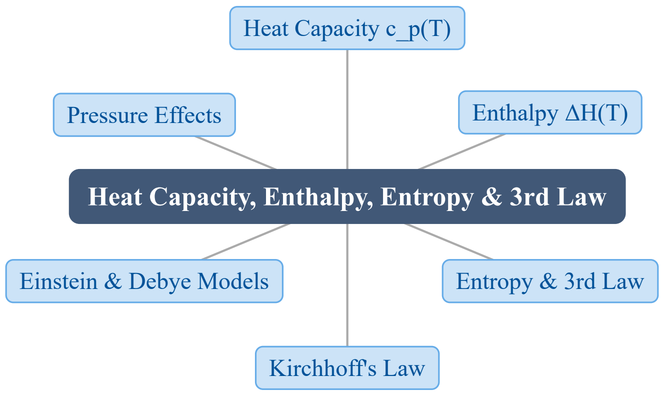
A $(\partial(G/T)/\partial T)_P = -H/T^2$.

13. Exam Tips

- **High priority:** deriving Maxwell relations from a fundamental equation, and using them to simplify $(\partial U/\partial V)_T$, $(\partial H/\partial P)_T$, or a TdS equation.
- Memorize the thermodynamic square — it regenerates all four fundamental equations **and** all four Maxwell relations in seconds.
- Gibbs-Helmholtz and the Clausius-Clapeyron equation are exam favorites; know how to get each from G.
- Always state your constraints (T, P, or V fixed) **before** choosing a potential or writing $\Delta H = q_P$.

6 Heat Capacity, Enthalpy, Entropy, and the Third Law of Thermodynamics

1. Chapter Overview



Heat capacity $c_{p(T)}$ is the bridge between atomic vibrations and the macroscopic quantities H and S. Integrating c_p over temperature gives the enthalpy and entropy changes that drive phase equilibria, reaction thermodynamics, and materials processing; the Third Law fixes the absolute entropy scale.

2. Learning Objectives

After studying this chapter, students should be able to:

- Define heat capacity c_p and c_v and state the Dulong-Petit high-temperature limit $c_v \approx 3R$.
- Explain the Einstein and Debye models, and the Debye T^3 law for $T \ll \theta_D$.
- Integrate $c_{p(T)}$ to obtain $\Delta H(T)$ and apply Kirchhoff's law across temperatures and phase changes.
- Use $\Delta S = \int c_p/T dT$ to compute entropy and apply the Third Law, $S(0) = 0$.
- Evaluate the pressure dependence of H and S from the Maxwell relations.

3. Why This Matters?

Heat capacity is the single quantity that connects a material's microscopic lattice and electronic structure to everything we compute in materials thermodynamics:

- **Phase diagrams & CALPHAD** — $G(T)$ curves from $H(T)$ and $S(T)$ built on c_p data.
- **Reaction equilibria** — Kirchhoff's law moves a known $\Delta H°$ to any temperature of interest.
- **Materials processing** — latent heats and sensible heats set furnace schedules and quench paths.
- **Absolute entropies** — the Third Law gives the $S°(T)$ values behind Ellingham and free-energy diagrams.

A small error in $c_{p(T)}$ propagates into every H and S you integrate, so the model you choose matters.

4. Intuition First

Example. Heat a block of metal: at high temperature each added joule raises the temperature by a steady amount, but near absolute zero the same joule barely moves the thermometer. **Picture.** Think of the atoms as masses on springs. Heating pumps energy into vibrations, and more vibrations need more energy — except quantum mechanics only allows discrete steps, so at low T most oscillators are frozen out and $c_v \to 0$. **Concept.** Classical (Dulong-Petit) theory gives a constant $3R$; quantum models (Einstein, Debye) capture the fall to zero. Once you know $c_{p(T)}$, **everything else** follows by integration. **Math.**

$$c_v \approx 3R \quad \text{and} \quad c_v = 3R(\theta_E/T)^2 \frac{e^{\theta_E/T}}{\left(e^{\theta_E/T} - 1\right)^2}$$

5. Visual Explanation

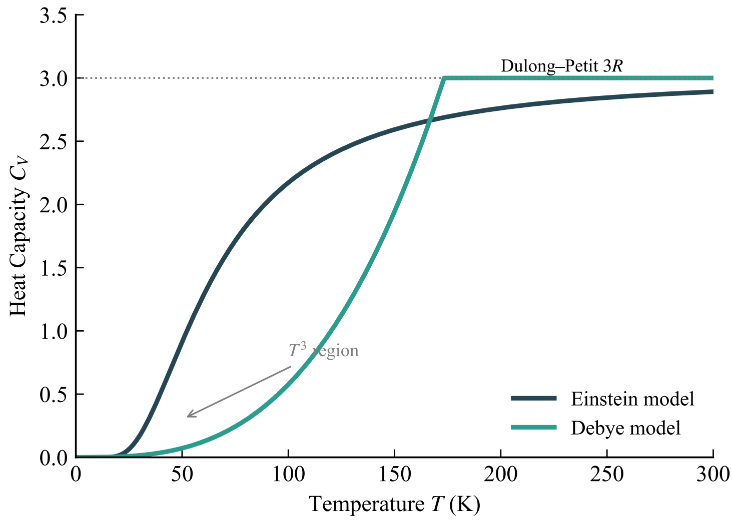


Fig. 11: Temperature dependence of the heat capacities c_p and c_v, showing the Dulong-Petit high-temperature plateau and the quantum fall-off toward $T = 0$.

6. Memory Box

Key results at a glance:

Model / Law	Formula	Valid when
Dulong-Petit	$c_v \approx 3R$	High T, solids
Einstein	$c_v = 3R(\theta_E/T)^2 e^{\theta_E/T}/\left(e^{\theta_E/T} - 1\right)^2$	Single frequency
Debye T^3	$c_v = 234R(T/\theta_D)^3$	$T \ll \theta_D$
Kirchhoff	$\Delta H(T_2) = \Delta H(T_1) + \int_{T_1}^{T_2} \Delta c_p dT$	Const P
Third Law	$S(T) = \int_0^T c_p/T\,dT$	$S(0) = 0$

Mnemonic: high-T solids remember **three are** $3R$; cold solids remember **cubed** T^3; perfect crystals remember **nothing** at zero kelvin ($S = 0$).

7. Common Mistakes

- **c_p vs. c_v.** Use c_p for constant-pressure heating ($\Delta H = q_P$); c_v for the Dulong-Petit/ Einstein/Debye solid-state models. They differ by $T(\partial P/\partial T)_V(\partial V/\partial T)_P$.
- **Dulong-Petit is not universal.** It holds only at high T; at low T quantum freezing makes $c_v \to 0$, so applying $3R$ there badly overestimates heat capacity.

- **The Debye T^3 constant.** The correct form is $234R(T/\theta_D)^3$. Writing $1944R(T/\theta_D)^3$ double-counts R (1944 already equals $234R$).
- **Third-Law exceptions.** Glasses, solid solutions, and molecules with residual orientational disorder (e.g. CO) keep a frozen-in entropy at 0 K: the Third Law needs a **perfect crystal**.
- **Forgetting latent heat.** Integrating c_p across a melting or transition temperature misses the ΔH_{trans} step — add it explicitly at the fixed transition temperature.
- **Extrapolating empirical c_p.** The $a + bT + c/T^2$ fit is valid only in its stated range; never extrapolate across a phase change or polymorphism boundary.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$c_p = a + bT + \frac{c}{T^2}$	Empirical heat-capacity fit	a, b, c, T	J/(mol·K)	Stated range only
$c_v \approx 3R$	Dulong-Petit high-T limit	R, T	J/(mol·K)	High T, solids
$c_v = 3R\left(\frac{\theta_E}{T}\right)^2 \frac{e^{\frac{\theta_E}{T}}}{\left(e^{\frac{\theta_E}{T}}-1\right)^2}$	Einstein model	θ_E, T, R	J/(mol·K)	Single frequency
$c_v = 234R\left(\frac{T}{\theta_D}\right)^3$	Debye T^3 law	θ_D, T, R	J/(mol·K)	$T \ll \theta_D$
$\Delta H = \int_{T_1}^{T_2} c_p dT$	Enthalpy on heating	c_p, T	J/mol	Const P
$\Delta H(T_2) = \Delta H(T_1) + \int_{T_1}^{T_2} \Delta c_p dT$	Kirchhoff's law	$\Delta c_p, T$	J/mol	Const P
$\Delta S = \int_{T_1}^{T_2} \frac{c_p}{T} dT$	Entropy on heating	c_p, T	J/(mol·K)	Const P
$S(T) = \int_0^T \frac{c_p}{T} dT$	Third-Law absolute entropy	c_p, T	J/(mol·K)	$S(0) = 0$
$\frac{\partial S}{\partial P}\vert_T = -\frac{\partial V}{\partial T}\vert_P$	Pressure effect on S	S, V, P, T	—	Maxwell

9. Worked Examples

Example 1 — Enthalpy of formation at 1000 K (Kirchhoff's law)

Problem The standard enthalpy of formation of Al_2O_3 at 298 K is −1675.7 kJ/mol. With $c_{p(\mathrm{Al_2O_3})} = 79.0 + 0.018T - 17.4 \times 10^5 T^{-2}$, $c_{p(\mathrm{Al})} = 20.7 + 0.0124T$, and $c_{p(\mathrm{O_2})} = 29.4$ (all J/(mol·K)), find $\Delta H_f°$ at 1000 K for $2\ \mathrm{Al} + \frac{3}{2}\mathrm{O_2} \rightarrow \mathrm{Al_2O_3}$.

Thinking Process Compute $\Delta c_p = c_{p(\mathrm{Al_2O_3})} - 2c_{p(\mathrm{Al})} - \frac{3}{2}c_{p(\mathrm{O_2})}$, then integrate Δc_p from 298 to 1000 K and add to $\Delta H(298)$.

Solution

$$\Delta c_p = (79.0 + 0.018T - 17.4 \times 10^5 T^{-2}) - 2(20.7 + 0.0124T) - 1.5(29.4)$$

$$\Delta c_p = -6.5 - 0.0068T - 17.4 \times 10^5 T^{-2}$$

$$\int_{298}^{1000} \Delta c_p dT = [-6.5T - 0.0034T^2 + 17.4 \times 10^5 T^{-1}]_{298}^{1000}$$

$$= (-6500 - 3400 + 1740)$$
$$-(-1937 - 302 + 5840) = -8160 - 3601 = -11761 \quad \mathrm{J/mol}$$

$$\Delta H(1000) = -1675700 - 11761 = -1687461 \quad \mathrm{J/mol}$$

Answer $\Delta H_f°(1000) = -1687.5$ kJ/mol.

Interpretation The heat of formation is slightly **more** negative at high T because $\Delta c_p < 0$; the correction is only about 12 kJ/mol over 700 K — small but not negligible for accurate diagrams.

Example 2 — Third-Law absolute entropy of Cu at 50 K

Problem For Cu with $T < 50$ K, $c_p = \gamma T + \beta T^3$ where $\gamma = 0.70 \times 10^{-3}$ and $\beta = 0.95 \times 10^{-5}$ (units J/(mol·K²) and J/(mol·K⁴)). Find $S(50K)$ given $S(0) = 0$.

Thinking Process Use the Third-Law integral $S(T) = \int_0^T c_p/T dT$; the T in c_p cancels the denominator, leaving a simple polynomial integral.

Solution

$$S(50) = \int_0^{50} \frac{\gamma T + \beta T^3}{T} dT = \int_0^{50} (\gamma + \beta T^2) dT$$

$$= \left[\gamma T + \frac{\beta T^3}{3}\right]_0^{50} = 0.70 \times 10^{-3}(50) + \frac{0.95 \times 10^{-5}(50)^3}{3}$$

$$= 0.035 + \frac{1.1875}{3} = 0.035 + 0.396 = 0.431 \quad \mathrm{J/(mol{\cdot}K)}$$

Answer $S(50 \quad \mathrm{K}) = 0.431$ J/(mol·K).

Interpretation The electronic (γT) and phonon (βT^3) parts contribute 0.035 and 0.396 respectively; at very low T the phonon T^3 term dominates the entropy of a metal.

Example 3 — Heating integral for ΔH and ΔS

Problem A solid has $c_p = 25 + 0.010T$ J/(mol·K) between 298 K and 1000 K, with no phase change. Find the enthalpy and entropy changes on heating from 298 to 1000 K.

Thinking Process Integrate c_p for ΔH and c_p/T for ΔS; split the latter into a logarithmic part and a linear part.

Solution

$$\Delta H = \int_{298}^{1000} (25 + 0.010T) dT = [25T + 0.005T^2]_{298}^{1000}$$

$$= (25000 + 5000) - (7450 + 444) = 22106 \quad \mathrm{J/mol}$$

$$\Delta S = \int_{298}^{1000} \left(\frac{25}{T} + 0.010\right) dT = [25 \ln T + 0.010T]_{298}^{1000}$$

$$= 25 \ln(1000/298) + 0.010(1000 - 298)$$
$$= 30.27 + 7.02 = 37.29 \quad \mathrm{J/(mol{\cdot}K)}$$

Answer $\Delta H = 2.21 \times 10^4$ J/mol, $\Delta S = 37.3$ J/(mol·K).

Interpretation Both integrals are the workhorses of thermochemistry: every tabulated $H°(T)$ and $S°(T)$ in a database comes from exactly this kind of integration over a fitted $c_{p(T)}$.

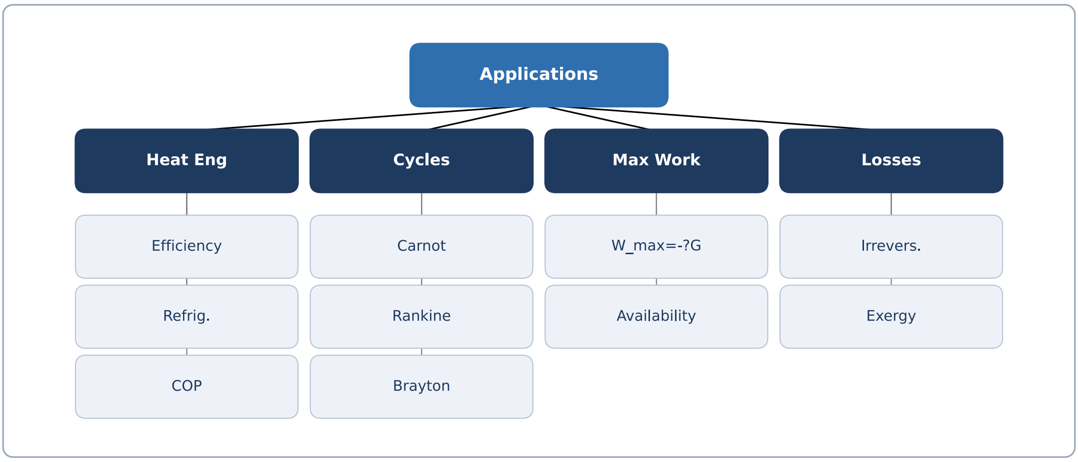
10. Quick Check

1. State the Dulong-Petit limit for the molar heat capacity c_v of a solid.
2. In the Debye model, how does c_v depend on T at very low temperature?
3. Write Kirchhoff's law relating ΔH at two temperatures.
4. According to the Third Law, what is the entropy of a perfect crystal at 0 K?
5. True or False: $\Delta H = q$ for any process.

Answers: 1. $c_v \approx 3R \approx 24.9$ J/(mol·K). 2. $c_v \propto T^3$, i.e. $c_v = 234R(T/\theta_D)^3$. 3. $\Delta H(T_2) = \Delta H(T_1) + \int_{T_1}^{T_2} \Delta c_p dT$. 4. $S(0) = 0$. 5. False — only at constant pressure, $\Delta H = q_P$.

11. Chapter Summary

- Heat capacity $c_{p(T)}$ is the bridge to $H(T)$ and $S(T)$; Dulong-Petit gives $c_v \approx 3R$ at high T.
- Einstein and Debye models explain the quantum fall of c_v to zero; Debye T^3 law: $c_v = 234R(T/\theta_D)^3$.
- Enthalpy change on heating is $\Delta H = \int c_p dT$ (plus latent heat); Kirchhoff's law shifts it between temperatures.
- Entropy change is $\Delta S = \int c_p/T dT$; the Third Law sets $S(0) = 0$ for a perfect crystal.
- Third-Law exceptions (glasses, solid solutions, residual disorder) retain frozen-in entropy at 0 K.
- Pressure effects: $(\partial S/\partial P)_T = -(\partial V/\partial T)_P$ (small for condensed phases).



12. Flash Cards

Q What is the Dulong-Petit law?

A $c_v \approx 3R$; the high-temperature molar heat capacity of a solid element.

Q What does the Einstein model assume?

A All $3N$ atomic oscillators share one frequency ν; gives $c_v \to 0$ as $T \to 0$.

Q State the Debye T^3 law.

A $c_v = 234R(T/\theta_D)^3$ for $T \ll \theta_D$; the low-T phonon limit.

Q What is Kirchhoff's law?

A $\Delta H(T_2) = \Delta H(T_1) + \int \Delta c_p dT$ — corrects reaction heats across T.

Q State the Third Law.

A The entropy of a perfect crystal at absolute zero is zero: $S(0) = 0$.

Q Pressure effect on entropy (Maxwell)?

A $(\partial S/\partial P)_T = -(\partial V/\partial T)_P$ — usually small for solids/liquids.

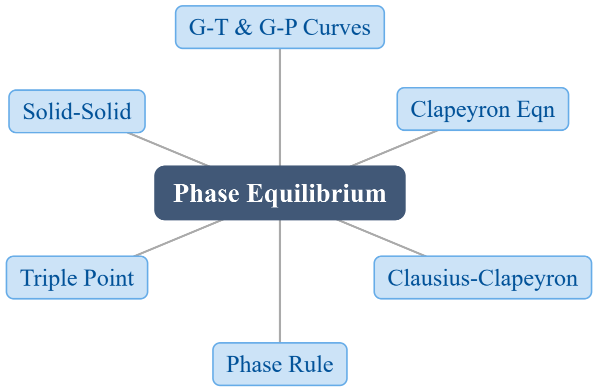
13. Exam Tips

- **High priority:** integrating $c_{p(T)}$ for ΔH and ΔS, and Kirchhoff's law between a known reference temperature and the one asked.

- Memorize the Dulong-Petit limit $3R$, the Debye T^3 form $234R(T/\theta_D)^3$, and the Einstein expression — exam questions probe the **low-** and **high-T** limits, not the full integrals.
- Always add latent heat explicitly when the path crosses a phase-transition temperature; a bare c_p integral silently misses it.
- For Third-Law entropy, set $S(0) = 0$ and integrate c_p/T; do not forget the electronic γT term for metals at very low T.
- State constraints (P fixed) before writing $\Delta H = q_P$, and use c_p (not c_v) for ordinary heating at atmospheric pressure.

7 Phase Equilibrium in a One-Component System

1. Chapter Overview



At fixed T, P, the stable phase of a one-component system is the one with the lowest Gibbs free energy G. Two phases are in equilibrium when their G values are equal. By examining how G varies with T and P we obtain the equations that describe every phase boundary on a P-T diagram.

2. Learning Objectives

After studying this chapter, students should be able to:

- Sketch the G vs. T and G vs. P curves and explain their slopes and curvature in terms of S and V.
- Derive the Clapeyron equation $(\partial P/\partial T) = (\Delta H)/(T\Delta V)$ from $dG^{\alpha} = dG^{\beta}$.
- Apply the Clausius-Clapeyron equation to vapor-pressure and boiling-point problems.
- Use the one-component phase rule $F = 3 - \varphi$ and locate the triple point.
- Explain why water's melting curve has a negative slope and how solid-solid transitions behave.

3. Why This Matters?

Phase equilibrium is the foundation of **every** phase diagram used in materials science and metallurgy:

- **Steelmaking** — the iron allotropes α (BCC), γ (FCC), δ (BCC) are mapped by their equilibrium lines; heat treatment is navigation of this map.
- **Processing** — vapor-pressure curves govern distillation, drying, and sintering atmospheres.
- **Earth & climate** — the abnormal slope of water's melting line controls glaciers, icebergs, and freezing lakes.

The Clapeyron and Clausius-Clapeyron equations turn a qualitative phase diagram into **quantitative** predictions of how a boundary shifts with T and P.

4. Intuition First

Example: Imagine heating a block of ice at 1 atm. Below $0°C$ the solid has the lower G; above it the liquid does. At exactly $0°C$ the two curves cross — that crossing **is** the melting point.

Picture: Each phase owns a surface $G(T, P)$. The lab is a point (T, P). Whichever surface is lowest there is the stable phase. Two surfaces touch along a **line** (the phase boundary); three touch at a single **point** (the triple point).

Concept: Equilibrium means equal G of the coexisting phases, not equal slopes. Along a boundary $G^\alpha = G^\beta$, so $dG^\alpha = dG^\beta$ — and that single equality produces the slope of the boundary.

Math: With $dG = -SdT + VdP$ for each phase,

$$-S^\alpha dT + V^\alpha dP = -S^\beta dT + V^\beta dP \quad \longrightarrow \quad (\partial P/\partial T) = \left(S^\beta - S^\alpha\right)/\left(V^\beta - V^\alpha\right)$$

5. Visual Explanation

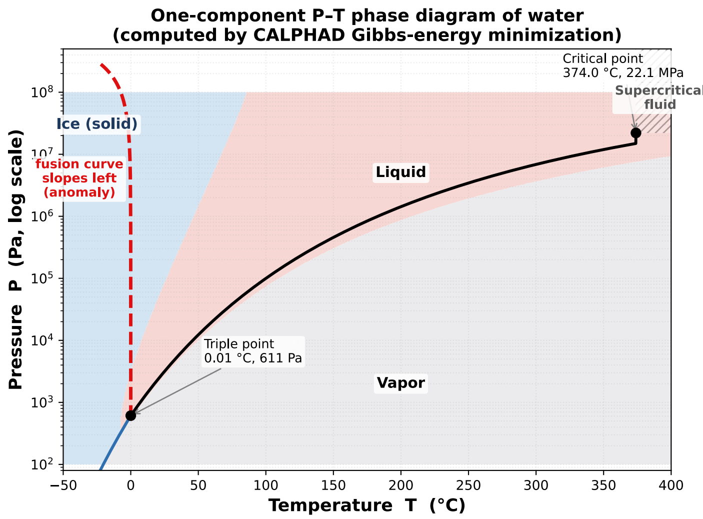


Fig. 12: One-component P-T phase diagram for water: solid, liquid, and vapor regions meet at the triple point O. The fusion curve (red) slopes to the **left** with rising pressure — the anomalous negative $(\partial P/\partial T)_G$ of ice — because liquid water is denser than ice.

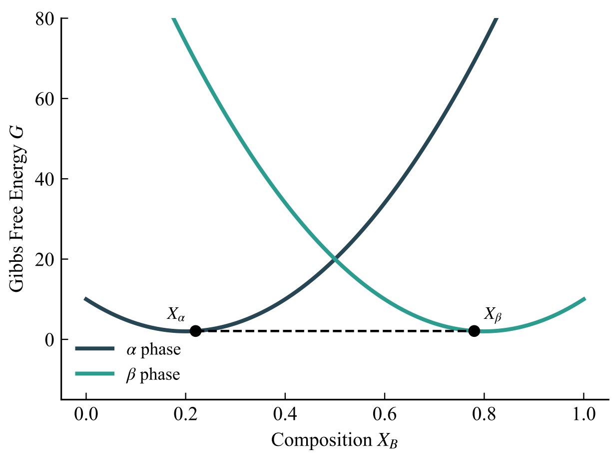


Fig. 13: Relative G of phases versus T at fixed P; the lowest curve is stable and crossings mark transition temperatures.

6. Memory Box

Curves of G:

Quantity	G vs. T (const P)	G vs. P (const T)
slope	$(\partial G/\partial T)_P = -S < 0$	$(\partial G/\partial P)_T = V > 0$
curvature	$(\partial^2 G/\partial T^2) = -c_p/T < 0$	$(\partial^2 G/\partial P^2) = -V\beta_T < 0$
shape	concave downward	concave downward

Two boundary equations: Clapeyron (exact, any transition) vs. Clausius-Clapeyron (vapor only, ideal gas). **Phase rule:** $F = C + 2 - \varphi$; for one component $F = 3 - \varphi$ (max 3 phases, $F = 0$, at the triple point).

7. Common Mistakes

- **Mixing up Clapeyron and Clausius-Clapeyron.** Clapeyron $(\partial P/\partial T) = (\Delta H)/(T\Delta V)$ is exact and uses the **full** ΔV. Clausius-Clapeyron approximates $\Delta V \approx V_{\text{vapor}}$ and needs the ideal-gas law — it applies only to a vapor boundary.
- **Sign error for water.** Most substances expand on melting, so $\Delta V_{\text{melt}} > 0$ and the slope is positive. Water **contracts** on freezing ($\Delta V_{\text{melt}} < 0$), giving a **negative** melting slope.
- **Treating ΔH as constant in Clausius-Clapeyron.** The integrated form assumes $\Delta c_p = 0$; over a wide T range the $\ln P$ vs. $1/T$ plot curves.
- **Confusing triple point with 1-atm transitions.** The triple point is at the substance's own vapor pressure (water: 0.006 atm); normal melting/boiling are defined at 1 atm and lie on different points.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$dG = -SdT + VdP$	Fundamental eqn (closed, PV work)	S, T, V, P	J	Only PV work
$(\partial G/\partial T)_P = -S$	Slope of G vs. T	G, S, T	J/K	Const P
$(\partial G/\partial P)_T = V$	Slope of G vs. P	G, V, P	m^3	Const T
$(\partial P/\partial T) = (\Delta H)/(T\Delta V)$	Clapeyron equation	H, V, T	Pa/K	Two-phase eq.
$(\partial(\ln P))/(\partial T) = (\Delta H)/(RT^2)$	Clausius-Clapeyron (diff.)	P, H, T, R	—	Vapor, ideal gas

Formula	Meaning	Variables	Units	Conditions / Limits
$\ln(P/P_0) = -(\Delta H/R)(1/T - 1/T_0)$	Clausius-Clapeyron (integrated)	P, H, T, R	—	ΔH const
$F = C + 2 - \varphi$	Gibbs phase rule	C, φ	—	C comp, φ phases
$\Delta S_{\text{melt}} = (\Delta H_m)/T_m$	Entropy of melting	H_m, T_m	J/K	1st-order transition

9. Worked Examples

Example 1 — Boiling point of water at 0.5 atm

Problem The normal boiling point of water is $T_0 = 373.15$ K at $P_0 = 1$ atm, with $\Delta H_{\text{vap}} = 40.7$ kJ/mol. Estimate the boiling point at $P = 0.5$ atm using the integrated Clausius-Clapeyron equation.

Thinking Process Use $\ln(P/P_0) = -\left(\Delta H_{\text{vap}}/R\right)(1/T - 1/T_0)$ and solve for $1/T$. Assume ΔH_{vap} is constant over this range.

Solution

$$\ln(0.5) = -((40700)/(8.314))(1/T - 1/373.15)$$

$$-0.6931 = -(4895)(1/T - 0.002680)$$

$$1/T = 0.002680 + 0.6931/4895 = 0.002680 + 0.0001416 = 0.002822$$

$$T = 354.4 \quad \text{K} = 81.2°C$$

Answer $T \approx 354.4$ K, i.e. $81.2°C$ at 0.5 atm.

Interpretation Lower pressure lowers the boiling point — at high altitude water boils below $100°C$, lengthening cooking times. The linear $\ln P$ vs. $1/T$ relation is the standard way to measure ΔH_{vap}.

Example 2 — Clapeyron slope of a solid-solid transition

Problem An α-β transition has $\Delta H_{\text{trs}} = 2.5$ kJ/mol, $\Delta V_{\text{trs}} = 0.5$ cm³/mol, and $T_{\text{trs}} = 500$ K. Find dP/dT and the new transition temperature when the pressure rises by 100 atm.

Thinking Process Apply the Clapeyron equation directly, converting units: 1 cm³/mol $= 10^{-6}$ m³/mol and 1 atm $= 1.013 \times 10^5$ Pa.

Solution

$$(dP/dT) = (\Delta H)/(T\Delta V) = (2500)/((500)(0.5 \times 10^{-6})) = 1.0 \times 10^7 \quad \text{Pa/K}$$

$$1.0 \times 10^7 \quad \text{Pa/K} \times (9.869 \times 10^{-6} \quad \text{atm/Pa}) = 98.7 \quad \text{atm/K}$$

$$\Delta T = (\Delta P)/(dP/dT) = 100/98.7 = 1.01 \quad \text{K}$$

$$T_{\text{trs}'} = 500 + 1.01 = 501.0 \quad \text{K}$$

Answer $dP/dT \approx 98.7$ atm/K; raising P by 100 atm raises T_{trs} by only 1.0 K to 501 K.

Interpretation Solid-solid boundaries are nearly vertical: huge pressure changes are needed to shift the transition temperature. This is why pressure has only a modest effect on allotropic transformations.

Example 3 — Why ice floats (Clapeyron slope of melting)

Problem Water expands on freezing: $\rho_{\text{ice}} < \rho_{\text{water}}$, so $\Delta V_{\text{melt}} = V_{\text{liq}} - V_{\text{solid}} < 0$. With $\Delta H_{\text{melt}} = 334$ kJ/mol at $T_m = 273$ K and $\Delta V_{\text{melt}} = -1.63 \times 10^{-6}$ m³/mol, find the Clapeyron slope of the melting curve.

Thinking Process Insert ΔH and ΔV into the Clapeyron equation; the sign of ΔV carries the result.

Solution

$$(dP/dT) = (\Delta H_{\text{melt}})/(T_m \Delta V_{\text{melt}}) = (334 \times 10^3)/((273)(-1.63 \times 10^{-6}))$$

$$(dP/dT) \approx -7.5 \times 10^5 \quad \text{Pa/K}$$

Answer $(dP/dT) \approx -7.5 \times 10^5$ Pa/K — a **negative** slope.

Interpretation Increasing pressure **lowers** the melting point of water. Ice is less dense than liquid water, so it floats and insulates lakes; if water behaved normally, lakes would freeze from the bottom up.

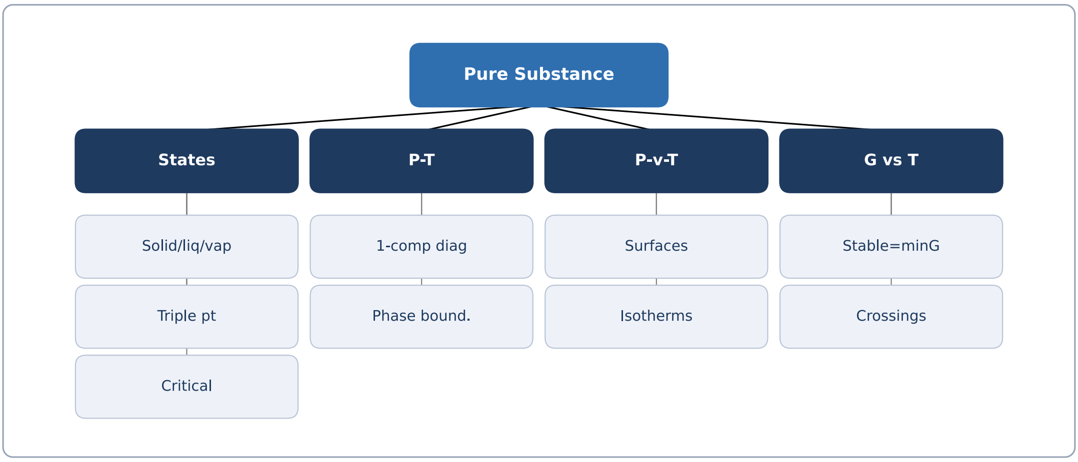
10. Quick Check

1. At fixed T, P, which phase is stable — the one with the higher or the lower G?
2. Write the Clapeyron equation for a two-phase boundary.
3. True or False: the Clausius-Clapeyron equation is exact for any solid-solid transition.
4. For a one-component system, what is the maximum number of coexisting phases, and what is F there?
5. Does water's melting curve slope up or down as pressure increases? Why?

Answers: 1. The phase with the **lower** G is stable. 2. $(\partial P/\partial T) = (\Delta H)/(T\Delta V)$. 3. False — it applies only to vapor boundaries (ideal-gas approximation). 4. Three phases; $F = 0$ (the triple point). 5. Down (negative slope), because $\Delta V_{\text{melt}} < 0$.

11. Chapter Summary

- At fixed T, P the stable phase has the lowest G; equilibrium means $G^\alpha = G^\beta$.
- Slopes: $(\partial G/\partial T)_P = -S < 0$, $(\partial G/\partial P)_T = V > 0$; both curves are concave downward.
- Clapeyron equation $(\partial P/\partial T) = (\Delta H)/(T\Delta V)$ for any two-phase boundary.
- Clausius-Clapeyron $\ln P = -\Delta H/(RT) + \text{const}$ for vapor equilibria; $\ln P$ vs. $1/T$ gives $-\Delta H/R$.
- Phase rule $F = C + 2 - \varphi$; for one component $F = 3 - \varphi$, with three phases meeting at the triple point ($F = 0$).
- Water's melting curve slopes negatively because $\Delta V_{\text{melt}} < 0$; solid-solid transitions are nearly vertical.



12. Flash Cards

Q What is the phase-equilibrium condition at fixed T, P?

A G of each coexisting phase is equal: $G^{\alpha} = G^{\beta}$.

Q What is the Clapeyron equation?

A $(\partial P/\partial T) = (\Delta H)/(T\Delta V)$ — exact slope of any phase boundary.

Q When does the Clausius-Clapeyron form apply?

A Vapor-condensed equilibrium with $\Delta V \approx V_{\text{vapor}}$ and ideal-gas behavior.

Q State the one-component phase rule.

A $F = 3 - \varphi$; at most three phases ($F = 0$) at the triple point.

Q Why is water's melting slope negative?

A $\Delta V_{\text{melt}} < 0$ (ice expands on freezing), so $(dP/dT) < 0$.

Q What is ΔS_{melt}?

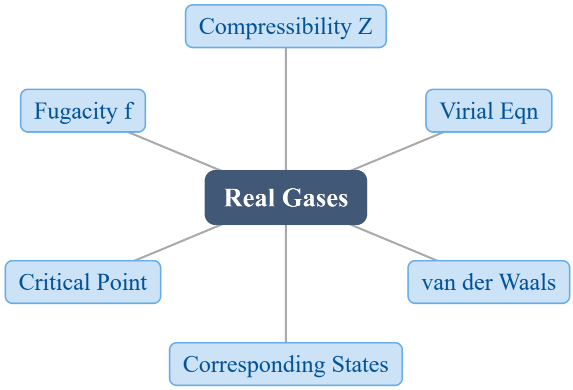
A $\Delta S_{\text{melt}} = (\Delta H_m)/T_m > 0$; the entropy jump at a first-order melting transition.

13. Exam Tips

- **High priority:** derive the Clapeyron equation from $dG^\alpha = dG^\beta$ — it appears in some form on nearly every exam.
- Memorize the Clausius-Clapeyron integrated form and the $\ln P$ vs. $1/T$ slope $-\Delta H/R$.
- Always note the sign of ΔV: for water $\Delta V_{\text{melt}} < 0$ flips the melting slope. Most other substances have $\Delta V > 0$ (positive slope).
- Apply the phase rule before counting degrees of freedom; for one component use $F = 3 - \varphi$.
- Watch unit conversions in numerical problems: $cm^3/mol \rightarrow m^3/mol$, atm $\rightarrow$ Pa, kJ $\rightarrow$ J.

8 The Behavior of Real Gases

1. Chapter Overview



The ideal gas law $PV = nRT$ is exact only in the limit of low pressure and high temperature. Real molecules have finite size and attract one another, so PV departs from nRT. This chapter quantifies that departure through the compressibility factor Z, the van der Waals equation, the law of corresponding states, and fugacity — the effective pressure that keeps the chemical-potential formula simple for real gases.

2. Learning Objectives

After studying this chapter, students should be able to:

- Define the compressibility factor Z and interpret $Z < 1$ versus $Z > 1$.
- Write the virial expansion and explain the meaning of the second virial coefficient $B(T)$.
- Use the van der Waals equation and extract the critical constants T_c, P_c, V_c and Z_c.
- Apply the law of corresponding states with the reduced variables T_R, P_R, V_R.
- Define fugacity f and the fugacity coefficient φ, and compute φ from Z data.

3. Why This Matters?

Real-gas behavior is not a textbook curiosity — it governs processes where pressure and density are high:

- **Gas storage & transport** — the capacity of a CO_2 or CH_4 pipeline depends on how far Z strays from 1.

- **Chemical & petroleum engineering** — reactor and separator design needs a correct equation of state at hundreds of atmospheres.
- **Phase equilibria** — fugacity is the quantity that must be equal across phases at equilibrium, replacing pressure for real fluids.
- **Extraction metallurgy & geochemistry** — high-pressure gas solubility and mineral-gas reactions.

Getting Z or φ wrong by even 20% can mean an undersized compressor or a runaway column.

4. Intuition First

Imagine pumping a gas into a cylinder. At low pressure the molecules rarely meet, so the ideal law holds. Push harder and two effects appear. **Finite size** removes free space, so the pressure climbs above the ideal value; **attraction** pulls molecules back, so the pressure falls below it.

The compressibility factor $Z = \frac{PV}{nRT}$ is the single number that reports the net effect: $Z < 1$ means attraction wins (most gases near room T), $Z > 1$ means size wins (high T or very dense). As temperature rises, attraction weakens and $Z \Rightarrow 1$.

The van der Waals equation captures both effects by subtracting a volume b for size and adding $\frac{a}{V_m^2}$ for attraction:

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

where $V_m = \frac{V}{n}$ is the molar volume. Fugacity f is then defined so the chemical potential keeps its ideal-looking logarithm: $\mu = \mu^\circ + RT\ln\left(\frac{f}{P^\circ}\right)$.

5. Visual Explanation

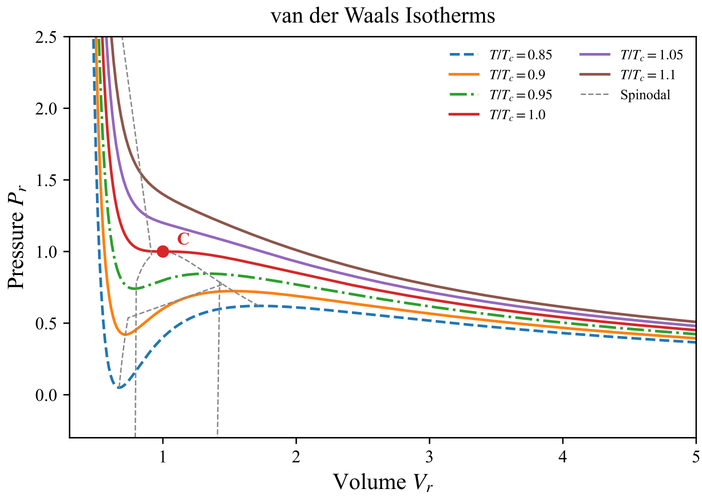


Fig. 14: van der Waals isotherms: pressure-volume curves above and below the critical point, with the spinodal line.

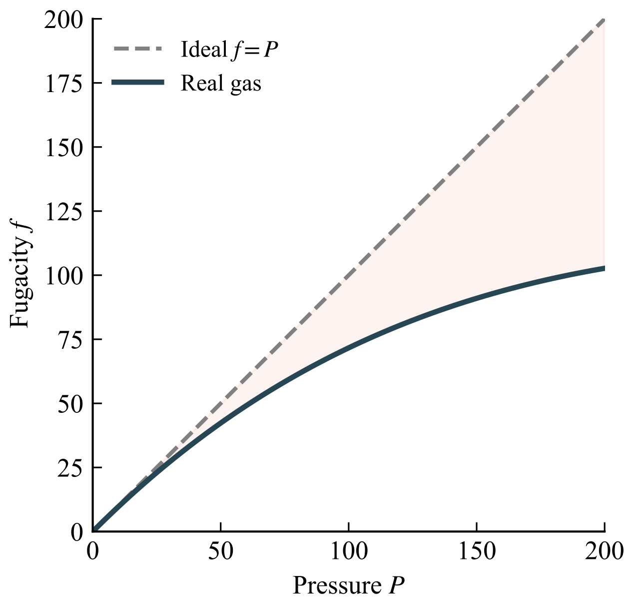


Fig. 15: Fugacity f versus pressure P: as $P \to 0$, $f \Rightarrow P$ so $\varphi \Rightarrow 1$.

6. Memory Box

Compressibility-factor regimes:

Regime	Cause	Example
$Z < 1$	Attraction dominates	Most gases near room T
$Z \approx 1$	Attraction $\approx$ repulsion	Above Boyle temperature
$Z > 1$	Repulsive size dominates	Dense or high-T gas

Critical constants from van der Waals: $V_c = 3b$, $T_c = \frac{8a}{27Rb}$, $P_c = \frac{a}{27b^2}$, so $Z_c = \frac{P_cV_c}{RT_c} = \frac{3}{8} = 0.375$. **Fugacity:** $\varphi = f/P$, with $\varphi \Rightarrow 1$ as $P \to 0$.

7. Common Mistakes

- **Treating $Z = 1$ as always valid.** A real gas at high pressure has Z far from 1; using the ideal law then gives large errors (up to about 25% in the worked examples).
- **Confusing P and f.** Fugacity is an effective pressure, not the actual pressure; only at $P \to 0$ do they coincide. Use P in the ideal gas law, f in $\mu = \mu^\circ + RT\ln(f/P^\circ)$.
- **Mixing molar and total volume in van der Waals.** Use $V_m = \frac{V}{n}$, or keep the n-form $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$ consistently.
- **Assuming $Z_c = 0.375$ for real gases.** That is the van der Waals prediction; real gases have $Z_c \approx 0.2$ to 0.3, exposing the model's quantitative limit.
- **Dropping the reference state in fugacity.** Fugacity is defined relative to P° (usually 1 bar); the chemical potential uses f/P°.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$Z = \frac{PV}{nRT}$	Compressibility factor; $Z = 1$ ideal	P, V, n, T	—	Any P, V, T
$\frac{PV}{RT} = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \ldots$	Virial expansion	B, C, V_m	—	$P \to 0$ truncation
$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$	van der Waals EOS	a, b, V_m, T	—	Real gas
$V_c = 3b, T_c = \frac{8a}{27Rb}, P_c = \frac{a}{27b^2}$	Critical constants	a, b, R	L, K, Pa	vdW only
$(\partial U/\partial V)_T = T(\partial P/\partial T)_V - P$	Internal pressure	U, V, T, P	Pa	Any EOS
$T_R = \frac{T}{T_c}, P_R = \frac{P}{P_c}, V_R = \frac{V}{V_c}$	Reduced variables	T, P, V, T_c, P_c, V_c	—	Corresponding states
$\mu_i = \mu_i^\circ + RT\ln(f_i/P^\circ)$	Real-gas chemical potential	μ, f, T	J/mol	Const T
$\ln\varphi = \frac{1}{RT}\int_0^P \left(V_m - \frac{RT}{P}\right)dP$	Fugacity from EOS	φ, P, V_m, T	—	Integrate to P

Formula	Meaning	Variables	Units	Conditions / Limits
$\ln\varphi = \int_0^P \left(\frac{Z-1}{P}\right) dP$	Fugacity from Z	φ, Z, P	—	$Z(P)$ data

9. Worked Examples

Example 1 — van der Waals versus ideal pressure (CO2)

Problem Find the pressure of 1 mol of CO_2 in a 0.5 L vessel at 300 K. Given $a = 3.658 \text{L}^2 \cdot \text{atm} \cdot \text{mol}^{-2}$, $b = 0.04286\ \text{L} \cdot \text{mol}^{-1}$.

Thinking Process Compute the ideal pressure $P = nR\frac{T}{V}$, then the van der Waals pressure using molar volume $V_m = \frac{V}{n}$. The attractive $\frac{a}{V_m^2}$ term lowers P.

Solution Ideal:

$$P_{\text{id}} = \frac{nRT}{V} = \frac{(1)(0.08206)(300)}{0.5} = 49.24\ \text{atm}$$

van der Waals ($V_m = 0.5$ L/mol):

$$P_{\text{vdW}} = \frac{RT}{V_m - b} - \frac{a}{V_m^2} = \frac{(0.08206)(300)}{0.5 - 0.04286} - \frac{3.658}{(0.5)^2}$$

$$= 53.85 - 14.63 = 39.22\ \text{atm}$$

Answer $P_{\text{id}} = 49.24$ atm, $P_{\text{vdW}} = 39.22$ atm — a 20% drop.

Interpretation The attraction term $\frac{a}{V_m^2}$ cuts the pressure well below ideal. At 600 K the gap shrinks to about 5%, confirming that high temperature restores near-ideal behavior.

Example 2 — Compressibility factor by virial iteration (N2)

Problem At 400 K the second virial coefficient of N_2 is $B = -0.0105\ \text{L} \cdot \text{mol}^{-1}$. Find Z and V_m at $P = 100$ atm, using $Z = 1 + \frac{B}{V_m}$.

Thinking Process Solve iteratively. Start from the ideal volume $V_m^0 = R\frac{T}{P}$, then update $V_m = \frac{RT}{P}\left(1 + \frac{B}{V_m}\right)$ until it converges, and compute $Z = P\frac{V_m}{RT}$.

Solution

$$V_m^0 = (0.08206)\frac{400}{100} = 0.3282 \text{ L/mol}$$

$$V_m^1 = 0.3282\left(1 + \frac{-0.0105}{0.3282}\right) = 0.3177 \text{ L/mol}$$

$$V_m^2 = 0.3282\left(1 + \frac{-0.0105}{0.3177}\right) = 0.3173 \text{ L/mol}$$

$$Z = \frac{PV_m}{RT} = (100)\frac{0.3173}{(0.08206)(400)} = 0.967$$

Answer $V_m = 0.3173$ L/mol, $Z = 0.967$.

Interpretation $Z < 1$ confirms attraction dominates; the volume is about 3.3% smaller than the ideal value. The iteration converges in two steps because $\frac{B}{V_m}$ is small.

Example 3 — Fugacity coefficient from corresponding states (CH4)

Problem Estimate φ for CH_4 at 300 K and 50 atm. Given $T_c = 190.6$ K, $P_c = 45.99$ atm.

Thinking Process Form the reduced variables $T_R = \frac{T}{T_c}$ and $P_R = \frac{P}{P_c}$, then read φ from the generalized fugacity chart for those reduced conditions.

Solution

$$T_R = \frac{300}{190.6} = 1.574, \quad P_R = \frac{50}{45.99} = 1.087$$

From the generalized chart at $T_R \approx 1.57$, $P_R \approx 1.09$: $\varphi \approx 0.93$.

$$f = \varphi P = (0.93)(50) = 46.5 \text{ atm}$$

Answer $\varphi \approx 0.93$, $f \approx 46.5$ atm.

Interpretation Only the critical constants are needed — no substance-specific EOS. Since $\varphi < 1$, attraction lowers f below P. For $T_R > 1.5$ and $P_R < 2$ the chart is usually within 5%.

10. Quick Check

1. What does $Z < 1$ indicate about intermolecular forces in a real gas?
2. True or False: the van der Waals constant b corrects for intermolecular attraction.
3. Write the three reduced variables used in the law of corresponding states.
4. Fill in the blank: fugacity f equals pressure P in the limit $P \to$ zero.
5. For a real gas, the chemical potential is $\mu = \mu^{\circ} + RT\ln(\_\_)$; what replaces P?

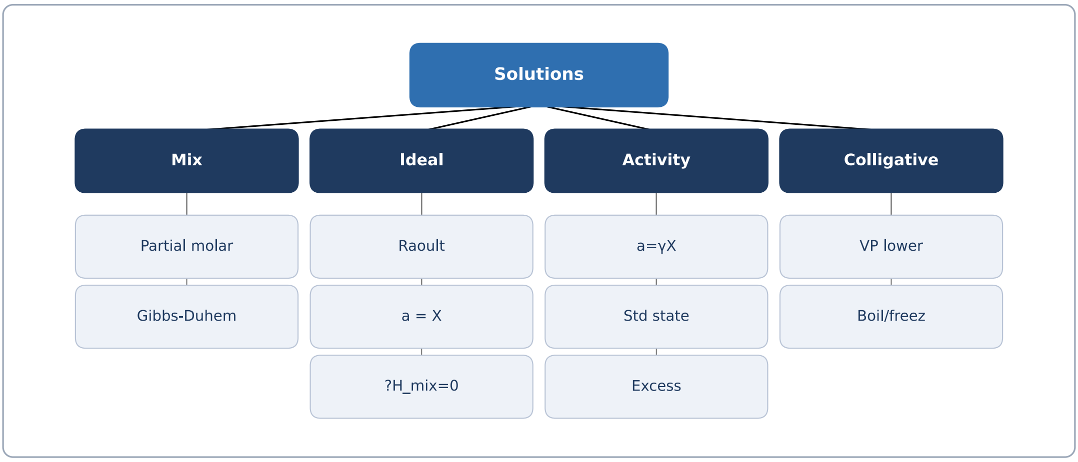
Answers: 1. Attraction dominates, so the volume is smaller than the ideal value. 2. False — b corrects for the finite molecular volume (repulsion); a corrects for attraction. 3. $T_R = \frac{T}{T_c}$, $P_R = \frac{P}{P_c}$, $V_R = \frac{V}{V_c}$. 4. 0 (i.e. $f \Rightarrow P$ as $P \to 0$). 5. f/P° — the fugacity over the standard pressure, not $\frac{P}{P^{\circ}}$.

11. Chapter Summary

- The ideal gas law is exact only at low pressure and high temperature; real gases depart via size and attraction.
- The compressibility factor $Z = \frac{PV}{nRT}$ reports the net departure: $Z < 1$ attraction, $Z > 1$ size.
- The virial expansion $Z = 1 + \frac{B}{V_m} + \ldots$ and the van der Waals EOS

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

  model it.
- van der Waals gives critical constants $V_c = 3b$, $T_c = 8\frac{a}{27Rb}$, $P_c = \frac{a}{27b^2}$, $Z_c = 0.375$.
- The law of corresponding states collapses fluids onto one curve using T_R, P_R, V_R.
- Fugacity $f = \varphi P$ keeps $\mu = \mu^{\circ} + RT\ln(f/P^{\circ})$; $\ln\varphi = \int\left(\frac{Z-1}{P}\right)dP$.



12. Flash Cards

Q What is the compressibility factor Z?

A $Z = \frac{PV}{nRT}$; $Z = 1$ for an ideal gas, $Z \neq 1$ for a real one.

Q Interpret $Z < 1$ and $Z > 1$.

A $Z < 1$: attraction dominates; $Z > 1$: repulsive molecular size dominates.

Q Write the van der Waals equation.

A $\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$, correcting for attraction (a) and volume (b).

Q What are the van der Waals critical constants?

A $V_c = 3b$, $T_c = 8\frac{a}{27Rb}$, $P_c = \frac{a}{27b^2}$, giving $Z_c = \frac{3}{8}$.

Q Define fugacity and the fugacity coefficient.

A f is effective pressure; $\varphi = f/P$, with $\varphi \Rightarrow 1$ as $P \to 0$.

Q How is φ found from compressibility data?

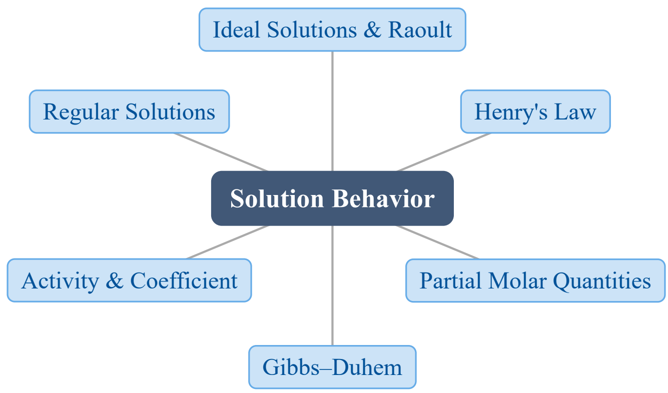
A $\ln\varphi = \int_0^P \left(\frac{Z-1}{P}\right) dP$.

13. Exam Tips

- **High priority:** computing Z and pressure from the virial or van der Waals EOS; extracting T_c, P_c, V_c, Z_c.
- Always state whether you are using molar volume V_m or total V in van der Waals — mixing them is the most common algebraic slip.
- Know the law of corresponding states: reduced variables need only T_c and P_c, no a, b.
- Fugacity is an exam favorite: remember $\mu = \mu^\circ + RT\ln(f/P^\circ)$ and $\ln\varphi = \int\left(\frac{Z-1}{P}\right)dP$.
- Do not quote $Z_c = 0.375$ as a real-gas value — it is the van der Waals prediction; real gases sit near 0.2 to 0.3.

9 The Behavior of Solutions

1. Chapter Overview



Real mixtures are rarely ideal. This chapter introduces the tools that describe nonideal solutions: Raoult's and Henry's laws for vapor pressures, partial molar quantities for mixing properties, the Gibbs–Duhem equation that links all components, and activities with regular-solution theory for deviations from ideality.

2. Learning Objectives

After studying this chapter, students should be able to:

- Define an ideal solution and apply Raoult's law to vapor–liquid equilibria.
- State Henry's law and explain when it applies to a dilute solute.
- Define a partial molar quantity and relate a total property to the sum over components.
- Use the Gibbs–Duhem equation to relate activity coefficients in a binary solution.
- Define activity and activity coefficient, and relate them to deviations from ideality.
- Apply the regular-solution model to compute activity coefficients and predict a miscibility gap.

3. Why This Matters?

Solution behavior governs materials processing and design:

- **Liquid-metal extraction** — Sieverts' law controls dissolved gases; activities set slag/metal equilibria.

- **Phase diagrams** — activity–composition curves give the common tangent that fixes phase boundaries.
- **Alloying** — activity coefficients predict segregation, deoxidation, and carburization in steels.
- **Distillation & separation** — Raoult/Henry deviations determine relative volatility.

A single activity value collapses complex intermolecular interactions into a number usable in any equilibrium calculation.

4. Intuition First

Picture a bottle of benzene and toluene. Benzene evaporates more eagerly, so a mixture's vapor is richer in benzene than the liquid. That observation **is** Raoult's law: each component pushes its own vapor proportional to how much of it is present (X_i) and how volatile it is alone (p_i°).

A **partial molar quantity** answers "what happens to the **whole** solution if I add one more mole of A, while T, P and everything else stay fixed?" It is the true marginal property of mixing — not the property of pure A.

When intermolecular forces differ, real mixtures deviate. We fix this by replacing X_i with an **activity** $a_i = \gamma_i X_i$:

$$\mu_i = \mu_i^\circ + RT \ln a_i = \mu_i^\circ + RT \ln X_i + RT \ln \gamma_i$$

The extra term $RT \ln \gamma_i$ is exactly the nonideal correction.

5. Visual Explanation

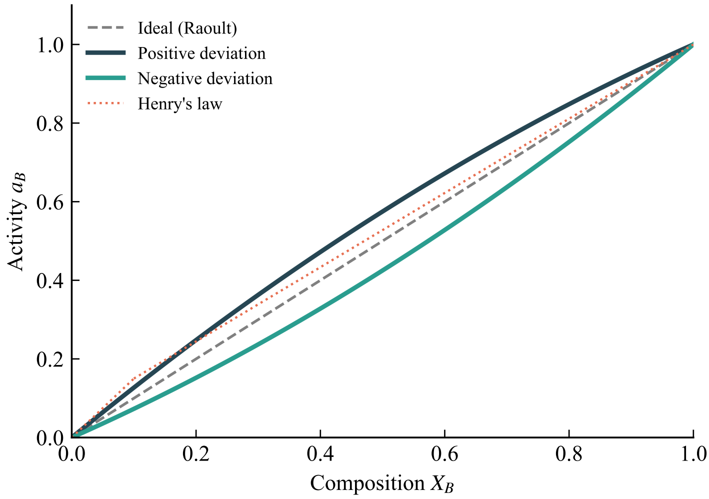


Fig. 16: Activity–composition relation: ideal ($\gamma = 1$), positive deviation ($\gamma > 1$) and negative deviation ($\gamma < 1$).

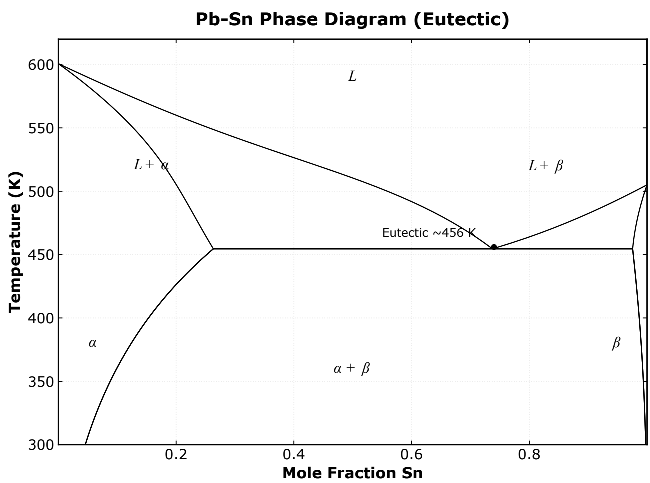


Fig. 17: A binary phase diagram — the miscibility gap and phase boundaries that activities help construct.

6. Memory Box

Regime	Law	Coefficient
Solvent, high X_i	Raoult: $p_i = X_i p_i^\circ$	p_i° (pure vapor pressure)
Solute, low X_i	Henry: $p_i = K_i X_i$	K_i (Henry constant)

Symmetry in a regular solution: $\ln\gamma_A = \alpha X_B^2$ and $\ln\gamma_B = \alpha X_A^2$. **Critical miscibility:** phase separation occurs when $\Omega > 2RT$, i.e. $T_c = \Omega/(2R)$.

7. Common Mistakes

- **Confusing Raoult's and Henry's regions.** Raoult's law describes the **solvent** (large X_i); Henry's law describes the **solute** in the infinite-dilution limit. Using p_i° for a dilute solute is wrong.
- **Assuming $\gamma_i = 1$ always.** Real solutions deviate; $\gamma_i > 1$ (repulsion, positive deviation) and $\gamma_i < 1$ (attraction, negative deviation).
- **Forgetting the Gibbs–Duhem coupling.** You cannot choose activities of two components independently; knowing $a_{A(X)}$ fixes $a_{B(X)}$.
- **Treating $\Omega/(RT)$ blindly.** The miscibility criterion is $\Omega/(RT) > 2$; at high T even a positive Ω gives a single phase.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$p_i = X_i p_i^\circ$	Raoult's law (ideal solvent)	X_i, p_i°	—	Ideal or solvent
$p_i = K_i X_i$	Henry's law (dilute solute)	X_i, K_i	—	Infinite dilution
$\bar{Y}_i = (\partial Y/\partial n_i)_{T,P,n_{j\neq i}}$	Partial molar quantity	Y, n_i	units of Y/mol	Const T, P
$\mu_i = \mu_i^\circ + RT\ln a_i$	Chemical potential w/ activity	R, T, a_i	J/mol	Any solution
$a_i = \gamma_i X_i$	Activity–coefficient def.	γ_i, X_i	—	—
$\sum_i X_i d\mu_i = 0$	Gibbs–Duhem (const T, P)	X_i, μ_i	—	Const T, P
$\ln\gamma_A = (\Omega/(RT))X_B^2$	Regular solution coeff.	Ω, R, T	—	$\Omega > 0$ pos. dev.
$T_c = \Omega/(2R)$	Critical miscibility temp.	Ω, R	K	$\Delta S_{\text{mix}} = \Delta S^{\text{id}}$

9. Worked Examples

Example 1 — Raoult's law and vapor–liquid equilibrium

Problem Benzene (A) and toluene (B) form an ideal solution. At 100°C, $p_A^\circ = 1350$ mmHg and $p_B^\circ = 556$ mmHg. Find the total vapor pressure and the vapor composition when $X_A = 0.4$.

Thinking Process Apply Raoult's law to each component for its partial pressure, sum for the total, then use Dalton's law to convert partial pressures into vapor mole fractions $y_i = p_i/P_{\text{total}}$.

Solution

$$p_A = X_A p_A^\circ = 0.4 \times 1350 = 540 \text{ mmHg}$$

$$p_B = X_B p_B^\circ = 0.6 \times 556 = 334 \text{ mmHg}$$

$$P_{\text{total}} = p_A + p_B = 874 \text{ mmHg}$$

$$y_A = p_A/P_{\text{total}} = 540/874 = 0.618, \quad y_B = 334/874 = 0.382$$

Answer $P_{\text{total}} = 874$ mmHg, $y_A = 0.618$, $y_B = 0.382$.

Interpretation The vapor is richer in benzene (0.618 > 0.4) because benzene is more volatile — the physical basis of fractional distillation.

Example 2 — Regular-solution coefficients and miscibility

Problem A binary regular solution has $\Omega = 12000$ J mol^{-1}. Find γ_A at $X_A = 0.3$ and 800 K, and decide whether the solution separates.

Thinking Process Use $\ln \gamma_A = (\Omega/(RT))X_B^2$ and $\ln \gamma_B = (\Omega/(RT))X_A^2$. Compare $\Omega/(RT)$ with the critical value 2 to test for a miscibility gap.

Solution

$$\Omega/(RT) = 12000/(8.314 \times 800) = 1.804$$

$$\ln \gamma_A = 1.804 \times 0.7^2 = 0.884 \Rightarrow \gamma_A = \exp(0.884) = 2.42$$

$$\ln \gamma_B = 1.804 \times 0.3^2 = 0.162 \Rightarrow \gamma_B = \exp(0.162) = 1.18$$

$$T_c = \Omega/(2R) = 12000/(2 \times 8.314) = 722 \text{ K}$$

Answer $\gamma_A = 2.42$, $\gamma_B = 1.18$; at 800 K there is no separation ($800 > 722$ K).

Interpretation $\gamma_A > 1$ signals a positive deviation. Since 800 K $> T_c = 722$ K, $\Omega/(RT) < 2$ and the liquid stays as one phase. Cooling below 722 K would open a miscibility gap.

Example 3 — Gibbs–Duhem integral relation

Problem Measurements give $\ln \gamma_A = \alpha X_B^2$ (α constant) for a binary A–B solution. Derive $\ln \gamma_B$.

Thinking Process Start from the binary Gibbs–Duhem at constant T, P: $X_A d\mu_A + X_B d\mu_B = 0$. Substitute $\mu_i = \mu_i^\circ + RT(\ln \gamma_i + \ln X_i)$ and use $dX_A + dX_B = 0$.

Solution

$$X_A d(\ln \gamma_A) + X_B d(\ln \gamma_B) = 0 \Rightarrow d(\ln \gamma_B) = -(X_A/X_B) d(\ln \gamma_A)$$

With $\ln \gamma_A = \alpha X_B^2$, $d(\ln \gamma_A) = 2\alpha X_B dX_B = -2\alpha X_B dX_A$:

$$d(\ln \gamma_B) = -(X_A/X_B)(-2\alpha X_B dX_A) = 2\alpha X_A dX_A$$

Integrating and using $\gamma_B \to 1$ as $X_B \to 1$ gives $\ln \gamma_B = \alpha X_A^2$.

Answer $\ln \gamma_B = \alpha X_A^2$, symmetric with $\ln \gamma_A = \alpha X_B^2$.

Interpretation The Gibbs–Duhem equation enforces the well-known symmetry of regular-solution activity coefficients — specifying one component fully determines the other.

10. Quick Check

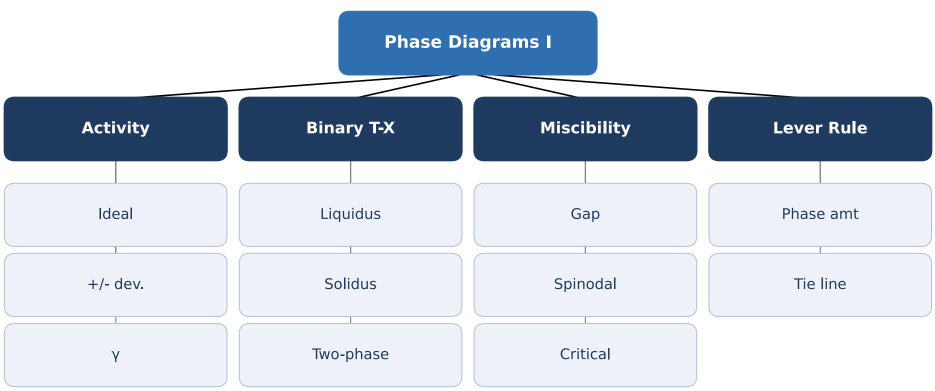
1. State Raoult's law and name the component it describes.
2. For $\gamma_i > 1$, is the deviation positive or negative?
3. Write the binary Gibbs–Duhem equation at constant T and P.

4. What condition on $\Omega/(RT)$ signals a miscibility gap?
5. Define a partial molar quantity Y_i.

Answers: 1. $p_i = X_i p_i^\circ$; describes the **solvent** (high mole fraction). 2. Positive deviation (repulsion, $a_i > X_i$). 3. $X_A d\mu_A + X_B d\mu_B = 0$. 4. Phase separation when $\Omega/(RT) > 2$, i.e. $T < T_c = \Omega/(2R)$. 5. $Y_i = (\partial Y/\partial n_i)_{T,P,n_{j\neq i}}$, the change in Y on adding 1 mol of i.

11. Chapter Summary

- Ideal solutions obey $\Delta V = 0$, $\Delta H = 0$; Raoult's law $p_i = X_i p_i^\circ$ applies to the solvent.
- Henry's law $p_i = K_i X_i$ describes a dilute solute in the infinite-dilution limit.
- A partial molar quantity $Y_i = (\partial Y/\partial n_i)$ is the marginal contribution of component i.
- The Gibbs–Duhem equation couples components: $X_A d\mu_A + X_B d\mu_B = 0$ at constant T, P.
- Activity $a_i = \gamma_i X_i$ and $\mu_i = \mu_i^\circ + RT \ln a_i$ handle nonideality; $\gamma_i \neq 1$ measures the deviation.
- Regular solutions give $\ln \gamma_A = (\Omega/(RT)) X_B^2$ and a miscibility gap when $\Omega > 2RT$.



12. Flash Cards

Q What is an ideal solution?

A $\Delta V = 0$ and $\Delta H = 0$ on mixing; Raoult's law holds for every component.

Q State Raoult's law.

A $p_i = X_i p_i^\circ$ — partial pressure of component i in an ideal solution.

Q What is Henry's law?

A $p_i = K_i X_i$ for a dilute solute; K_i is the Henry constant, $\neq p_i^\circ$.

Q Define a partial molar quantity.

A $\bar{Y}_i = (\partial Y/\partial n_i)_{T,P,n_{j\neq i}}$ — effect of adding 1 mol of i.

Q Define activity and activity coefficient.

A $a_i = \gamma_i X_i$, $\mu_i = \mu_i^\circ + RT \ln a_i$; γ_i measures nonideality.

Q When does a regular solution phase-separate?

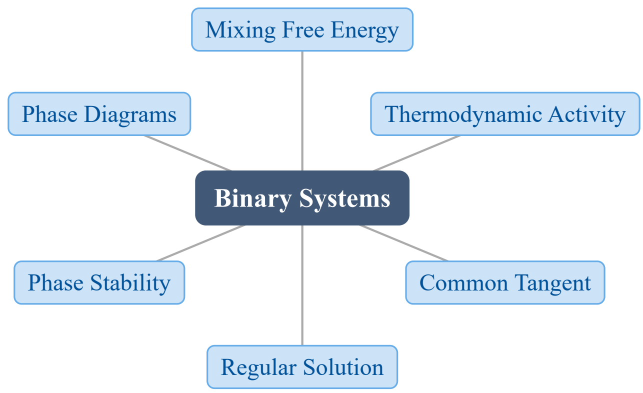
A When $\Omega > 2RT$ (below $T_c = \Omega/(2R)$) a miscibility gap opens.

13. Exam Tips

- **High priority:** Raoult vs. Henry regions, and converting X_i to vapor composition via Dalton's law.
- Always state the standard state (Raoult for solvent, Henry for solute) before writing a_i.
- Use the Gibbs–Duhem equation to derive one activity coefficient from the other — a classic exam derivation.
- For regular solutions, check $\Omega/(RT)$ against 2 to decide single phase vs. miscibility gap.

10 Gibbs Free Energy, Composition and Phase Diagrams of Binary Systems

1. Chapter Overview



A binary phase diagram is the central tool of materials thermodynamics. This chapter links Gibbs free energy, composition, and the phase diagram: by plotting free-energy-composition curves and locating the common tangents, the entire temperature-composition phase diagram can be constructed. The free-energy curve, the activity-composition curve, and the phase diagram are three faces of the same thermodynamic information.

2. Learning Objectives

After studying this chapter, students should be able to:

- Relate the Gibbs free energy of mixing to the activities: $\Delta G_M = RT \sum X_i \ln a_i$.
- Use tangent intercepts on the ΔG_M curve to read off the activity of each component.
- Apply the common-tangent condition to find coexisting-phase compositions and the liquidus/solidus.
- Use the regular-solution model: $\Delta G_M = \Omega X_A X_B + RT(X_A \ln X_A + X_B \ln X_B)$.
- State the stability criterion $\Omega > 2RT$ for a miscibility gap and locate the spinodal.
- Apply the lever rule to determine phase fractions in a two-phase region.

3. Why This Matters?

Binary phase diagrams decide **what you can make** in metallurgy, ceramics, and semiconductors:

- **Alloy design** — a T–X diagram tells you which single-phase or two-phase field a given composition lands in, and how to heat-treat it.
- **Solidification** — the liquidus and solidus set the freezing path and microstructure (dendrites, eutectics, peritectics).
- **Extractive & processing metallurgy** — activity drives slag/metal equilibria and refining.
- **Microstructure control** — miscibility gaps and spinodal decomposition produce precipitates and modulated structures.

Understanding the free-energy basis turns a diagram you **memorize** into one you can **derive**.

4. Intuition First

Imagine two curves, one for the liquid and one for the solid, each plotting G versus composition X_B. At a fixed T the system minimizes its total G, so equilibrium picks the **lowest** envelope of the two curves.

Picture A stretched string laid across the two curves touches each at a **common tangent**; the touching points are the coexisting liquid and solid compositions.

Concept Phase equilibrium requires the chemical potential (the partial molar G) of every component to be equal in every phase:

$$G_A^L = G_A^S \quad \text{and} \quad G_B^L = G_B^S$$

which is exactly the common-tangent condition.

Math For a regular solution the mixing free energy is a competition between an ideal entropy term (favoring mixing) and an enthalpy term $\Omega X_A X_B$ (favoring separation):

$$\Delta G_M = \Omega X_A X_B + RT(X_A \ln X_A + X_B \ln X_B)$$

When the Ω term dominates ($\Omega > 2RT$), the curve sags into a double well and the alloy splits into A-rich and B-rich phases.

5. Visual Explanation

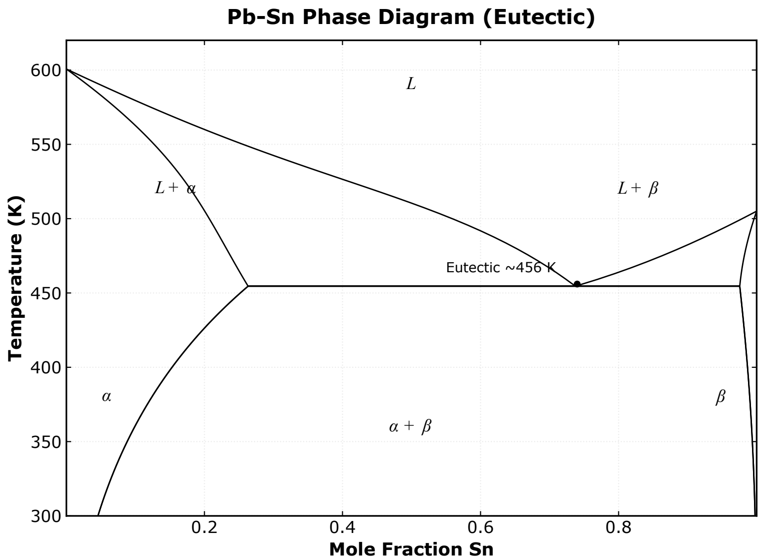


Fig. 18: A binary T–X phase diagram: liquidus, solidus, and the two-phase region between them.

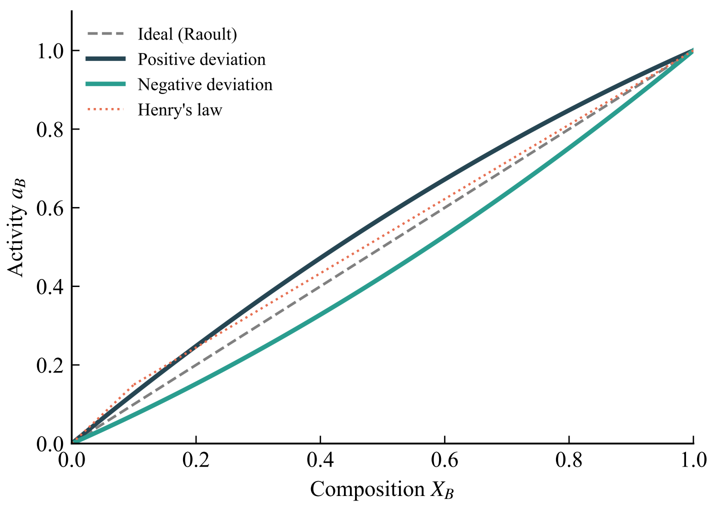


Fig. 19: Activity versus composition for ideal and non-ideal (positive/negative deviation) solutions; the activity curve encodes the same information as the free-energy curve.

6. Memory Box

Three equivalent representations of one binary system:

View	What you read	Key relation
Free-energy curve	Phase stability, common tangent	$dG = 0$, G_i equal
Activity curve	Non-ideality, γ_i	$RT \ln a_i =$ intercept
T–X diagram	Phase fields, liquidus/solidus	Common tangent at each T

Regular-solution checklist: $\Delta H_M = \Omega X_A X_B$, $\Delta S_M = \Delta S_{M,\text{id}} = -R(X_A \ln X_A + X_B \ln X_B)$; critical (spinodal) temperature $T_c = \Omega/(2R)$.

7. Common Mistakes

- **Forgetting the chemical potential is what is equalized, not the free energy.** At equilibrium G_A and G_B are equal across phases; the **total** G is merely minimized, not equal phase to phase.
- **Applying the lever rule outside the two-phase region.** If X_B lies outside the tie-line limits, the system is a single phase: f_L and f_S would fall outside $[0, 1]$.
- **Mislabeling T_c.** $T_c = \Omega/(2R)$ is the **spinodal / miscibility-gap** temperature, not a melting point. The miscibility gap opens at $T < T_c$.
- **Assuming ideal mixing.** Real solutions have $a_i = \gamma_i X_i$; using $a_i = X_i$ ignores the enthalpy term and predicts no miscibility gap.
- **Confusing Raoult's and Henry's limits.** As $X_i \to 1$, $a_i \to X_i$ (Raoult); as $X_i \to 0$, $a_i \to \gamma_i^\circ X_i$ (Henry).

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$\Delta G_M = RT(X_A \ln a_A + X_B \ln a_B)$	Mixing free energy from activities	X_i, a_i, T	J/mol	Any solution
$\Delta G_{M,\text{id}} = RT(X_A \ln X_A + X_B \ln X_B)$	Ideal mixing free energy	X_i, T	J/mol	$\Delta H_M = 0$
$\Delta G_{m(i)} = \Delta H_{m(i)}\left(1 - \frac{T}{T_{m(i)}}\right)$	Free energy of fusion	$T_{m(i)}, T$	J/mol	Pure component i
$\Delta G_M = \Omega X_A X_B + RT(X_A \ln X_A + X_B \ln X_B)$	Regular-solution mixing	Ω, X_i, T	J/mol	Regular solution
$\frac{\partial^2 \Delta G_M}{\partial X_B^2} = 0$	Spinodal boundary	Ω, T	—	$\Omega = 2RT$ at T_c

Formula	Meaning	Variables	Units	Conditions / Limits
$\ln \gamma_A = \frac{\Omega}{RT} X_B^2$	Regular-solution activity coeff	Ω, X_B, T	—	γ_B swaps X_A, X_B
$f_L = \frac{X_B - X_B^S}{X_B^L - X_B^S}$	Lever rule (liquid fraction)	X_B, X_B^{ph}	—	In two-phase region

9. Worked Examples

Example 1 — Lever rule in a two-phase region

Problem A binary A–B system is at liquid–solid equilibrium, 500 K. Liquid is $X_B^L = 0.35$, solid is $X_B^S = 0.65$. Find the phase fractions for an overall $X_B = 0.50$.

Thinking Process The lever rule places the overall composition on a tie line between the two phase compositions; the fraction of each phase is the opposite segment length divided by the total length.

Solution

$$f_L = \frac{X_B - X_B^S}{X_B^L - X_B^S} = \frac{0.50 - 0.65}{0.35 - 0.65} = \frac{-0.15}{-0.30} = 0.50$$

$$f_S = 1 - f_L = 0.50$$

Answer $f_L = 0.50$, $f_S = 0.50$ — equal liquid and solid amounts.

Interpretation The lever rule quantifies microstructure: a 50/50 two-phase mixture at this overall composition. If instead $X_B = 0.20$ (inside the liquid field), $f_L = 1$ and the system is a single homogeneous liquid.

Example 2 — Miscibility gap of a regular solution

Problem A regular solution has $\Omega = 15000 \text{ J/mol}$. Does it phase-separate at $T = 900 \text{ K}$? If so, find the coexisting compositions.

Thinking Process Phase separation occurs when $\Omega > 2RT$, i.e. below the critical temperature $T_c = \Omega/(2R)$. The coexisting compositions sit at the two minima of ΔG_M, found from $d(\Delta G_M)/dX_B = 0$.

Solution

$$T_c = \frac{\Omega}{2R} = \frac{15000}{(2)(8.314)} \approx 902 \quad \text{K}$$

Since $900 < 902 \quad \text{K}$, the alloy is in the two-phase region. Set the derivative to zero:

$$\Omega(1 - 2X_B) + RT \ln\left(\frac{X_B}{1 - X_B}\right) = 0$$

$$15000(1 - 2X_B) + (8.314)(900) \ln\left(\frac{X_B}{1 - X_B}\right) = 0$$

Numerical solution gives $X_{B'} \approx 0.19$ and $X_{B''} \approx 0.81$.

Answer Phase separation occurs; coexisting phases are A-rich $X_B \approx 0.19$ and B-rich $X_B \approx 0.81$.

Interpretation Just below T_c the gap is narrow; the two minima are the compositions connected by the common tangent. This is the thermodynamic origin of precipitates and spinodal microstructures.

Example 3 — Activity and stability from Ω/RT

Problem A regular solution has $\Omega = 10000 \quad \text{J/mol}$. At $T = 600 \quad \text{K}$, sketch $\ln a_A$ vs. X_A and decide whether the solution is stable.

Thinking Process Compute $\Omega/(RT)$ and compare with 2. The regular-solution activity is $\ln a_A = \ln X_A + (\Omega/RT)X_B^2$, so $\ln a_A$ is S-shaped whenever $\Omega/(RT) > 2$, signaling phase separation.

Solution

$$\frac{\Omega}{RT} = \frac{10000}{(8.314)(600)} = 2.004 > 2$$

Activity coefficients:

$$\ln \gamma_A = 2.004X_B^2, \quad \ln \gamma_B = 2.004X_A^2$$

$$a_A = X_A \exp(2.004X_B^2), \quad a_B = X_B \exp(2.004X_A^2)$$

Answer $\Omega/RT = 2.004 > 2$, so the solution is **unstable** and separates.

Interpretation An S-shaped $\ln a_A$ vs. X_A is the activity-curve signature of a miscibility gap. The common tangent joins the two minima — the same two compositions revealed by the free-energy view.

10. Quick Check

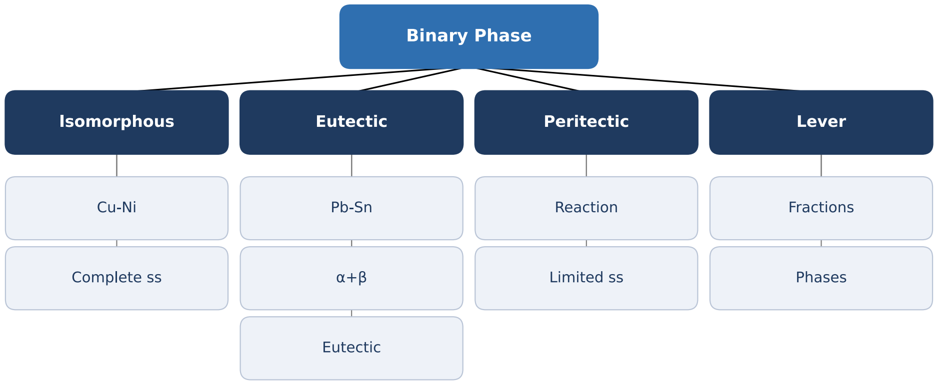
1. What condition must hold for phase equilibrium between liquid and solid in a binary system?
2. For an ideal solution, how is activity a_i related to mole fraction X_i?
3. State the regular-solution criterion for a miscibility gap to appear.
4. Write the lever rule for the liquid fraction f_L in terms of X_B, X_B^L, X_B^S.
5. In the dilute limit $X_i \to 0$, which law governs a_i: Raoult's or Henry's?

Answers: 1. Equal chemical potentials: $G_A^L = G_A^S$ and $G_B^L = G_B^S$ (common tangent). 2. $a_i = X_i$ (Raoult's law, ideal). 3. $\Omega > 2RT$, i.e. below $T_c = \Omega/(2R)$. 4. $f_L = (X_B - X_B^S)/(X_B^L - X_B^S)$. 5. Henry's law: $a_i \to \gamma_i^\circ X_i$ as $X_i \to 0$.

11. Chapter Summary

- Binary phase diagrams are derived from free-energy-composition curves via the common-tangent construction.
- Mixing free energy and activity: $\Delta G_M = RT(X_A \ln a_A + X_B \ln a_B)$; tangent intercepts give $RT \ln a_i$.
- Free energy of fusion: $\Delta G_{m(i)} = \Delta H_{m(i)}\left(1 - \frac{T}{T_{m(i)}}\right)$ shifts liquid vs. solid curves.
- Regular solution: $\Delta H_M = \Omega X_A X_B$, $\Delta S_M = \Delta S_{M,\text{id}}$; $\Delta G_M = \Omega X_A X_B + RT \sum X_i \ln X_i$.
- Phase stability: miscibility gap when $\Omega > 2RT$; spinodal at $\Omega = 2RT$.

- Activity limits: Raoult ($X_i \to 1$) and Henry ($X_i \to 0$); the lever rule gives phase fractions.



12. Flash Cards

Q What links ΔG_M to activities?

A $\Delta G_M = RT(X_A \ln a_A + X_B \ln a_B)$; ideal gives $a_i = X_i$.

Q What is the common-tangent condition?

A Equal chemical potentials of each component in both phases: $G_A^L = G_A^S$, $G_B^L = G_B^S$.

Q What is a regular solution?

A $\Delta H_M = \Omega X_A X_B$ with ideal entropy of mixing; $\Delta G_M = \Omega X_A X_B + RT \sum X_i \ln X_i$.

Q Criterion for a miscibility gap?

A $\Omega > 2RT$; the spinodal (critical) temperature is $T_c = \Omega/(2R)$.

Q State the lever rule.

A $f_L = (X_B - X_B^S)/(X_B^L - X_B^S)$; gives phase fractions in a two-phase field.

Q Raoult vs. Henry in the dilute limit?

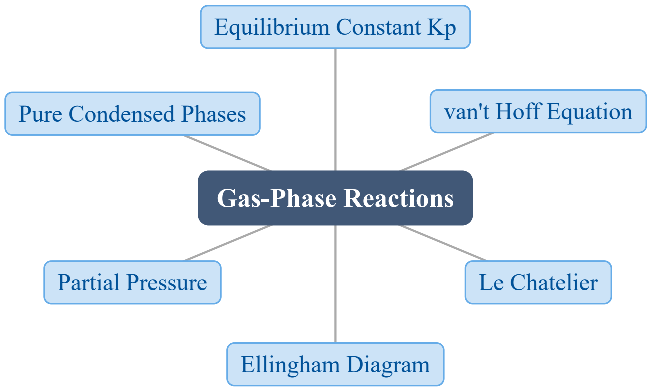
A $X_i \to 1$: $a_i \to X_i$ (Raoult); $X_i \to 0$: $a_i \to \gamma_i^\circ X_i$ (Henry).

13. Exam Tips

- **High priority:** the common-tangent construction and the lever rule — they appear on every phase-diagram problem. Draw the G–X curves, then the tangent.
- Memorize $T_c = \Omega/(2R)$ and the regular-solution ΔG_M; most stability questions are one substitution away from it.
- Always check $\Omega/(RT)$ against 2 before deciding "stable mixture" vs. "miscibility gap".
- Know the reaction vocabulary: eutectic $L \to \alpha + \beta$, eutectoid $\gamma \to \alpha + \mathrm{Fe_3\ C}$, peritectic $\alpha + L \to \beta$ — and where each sits on the diagram.

11 Reactions Involving Gases

1. Chapter Overview



A gas-phase reaction at constant T, P settles at the composition that minimizes the total Gibbs energy. The outcome is captured by a single number, K_p, which follows directly from $\Delta G°$ and shifts with temperature (van't Hoff), total pressure, and the presence of condensed phases.

2. Learning Objectives

After studying this chapter, students should be able to:

- Write $\Delta G° = -RT \ln \mathrm{K}_p$ and express K_p in partial pressures for a general gas reaction.
- Apply the van't Hoff equation to find how K_p changes with temperature from $\Delta H°$.
- Predict the pressure dependence of the equilibrium composition using Le Chatelier's principle when $\Delta n_g \neq 0$.
- Use an Ellingham diagram to judge the stability of oxides and the direction of oxidation-reduction.
- Set up equilibrium for a reaction between a pure condensed phase and a gas (e.g. $M + \frac{1}{2}O_2 = \mathrm{MO}$) using oxygen partial pressure as the control variable.

3. Why This Matters?

Gas-phase equilibrium is the backbone of industrial materials processing:

- **Steelmaking & extraction** ??oxidation of C, Si, P and reduction of iron oxides by $CO\ /H_2$ are all gas equilibria.
- **Sintering, annealing atmospheres** ??the $H_2/H_2\ O$ and $CO\ /CO_2$ buffers set the redox potential of the furnace.
- **Ellingham diagrams** ??$\Delta G°$ vs. T decide which metal oxide is stable, guiding carbothermic reduction and refining.
- **Catalysis & combustion** ??equilibrium yield (e.g. NH_3, SO_3) is bounded by K_p at the operating T, P.

Mastering K_p lets you predict conversion before building a reactor.

4. Intuition First

Example: burn methane in air ??at high temperature the products are mostly CO_2 and $H_2\ O$, but close the throttle and the balance shifts.

Picture: imagine a hill of total Gibbs energy G' as a function of the reaction extent ξ; equilibrium sits at the bottom of the valley. Change T or P and the valley moves.

Concept: each gas species contributes its molar Gibbs energy $G_i = G_i^\circ + RT \ln p_i$. Minimizing the sum $\sum_i n_i G_i$ with respect to ξ forces the "reactant side" and "product side" chemical potentials into balance. The constants G_i° cancel into one number ?? $\Delta G°$??and the pressure terms collect into K_p.

Math: for $A + B = 2C$,

$$G_A + G_B = 2G_C \quad \Rightarrow \quad \Delta G^\circ = -RT \ln\bigl(p_C^2/(p_A p_B)\bigr) = -RT \ln \mathrm{K}_p$$

raising T for an endothermic reaction deepens the product side of the valley, so K_p grows.

5. Visual Explanation

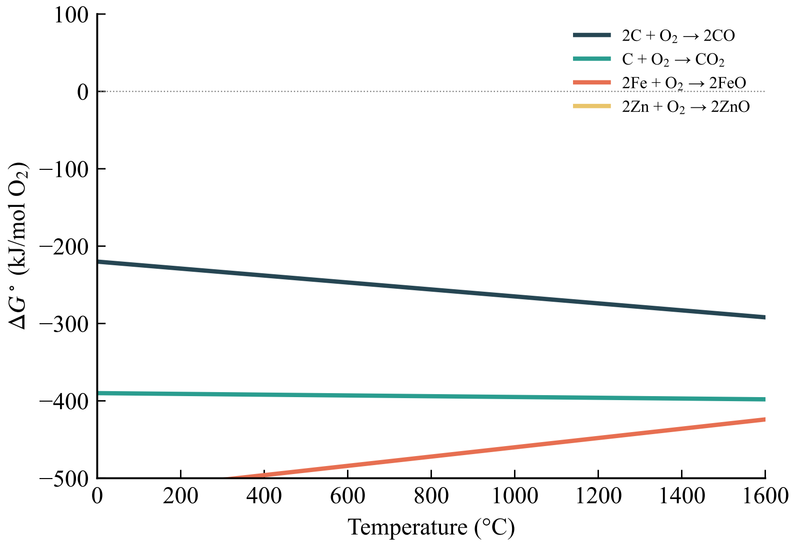


Fig. 20: An Ellingham diagram: $\Delta G°$ of oxide formation vs. T. Lines lower on the plot denote more stable oxides; the slope is $-\Delta S°$ and crossing points reveal reduction temperatures.

6. Memory Box

Key relations at a glance:

Quantity	Expression
Equilibrium constant	$\Delta G° = -RT \ln \mathrm{K}_p$
van't Hoff (integrated)	$\ln(K_2/K_1) = -\frac{\Delta H°}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$
Temperature slope	$\frac{d \ln \mathrm{K}_p}{dT} = \Delta H°/(RT^2)$
Gas activity	$a_i = p_i/p^\circ$ (take $p^\circ = 1$ atm)
Pure condensed phase	$a = 1$ (independent of P near 1 atm)
$M + \frac{1}{2}O_2 = \mathrm{MO}$	$\mathrm{K} = 1/p_{O_2}^{\frac{1}{2}}$

Sign rule: $\Delta H° > 0$ (endothermic) $\Rightarrow$ higher T raises K_p; $\Delta H° < 0$ (exothermic) $\Rightarrow$ higher T lowers K_p.

7. Common Mistakes

- **Forgetting p°.** K_p uses dimensionless activities p_i/p°; with $p^\circ = 1$ atm the numbers coincide, but K_p is not "atm to some power" ??it is dimensionless.
- **Confusing K_p with composition.** K_p depends only on T (and the chosen standard pressure); the **composition** shifts with total P when $\Delta n_g \neq 0$.
- **Wrong Le Chatelier sign.** If $\Delta n_g > 0$ (more gas moles on the product side), raising P shifts equilibrium toward reactants; if $\Delta n_g < 0$ the opposite.
- **Dropping the condensed phase.** Its activity is 1, but its presence still fixes the exponent on the gas partial pressure ??for $\frac{1}{2}O_2$ it is $p_{O_2}^{\frac{1}{2}}$, not p_{O_2}.
- **Treating ΔH° as constant over a wide range.** This is valid only when $\Delta C_P \approx 0$; otherwise the van't Hoff plot of $\ln K_p$ vs. $\frac{1}{T}$ curves.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$\Delta G^\circ = -RT \ln K_p$	Equilibrium constant from standard free energy	$\Delta G^\circ, R, T, K_p$	J/mol	Any T
$K_p = \prod_i p_i^{\nu_i}$	Mass-action expression in partial pressures	p_i, ν_i	?�	Gas-phase rxn
$\frac{d \ln K_p}{dT} = \frac{\Delta H^\circ}{RT^2}$	van't Hoff differential form	$K_p, \Delta H^\circ, R, T$	?�	Const ΔH°
$\ln\left(\frac{K_2}{K_1}\right) = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$	van't Hoff integrated form	$K_1, K_2, \Delta H^\circ, T$	?�	Const ΔH°
$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$	Free energy split into enthalpy and entropy	$\Delta G^\circ, \Delta H^\circ, \Delta S^\circ, T$	J/mol	Const ΔC_P

9. Worked Examples

Example 1 ??Equilibrium constant from ΔG°

Problem For $SO_2 + \frac{1}{2}O_2 = SO_3$, $\Delta G^\circ = -30.0$ kJ/mol at 800 K. Find K_p and the conversion from an initial $SO_2 : O_2 = 2 : 1$ mixture at 1 atm.

Thinking Process Convert ΔG° to J/mol, use $\ln K_p = -\Delta G^\circ/(RT)$, then set up an extent-of-reaction balance.

Solution

$$\ln K_p = -\frac{\Delta G^\circ}{RT} = -\frac{-30000}{(8.314)(800)} = 4.510$$

$$K_p = \exp(4.510) = 90.9$$

Start $n_{SO_2} = 2$, $n_{O_2} = 1$, $n_{SO_3} = 0$; after extent ξ:

$$n_{SO_2} = 2 - \xi, \quad n_{O_2} = 1 - \frac{\xi}{2}, \quad n_{SO_3} = \xi, \quad n_{tot} = 3 - \frac{\xi}{2}$$

With $P = 1\ \ \text{atm}$, $K_p = \dfrac{\frac{\xi}{3-\frac{\xi}{2}}}{\left(\frac{2-\xi}{3-\frac{\xi}{2}}\right)\left(\frac{1-\frac{\xi}{2}}{3-\frac{\xi}{2}}\right)^{\frac{1}{2}}}$. Iterating gives $\xi \approx 1.76$.

Answer $K_p = 90.9$; equilibrium moles $n_{SO_2} = 0.24$, $n_{O_2} = 0.12$, $n_{SO_3} = 1.76$, i.e. about 88% conversion.

Interpretation A strongly negative ΔG° gives a large K_p and near-complete conversion when the gas-mole count drops on the product side.

Example 2 ??van't Hoff temperature shift

Problem For $N_2 + 3H_2 = 2NH_3$, $K_p = 1.0 \times 10^4$ at 400 K and $\Delta H^\circ = -46.2\ \ \text{kJ/mol}$. Estimate K_p at 500 K, assuming ΔH° constant.

Thinking Process Use the integrated van't Hoff equation with $T_1 = 400$ K, $T_2 = 500$ K and the exothermic ΔH°.

Solution

$$\ln\left(\frac{K_2}{K_1}\right) = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$= -\frac{(-46200)}{8.314}\left(\frac{1}{500} - \frac{1}{400}\right) = 5556(-0.0005) = -2.778$$

$$K_2 = (1.0 \times 10^4)\exp(-2.778) = 6.22 \times 10^2$$

Answer $K_{p(500\ \ K)} = 6.22 \times 10^2$, down from 1.0×10^4 at 400 K.

Interpretation Because the synthesis is exothermic, raising T lowers K_p ??the classic reason the Haber process runs at a compromise temperature.

Example 3 ??Oxygen partial pressure for oxide formation

Problem For 2 Ni + O_2 = 2 NiO, $\Delta G° = -300$ kJ/mol at 1000 K. Find the equilibrium oxygen partial pressure.

Thinking Process Pure solids have unit activity, so $K = 1/p_{O_2}$; relate it to $\Delta G°$ via K_p.

Solution

$$K = \frac{a_{\mathrm{NiO}}^2}{a_{\mathrm{Ni}}^2 p_{O_2}} = \frac{1}{p_{O_2}}$$

$$\Delta G° = -RT \ln K = RT \ln p_{O_2}$$

$$\ln p_{O_2} = \frac{\Delta G°}{RT} = \frac{-300000}{(8.314)(1000)} = -36.08$$

$$p_{O_2} = \exp(-36.08) = 2.15 \times 10^{-16} \ \ \text{atm}$$

Answer $p_{O_2,=} = 2.15 \times 10^{-16}$ atm.

Interpretation An extraordinarily low equilibrium p_{O_2} means Ni oxidizes readily at 1000 K; the gas must be purged below 10^{-16} atm to keep Ni metallic.

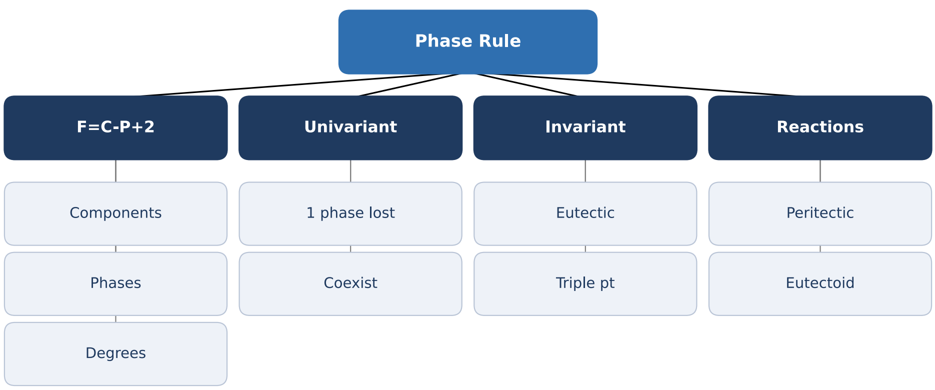
10. Quick Check

1. Write the relation between $\Delta G°$ and K_p.
2. For an exothermic reaction, does K_p increase or decrease as T rises?
3. A gas reaction has $\Delta n_g = +1$. What happens to product yield when total pressure is increased?
4. For $M(s) + \frac{1}{2}O_2(g) = \mathrm{MO}(s)$, what is K in terms of oxygen partial pressure?
5. True or False: the activity of a pure condensed phase is 1 and is essentially independent of pressure near 1 atm.

Answers: 1. $\Delta G^\circ = -RT \ln \mathrm{K}_p$ 2. Decreases (van't Hoff, since $\Delta H^\circ < 0$) 3. Yield drops ?? equilibrium shifts toward the fewer-mole (reactant) side (Le Chatelier) 4. $\mathrm{K} = 1/p_{O_2}^{\frac{1}{2}}$ 5. True

11. Chapter Summary

- Gas-phase equilibrium: $\Delta G^\circ = -RT \ln \mathrm{K}_p$ with $\mathrm{K}_p = \prod_i p_i^{\nu_i}$.
- van't Hoff: $\ln\left(\frac{K_2}{K_1}\right) = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$; endothermic raises K_p with T.
- K_p is a function of T only; total pressure changes composition when $\Delta n_g \neq 0$ (Le Chatelier).
- $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ explains equilibrium as an enthalpy-entropy compromise.
- Ellingham diagrams compare oxide stabilities via ΔG° vs. T.
- Pure condensed phase + gas: activity 1, so for $M + \frac{1}{2}O_2 = \mathrm{MO}$ equilibrium is fixed by p_{O_2}.



12. Flash Cards

Q What is K_p?

A $\mathrm{K}_p = \prod_i p_i^{\nu_i}$; linked to standard free energy by $\Delta G^\circ = -RT \ln \mathrm{K}_p$.

Q State the van't Hoff equation.

A $\frac{d \ln \mathrm{K}_p}{dT} = \Delta H^\circ/(RT^2)$; $\ln\left(\frac{K_2}{K_1}\right) = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$.

Q How does pressure affect K_p?

A K_p depends only on T; total pressure changes the **composition** only when $\Delta n_g \neq 0$.

Q What is an Ellingham diagram?

A A plot of $\Delta G°$ of oxide formation vs. T; lower lines mean more stable oxides.

Q Activity of a pure gas at pressure p?

A $a_i = p_i/p^{\circ}$, usually taking $p^{\circ} = 1\ \text{atm}$.

Q Equilibrium for $M + \frac{1}{2}O_2 = \text{MO}$?

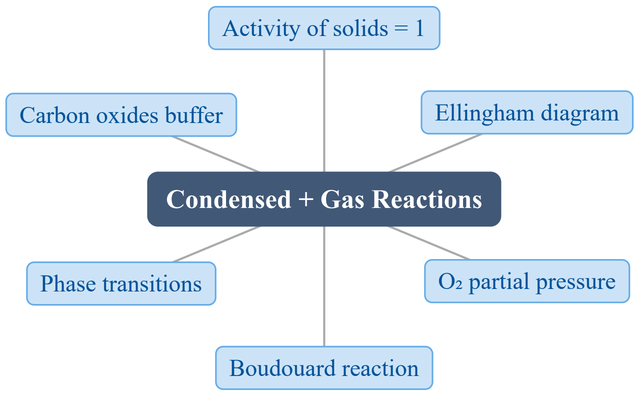
A Pure solids have $a = 1$, so $\text{K} = 1/p_{O_2}^{\frac{1}{2}}$; p_{O_2} is the control variable.

13. Exam Tips

- **High priority:** computing K_p from $\Delta G°$ and solving an extent-of-reaction balance for the equilibrium composition.
- Memorize the van't Hoff integrated form and the sign rule: exothermic $\Rightarrow$ K_p falls with T.
- Always state p° as 1 atm so K_p is dimensionless; never report it with pressure units.
- When a problem mentions a pure solid or liquid, set its activity to 1 immediately and watch the gas exponent it leaves behind.
- For $\Delta n_g \neq 0$, link Le Chatelier's principle to the pressure effect on composition, not to K_p.

12 Reactions Involving Pure Condensed Phases and a Gaseous Phase

1. Chapter Overview



When a reaction involves only pure solid (or liquid) phases plus a gas, the condensed phases have activity $a = 1$. The equilibrium is then fixed entirely by the gas-phase partial pressure (usually p_{O_2}), collapsing a multicomponent problem to a single variable. The Ellingham diagram plots $\Delta G°$ against T for such oxidations and is the central tool of extractive metallurgy.

2. Learning Objectives

After studying this chapter, students should be able to:

- State why the activity of a pure condensed phase is 1 and how this simplifies the equilibrium constant.
- Relate $\Delta G°$ to the equilibrium oxygen partial pressure p_{O_2} for a metal-oxide reaction.
- Read an Ellingham diagram: interpret the slope $(-\Delta S°)$ and intercept $(\Delta H°)$ per mole of O_2.
- Apply the linear approximation $\Delta G° = A + BT$ to compute p_{O_2} at a given T.
- Explain why the CO $/CO_2$ mixture is an oxygen-partial-pressure buffer.
- Predict the “elbow” in an Ellingham line at a phase-transition temperature.

3. Why This Matters?

Controlling the atmosphere is **the** lever of extractive metallurgy: whether iron ore becomes metal or stays as oxide is decided by T and p_{O_2}, not by chemistry alone. This chapter underlies:

- **Reduction of metal oxides** — blast furnaces, roasting, and direct-reduction processes.
- **Carbothermic reactions** — the Boudouard equilibrium $2\ \mathrm{CO} = \mathrm{CO_2} + \mathrm{C}$ sets the gas mix at the tuyere.
- **Sintering & annealing atmospheres** — avoiding unwanted oxidation of finished parts.
- **Ellingham diagrams** — a single plot that compares the stability of every oxide at once.

Getting the equilibrium pressure right means the difference between a usable metal and a pile of scale.

4. Intuition First

Why does a solid get to "drop out" of the equilibrium? A pure solid (say $\mathrm{MO}(s)$) has a fixed chemical potential at a given T — raising P from 0 to 1 atm changes it by only about -0.7 J for iron, utterly negligible next to hundreds of kJ. So we set its activity to 1.

Example (roasting): $2\ \mathrm{Cu}(s) + \mathrm{O_2}(g) = 2\ \mathrm{CuO}(s)$ — only p_{O_2} matters.

Picture: imagine the equilibrium as a see-saw balanced on p_{O_2}: the solids are fixed weights on the bench, and only the gas pressure can tip the balance toward metal or oxide.

Concept: with $a_{\text{solid}} = 1$, K depends only on the gas pressure, so ΔG° determines a single equilibrium pressure at each T.

Math: for $\mathrm{M}(s) + \frac{1}{2}\mathrm{O_2}(g) = \mathrm{MO}(s)$,

$$K = \frac{1}{p_{O_2}^{\frac{1}{2}}}, \quad \text{so} \quad p_{O_2\,(\mathrm{eq})} = (K)^{-2}.$$

5. Visual Explanation

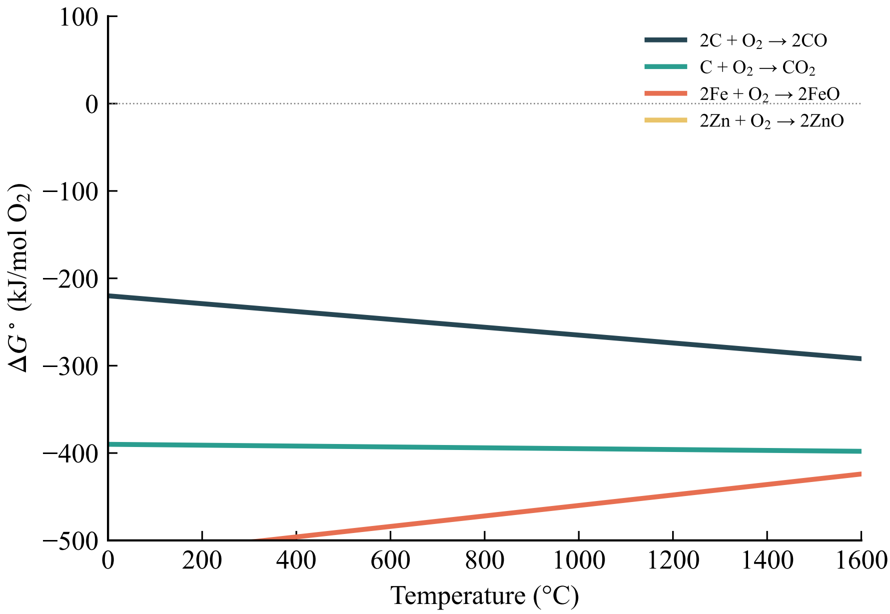


Fig. 21: Ellingham diagram: $\Delta G°$ vs. T for metal-oxidation and carbon-oxide reactions, normalized per mole of O_2. More negative $\Delta G°$ means a more stable oxide.

6. Memory Box

Key facts to keep:

Item	Value / Rule
Activity of pure solid/liquid	$a = 1$ (negligible P effect)
Equilibrium constant	$\Delta G° = -RT \ln K$
Ellingham slope	$-\Delta S°$ per mole O_2 (lines nearly parallel)
Ellingham intercept	$\Delta H°$ at $T \to 0$
CO / CO_2 system	Passes through $(0, 0)$; oxygen buffer
Phase transition	Elbow (bend) in the $\Delta G°$ line

Mnemonic: "$-\Delta S°$ is the **slope**, $\Delta H°$ is the **start**." For oxidations consuming 1 mol O_2, the slopes are all similar because the entropy change is dominated by losing one mole of gas.

7. Common Mistakes

- **Forgetting to normalize $\Delta G°$ per mole of O_2.** Ellingham lines are only comparable when $\Delta G°$ is reported per 1 mol of O_2 consumed; a $\frac{1}{2}O_2$ reaction must be doubled before plotting.
- **Treating K as unitless but mixing pressures.** Use the standard-state pressure (1 atm) consistently; p enters as a ratio, so p_{O_2} must be in atm relative to 1 atm.
- **Sign error in p_{O_2}.** From $\Delta G° = -RT\ln\left(\frac{1}{p_{O_2}^{\frac{1}{2}}}\right)$, one gets $\ln p_{O_2} = 2\Delta G\frac{°}{RT}$ — the factor 2 and the sign are easy to drop.
- **Assuming a linear Ellingham line holds through a phase change.** Melting or vaporization bends the line; the slope is $-(\Delta S° - \Delta S_{\text{trans}})$, not the low-$T$ slope.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$K = \exp(-\Delta G\frac{°}{RT})$	Equilibrium constant from standard free energy	$\Delta G°, R, T$	—	Any reaction
$p_{O_2\ (\text{eq})} = \exp(\frac{2\Delta G°}{RT})$	Equilibrium O_2 pressure, $\frac{1}{2}O_2$ reaction	$\Delta G°, R, T$	atm	Metal + $\frac{1}{2}\mathrm{O_2} = \mathrm{MO}$
$\Delta G° = A + BT$	Linear Ellingham approximation	A, B	J	$\Delta C_P \approx 0$
$B = -\Delta S°$	Slope of Ellingham line per mol O_2	$\Delta S°$	J/K	Const C_P
$K = \frac{p_{CO_2}}{p_{CO}^2}$	Boudouard equilibrium constant	p_{CO}, p_{CO_2}	atm	$2\ \mathrm{CO} = \mathrm{CO_2} + \mathrm{C}(s)$
$\Delta G° = \Delta H° - T\Delta S°$	Definition of standard free energy	$\Delta H°, \Delta S°, T$	J	Standard state

9. Worked Examples

Example 1 — Equilibrium p_{O_2} from $\Delta G°$

Problem For $\mathrm{M}(s) + \frac{1}{2}\mathrm{O_2} = \mathrm{MO}(s)$ with $\Delta G° = -400000 + 80T$ (J/mol). (a) Find p_{O_2} at 1500 K. (b) If $p_{O_2} = 10^{-10}$ atm, which phase is stable?

Thinking Process Write $K = \frac{1}{p_{O_2}^{\frac{1}{2}}}$, substitute into $\Delta G° = -RT\ln K$, and solve for $\ln p_{O_2}$. Then compare actual p_{O_2} with the equilibrium value.

Solution

$$\Delta G^\circ(1500) = -400000 + 80(1500) = -280000 \quad \mathrm{J/mol}$$

$$\Delta G^\circ = -RT \ln\left(\frac{1}{p_{O_2}^{\frac{1}{2}}}\right) = \frac{RT}{2} \ln p_{O_2}$$

$$\ln p_{O_2} = \frac{2\Delta G^\circ}{RT} = \frac{2(-280000)}{8.314 \cdot 1500} = -44.91$$

$$p_{O_2} = \exp(-44.91) = 8.5 \times 10^{-20} \quad \mathrm{atm}$$

For (b): $p_{O_2} = 10^{-10} > 8.5 \times 10^{-20}$, so $Q < K$ and the reaction drives right.

Answer $p_{O_2\ (eq)} = 8.5 \times 10^{-20}$ atm; at 10^{-10} atm metallic M is oxidized to MO.

Interpretation A vastly higher actual oxygen pressure than the equilibrium value means the oxide is the stable product — the solid simply "uses up" the available oxygen until equilibrium is reached.

Example 2 — Boudouard equilibrium gas mix

Problem For 2 $CO(g) = CO_2(g) + C(s)$, $\Delta G^\circ = -16900$ J/mol at 1000 K. Find the equilibrium $CO\ /CO_2$ ratio at total pressure 1 atm.

Thinking Process Compute K, then impose $K = \frac{p_{CO_2}}{p_{CO^2}}$ and $p_{CO} + p_{CO_2} = 1$ atm; solve the quadratic for p_{CO}.

Solution

$$K = \exp\left(-\Delta G\frac{\circ}{RT}\right) = \exp\left(\frac{16900}{8.314 \cdot 1000}\right) = \exp(2.033) = 7.64$$

Let $p_{CO} = x$, $p_{CO_2} = 1 - x$:

$$7.64 = \frac{1-x}{x^2} \Rightarrow 7.64x^2 + x - 1 = 0$$

$$x = \frac{-1 + \sqrt{1 + 4 \cdot 7.64}}{2 \cdot 7.64} = \frac{-1 + 5.618}{15.28} = 0.302 \quad \text{atm}$$

$$p_{CO_2} = 0.698 \quad \text{atm}$$

Answer $p_{CO} = 0.302$ atm, $p_{CO_2} = 0.698$ atm; CO is 30% of the gas.

Interpretation The equilibrium favors CO_2 at 1000 K. Raising T shifts it toward CO (endothermic), which is why higher furnace temperatures increase the reducing power of the gas.

Example 3 — Elbow at a phase transition

Problem For 2 $Cu(s) + O_2 = 2\ CuO(s)$, $\Delta G° = -315000 + 106T$ (J) over 298–1357 K. Cu melts at $T_m = 1357$ K with $\Delta H_m = 13.0$ kJ/mol. Find $\Delta G°$ for $T > T_m$.

Thinking Process Above T_m, Cu becomes liquid. Two moles of Cu gain $2\Delta H_m$ in enthalpy and $2\Delta \frac{H_m}{T_m}$ in entropy, which changes both A and B of the linear form.

Solution

$$\Delta H \text{ rises by } 2\Delta H_m = 26000 \quad \text{J}, \quad \Delta S \text{ rises by } \frac{26000}{1357} = 19.2 \quad \text{J/K}$$

$$A : -315000 \to -315000 + 26000 = -289000 \quad \text{J}$$

$$B : 106 \to 106 - 19.2 = 86.8 \quad \text{J/K}$$

$$\Delta G°(T > T_m) = -289000 + 86.8T \quad \text{J}$$

Answer $\Delta G° = -289000 + 86.8T$ (J) for $T > 1357$ K.

Interpretation The slope flattens from -106 to -86.8; the line bends upward at T_m. Liquid Cu is slightly less eager to react with oxygen than solid Cu — the elbow is a direct signature of the phase change.

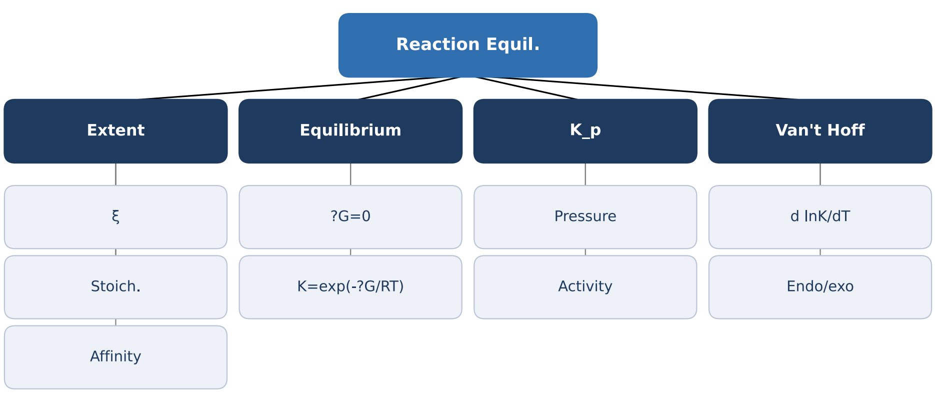
10. Quick Check

1. Why can the activity of a pure solid be set to 1 in these equilibria?
2. For $\mathrm{M} + \frac{1}{2}O_2 = \mathrm{MO}$, what is K in terms of p_{O_2}?
3. On an Ellingham diagram, what does the slope of a line represent?
4. Why does the CO $/\mathrm{CO}_2$ line pass through the origin $(0, 0)$?
5. What happens to an Ellingham line at the melting point of a reactant?

Answers: 1. Pressure has a negligible effect on the chemical potential of a condensed phase over 0–1 atm, so $a = 1$. 2. $K = \frac{1}{p_{\mathrm{O}_2}^{\frac{1}{2}}}$. 3. The slope is $-\Delta S°$ (per mole of O_2 consumed). 4. Because the reaction 2 $\mathrm{CO} + \mathrm{O}_2 = 2\mathrm{CO}_2$ gives $\Delta G° = 0$ at $p_{\mathrm{O}_2} = 1$ atm, so the line crosses the origin. 5. It bends upward (“elbow”): $\Delta H°$ and $\Delta S°$ jump, changing the slope.

11. Chapter Summary

- Pure condensed phases have activity $a = 1$; the equilibrium depends only on the gas partial pressure.
- For $\mathrm{M} + \frac{1}{2}O_2 = \mathrm{MO}$, $K = \frac{1}{p_{\mathrm{O}_2}^{\frac{1}{2}}}$ and $p_{\mathrm{O}_2\ (\mathrm{eq})} = \exp\big(2\Delta G\frac{\circ}{R}T\big)$.
- The Ellingham diagram plots $\Delta G°$ vs. T (per mol O_2); slope $= -\Delta S°$, intercept $\approx \Delta H°$.
- The linear approximation $\Delta G° = A + BT$ lets you compute equilibrium pressures by hand.
- Phase transitions create an elbow in the Ellingham line because $\Delta H°$ and $\Delta S°$ change.
- The CO $/\mathrm{CO}_2$ pair is an excellent oxygen-partial-pressure buffer passing through the origin.



12. Flash Cards

Q Why is the activity of a pure solid 1?

A Pressure changes its chemical potential by < 1 J over 0–1 atm, negligible; set $a = 1$.

Q Relation between $\Delta G°$ and p_{O_2}?

A For $\frac{1}{2}O_2$: $\ln p_{O_2} = 2\Delta G\frac{°}{RT}$.

Q What does an Ellingham slope mean?

A Slope $= -\Delta S°$ per mole of O_2; most oxidation lines are near-parallel.

Q What is the Boudouard reaction?

A $2\ \mathrm{CO} = \mathrm{CO_2} + \mathrm{C}(s)$; its equilibrium sets the CO $/\mathrm{CO_2}$ ratio.

Q Why is CO $/\mathrm{CO_2}$ a buffer?

A Its Ellingham line passes through $(0, 0)$, giving a precise p_{O_2} reference.

Q What causes the Ellingham elbow?

A A phase change (melt/vaporize) shifts $\Delta H°$ and $\Delta S°$, bending the line.

Q How to compare oxide stabilities?

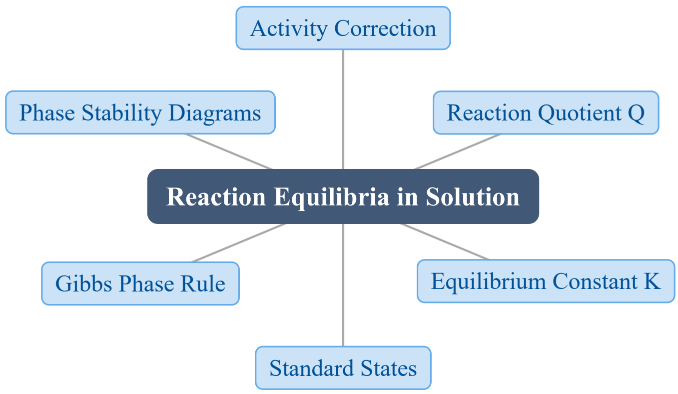
A Lower (more negative) $\Delta G°$ on the Ellingham diagram means the oxide is more stable.

13. Exam Tips

- **High priority:** converting a $\Delta G°(T)$ line into an equilibrium p_{O_2} — master the $\ln p_{O_2} = 2\Delta G\frac{°}{RT}$ step and the factor of 2 from the $\frac{1}{2}O_2$ stoichiometry.
- Always normalize $\Delta G°$ to 1 mol of O_2 before comparing two Ellingham lines or finding a crossing (transition) temperature.
- The Boudouard quadratic is a common exam calculation — practice solving $K = \frac{1-x}{x^2}$ cleanly.
- If a problem mentions melting or boiling, expect an elbow: recompute A and B using $2\Delta H_m$ and $2\Delta\frac{H_m}{T_m}$ for the changed species.

13 Reaction Equilibria in Systems Containing Components in Condensed Solution

1. Chapter Overview



When a reactant or product lives in a **condensed solution** rather than as a pure phase, its Gibbs energy carries an extra $RT \ln a_i$ term. This single correction shifts every reaction equilibrium: the equilibrium constant now reads $\Delta G° = -RT \ln K$ with $K = Q_{\text{eq}}$ built from **activities**. We choose a standard state (Raoultian, Henrian, or 1 wt%), count degrees of freedom with the Gibbs phase rule, and read off stable phases from phase-stability diagrams.

2. Learning Objectives

After studying this chapter, students should be able to:

- Write the equilibrium criterion $\Delta G' = \Delta G° + RT \ln Q$ and show $\Delta G° = -RT \ln K$ at equilibrium.
- Explain how the activity a_i of a component in solution shifts the equilibrium oxygen partial pressure.
- Distinguish the Raoultian, Henrian, and 1 wt% standard states and convert between them.
- Apply the Gibbs phase rule $F = C - \Phi + 2$ (and its constant-P form) to solution systems.
- Read a phase-stability diagram to identify the stable phase at a given T and p_{O_2}.
- Relate the Gibbs free energy of formation $\Delta G°_f$ of a compound to its stability in a binary diagram.

3. Why This Matters?

Almost no industrial reaction runs with pure phases — metals dissolve into alloys, oxides dissolve into slags, and every one of those components has an activity $a_i < 1$. Getting the $RT \ln a_i$ correction right decides real outcomes:

- **Steelmaking** — lowering a metal's activity by alloying changes how easily it oxidizes or is refined.
- **Slag metallurgy** — a lowered oxide activity ($a_{\mathrm{MO}} < 1$) promotes reduction and extraction.
- **Deoxidation & trace elements** — the Henrian and 1 wt% standard states are the working basis for C, N, O, S in steel.
- **Refractory & ceramic stability** — phase-stability diagrams predict which oxide survives at a given p_{O_2}.

The activity term is what connects tabulated $\Delta G°$ data to the **actual** temperature, pressure, and composition of a working furnace.

4. Intuition First

Take $\mathrm{SiO_2}(s) = \mathrm{Si} + \mathrm{O_2}(g)$. If both solids are **pure**, the equilibrium oxygen pressure is fixed by $\Delta G°$ alone — one number at each temperature. Now dissolve the $\mathrm{SiO_2}$ into an alumina–silica melt so its **activity** drops to $a_{\mathrm{SiO_2}} < 1$.

$$p'_{\mathrm{O}_2} = p^{\mathrm{pure}}_{\mathrm{O}_2} \cdot a_{\mathrm{SiO}_2}$$

Picture the reactant as being "diluted": the system has less pure $\mathrm{SiO_2}$ pushing the reaction forward, so it settles at a **lower** equilibrium oxygen pressure and the oxide becomes harder to decompose. The mathematics is just the chemical potential $G_i = G°_i + RT \ln a_i$ threaded through every species. Each activity that falls below 1 shifts the balance — lowering a reactant's activity stabilizes the products, lowering a product's activity destabilizes it.

5. Visual Explanation

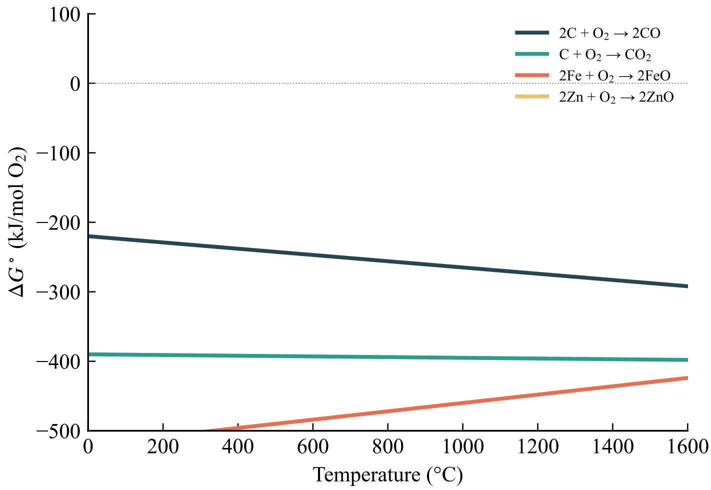


Fig. 22: Ellingham diagram ($\Delta G°$ vs. T). An $RT \ln a$ correction shifts a line's $T = 0$ intercept: $a_{\mathrm{M}} < 1$ enlarges the metal-stable region, $a_{\mathrm{MO}} < 1$ shrinks the oxide-stable region.

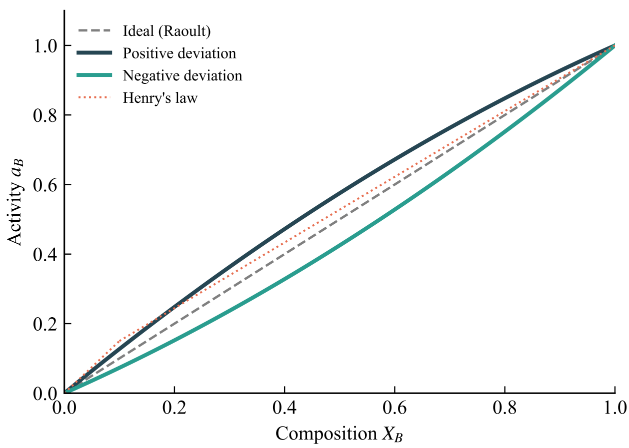


Fig. 23: Activity–composition behavior. Raoultian ($a \to X$ near $X = 1$) and Henrian ($a/X \to \gamma°$ near $X = 0$) standard states are the two limiting reference lines for a solute.

6. Memory Box

The one correction that runs the whole chapter:

$$G_i = G°_i + RT \ln a_i \quad \Rightarrow \quad \Delta G' = \Delta G° + RT \ln Q$$

Three standard states:

Standard state	Reference	Activity form
Raoultian (R)	Pure i, $X_i \to 1$	$a_i = \gamma_i X_i$
Henrian (H)	Infinite dilution, $X_i \to 0$	$a_i = \gamma°_i X_i$
1 wt% (w)	1 wt% ideal-dilute	$a_i = f_i(\text{wt\% } i)$

Sign compass: activity down on a **reactant** $\Rightarrow$ product side favored; activity down on a **product** $\Rightarrow$ reactant side favored. **Phase rule:** $F = C - \Phi + 2$ (or $C - \Phi + 1$ at fixed P).

7. Common Mistakes

- **Setting all activities to 1 out of habit.** $a_i = 1$ holds only for a **pure** condensed phase; a component in solution has $a_i < 1$ and the $RT \ln a_i$ term must be kept.
- **Confusing Q with K.** Q is the activity quotient at **any** state; $K = Q_{\text{eq}}$ only **at equilibrium**, and only then does $\Delta G° = -RT \ln K$.
- **Mixing standard states.** An activity is meaningless without its reference. Raoultian and Henrian activities of the same solute differ by the factor $\gamma°_i$ — never compare them directly.
- **Forgetting the constant-P phase rule.** At fixed total pressure use $F = C - \Phi + 1$; using $C - \Phi + 2$ then over-counts the degrees of freedom by one.
- **Treating a compound like a solution.** A stoichiometric compound has fixed composition and $a = 1$; its stability is set by $\Delta G°_{\text{f}}$, not by a varying mole fraction.

8. Formula Sheet

Formula	Meaning	Variables	Units	Conditions / Limits
$G_i = G°_i + RT \ln a_i$	Partial molar Gibbs energy in solution	G_i, a_i	J/mol	Component i in solution
$\Delta G' = \Delta G° + RT \ln Q$	Reaction Gibbs energy, any state	Q = activity quotient	J	Const T, P
$Q = (a_C^c a_D^d)/(a_A^a a_B^b)$	Reaction (activity) quotient	a_i = activities	—	For $aA + bB = cC + dD$
$\Delta G° = -RT \ln K$	Standard reaction Gibbs energy	$K = Q_{\text{eq}}$	J	At equilibrium, $\Delta G' = 0$
$p'_{O_2} = p_{O_2}^{\text{pure}} \cdot a_{\text{MO}}/a_M$	Activity-shifted eq. oxygen pressure	a_M, a_{MO}	atm	$\text{M} + \frac{1}{2}\text{O}_2 = \text{MO}$

Formula	Meaning	Variables	Units	Conditions / Limits
$a_B = \gamma^\circ{}_B X_B$	Henrian activity	$\gamma^\circ{}_B$ = Henry const.	—	Dilute solute, $X_B \to 0$
$\Delta G^\circ{}_{R\to H} = RT \ln \gamma^\circ{}_B$	Raoultian → Henrian conversion	$\gamma^\circ{}_B$	J/mol	Standard-state change
$F = C - \Phi + 2$	Gibbs phase rule	C, Φ	—	$C - \Phi + 1$ at fixed P

9. Worked Examples

Example 1 — Activity shifts the equilibrium oxygen pressure

Problem For $SiO_2(s) = Si + O_2(g)$, $\Delta G^\circ = 700$ kJ/mol at 1600 K. Find the equilibrium p_{O_2} (a) for pure phases, and (b) when $a_{Si} = 0.01$.

Thinking Process Write $K = \left(a_{Si} p_{O_2}\right)/a_{SiO_2}$ from $\Delta G^\circ = -RT \ln K$. For pure phases all $a = 1$; when Si is diluted, keep a_{Si} in the quotient and solve for p_{O_2}.

Solution

$$K = \exp\left(-\frac{\Delta G^\circ}{RT}\right) = \exp\left(-\frac{700000}{(8.314)(1600)}\right) = \exp(-52.6) = 1.4 \times 10^{-23}$$

(a) Pure phases, $a_{Si} = a_{SiO_2} = 1$:

$$p_{O_2} = K = 1.4 \times 10^{-23} \quad \text{atm}$$

(b) With $a_{Si} = 0.01$: $K = a_{Si} p_{O_2}$, so

$$p_{O_2} = \frac{K}{a_{Si}} = \frac{1.4 \times 10^{-23}}{0.01} = 1.4 \times 10^{-21} \quad \text{atm}$$

Answer (a) $p_{O_2} = 1.4 \times 10^{-23}$ atm; (b) $p_{O_2} = 1.4 \times 10^{-21}$ atm.

Interpretation Lowering the Si activity by a factor of 100 **raises** the equilibrium oxygen pressure by 100×: diluting the metal makes it oxidize more readily. This is why alloying to reduce a metal's activity reduces its oxidation resistance.

Example 2 — Displacement reaction in molten Fe–Mn

Problem At 2073 K, FeO + Mn = MnO + Fe has $\Delta G^\circ = -90100$ J. Assuming Raoultian ideal solutions, find the equilibrium composition ratio.

Thinking Process Get K from $\Delta G^\circ = -RT \ln K$. With ideal solutions each $a_i = X_i$, so K equals the mole-fraction quotient directly.

Solution

$$\ln K = -\frac{\Delta G^\circ}{RT} = \frac{90100}{(8.314)(2073)} = 5.23$$

$$K = e^{5.23} = 186 = \frac{a_{\mathrm{MnO}} a_{\mathrm{Fe}}}{a_{\mathrm{FeO}} a_{\mathrm{Mn}}}$$

Ideal solution ($a_i = X_i$):

$$\frac{X_{\mathrm{MnO}} X_{\mathrm{Fe}}}{X_{\mathrm{FeO}} X_{\mathrm{Mn}}} = 186$$

Answer $K = 186$; the ideal mole-fraction quotient equals 186.

Interpretation The large K shows Mn strongly displaces Fe from FeO — Mn has the greater oxygen affinity. The ratio fixes tie-lines and oxygen isobars on an isothermal section of the Fe–Mn–O diagram.

Example 3 — Gibbs phase rule in the Fe–O system

Problem Fe (s) and FeO (s) coexist via $2\ \mathrm{Fe} + \mathrm{O_2} = 2\ \mathrm{FeO}$ with a gas phase. Find F. Then repeat when Fe is an Fe–C alloy.

Thinking Process Count components C and phases Φ, apply $F = C - \Phi + 2$. Adding carbon raises C by one.

Solution Fe–O: $C = 2$ (Fe, O), $\Phi = 3$ (Fe, FeO, gas):

$$F = C - \Phi + 2 = 2 - 3 + 2 = 1$$

Fe–C alloy: $C = 3$, $\Phi = 3$:

$$F = 3 - 3 + 2 = 2$$

Answer Pure Fe–O: $F = 1$; Fe–C alloy: $F = 2$.

Interpretation With $F = 1$, fixing T alone pins the equilibrium p_{O_2}. Alloying adds a degree of freedom: now both T **and** the carbon activity must be set, so control of the process becomes more demanding.

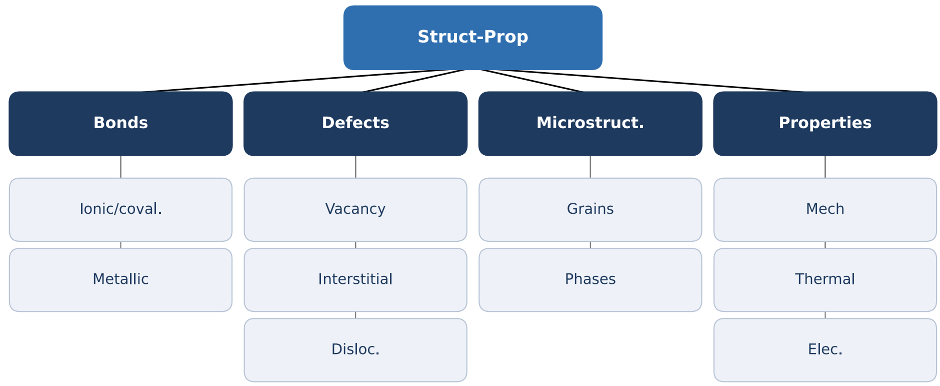
10. Quick Check

1. What extra term appears in G_i when component i is in solution?
2. When does Q equal K?
3. If a reactant's activity drops below 1, does the reaction shift toward products or reactants?
4. State the constant-pressure form of the Gibbs phase rule.
5. Which standard state uses infinite dilution as its reference?

Answers: 1. $RT \ln a_i$ 2. Only at equilibrium ($\Delta G' = 0$) 3. Toward products 4. $F = C - \Phi + 1$ 5. Henrian

11. Chapter Summary

- In solution each component carries $G_i = G°_i + RT \ln a_i$, so $\Delta G' = \Delta G° + RT \ln Q$.
- At equilibrium $Q = K$ and $\Delta G° = -RT \ln K$; activities $= 1$ shift the equilibrium and the eq. p_{O_2}.
- Lowering a reactant activity favors products; lowering a product activity favors reactants.
- Three standard states — Raoultian, Henrian ($a_B = \gamma°_B X_B$), and 1 wt% — with $\Delta G°_{R \to H} = RT \ln \gamma°_B$.
- The Gibbs phase rule $F = C - \Phi + 2$ (or $C - \Phi + 1$ at fixed P) counts the degrees of freedom.
- Phase-stability diagrams and $\Delta G°_f$ of compounds identify the stable phase at given T and p_{O_2}.



12. Flash Cards

Q Equilibrium criterion in solution?

A $\Delta G' = \Delta G^\circ + RT \ln Q$; at equilibrium $\Delta G^\circ = -RT \ln K$, $K = Q_{\text{eq}}$.

Q How does dilution change G_i?

A Adds $RT \ln a_i$ with $a_i < 1$, lowering the partial molar Gibbs energy.

Q Activity-shifted oxygen pressure?

A $p'_{O_2} = p^{\text{pure}}_{O_2} \cdot a_{\text{MO}}/a_M$.

Q Henrian activity and constant?

A $a_B = \gamma^\circ{}_B X_B$; $\gamma^\circ{}_B$ is the Henry's law constant (dilute limit).

Q Raoultian → Henrian conversion?

A $\Delta G^\circ{}_{R\to H} = RT \ln \gamma^\circ{}_B$.

Q Gibbs phase rule?

A $F = C - \Phi + 2$ (general); $F = C - \Phi + 1$ at fixed total pressure.

Q What sets a compound's stability?

A Its Gibbs energy of formation $\Delta G^\circ{}_{\text{f}} = -RT \ln K_{\text{f}}$; its activity is fixed at 1.

13. Exam Tips

- **High priority:** compute a shifted equilibrium p_{O_2} from $\Delta G°$ and given activities — always start from $\Delta G° = -RT\ln K$ and keep every $a_i = 1$ in Q.
- State the **standard state** before quoting any activity; conversion questions hinge on $\Delta G°_{R\to H} = RT\ln\gamma°_B$.
- For phase-rule problems, count the gas phase explicitly and switch to $F = C - \Phi + 1$ when P is fixed.
- On an Ellingham/stability diagram, the **lower** line (more negative $\Delta G°$, lower eq. p_{O_2}) marks the stronger oxygen affinity — that element oxidizes first.
- Watch the sign logic: reactant activity $\downarrow$ favors products; product activity $\downarrow$ favors reactants.